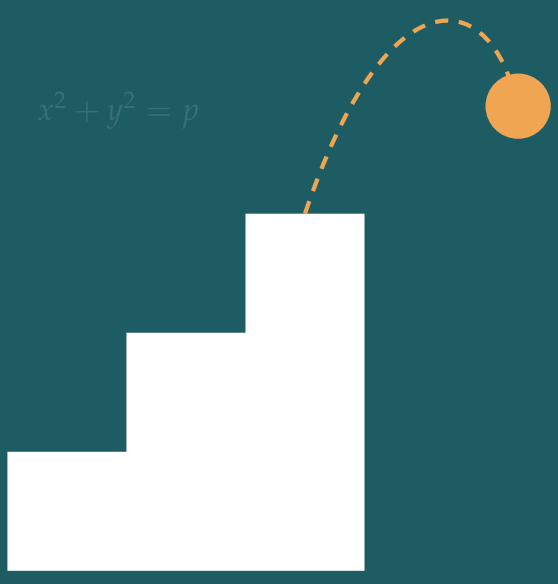
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2002 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2002 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2002 Worked Solutions

Worked solutions to all 42 questions of the 2002 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Circles and Ellipses Pure . . . 3
Q2 Curve Sketching Pure . . . 6
Q3 Inequalities by Calculus Pure . . . 10
Q4 Tangents and Inequalities Pure . . . 14
Q5 Integer Roots of Polynomials Pure . . . 18
Q6 Pyramid Geometry Pure . . . 21
Q7 Trigonometric Integrals Pure . . . 25
Q8 Compound Interest Loans Pure . . . 29
Q9 Toppling on an Incline Mechanics . . . 32
Q10 Elastic Collisions Mechanics . . . 36
Q11 Oblique Elastic Collisions Mechanics . . . 40
Q12 Geometric Probability Probability and Statistics . . . 43
Q13 Conditional Expectation Probability and Statistics . . . 47
Q14 Geometric Series in Probability Probability and Statistics . . . 50

STEP II

Q1 Integration by Substitution Pure . . . 53
Q2 Reciprocal Polynomial Equations Pure . . . 56
Q3 Fermat Numbers and Primes Pure . . . 59
Q4 Inequalities from Integrals Pure . . . 62
Q5 Logistic Map Iteration Pure . . . 65
Q6 Lines in Inclined Planes Pure . . . 69
Q7 Vector Geometry of Lines Pure . . . 72
Q8 First-Order Differential Equations Pure . . . 76
Q9 Projectile Motion Mechanics . . . 80
Q10 Kinematics and Optimisation Mechanics . . . 84
Q11 Friction and Limiting Equilibrium Mechanics . . . 88
Q12 Normal and Poisson Approximations Probability and Statistics . . . 92
Q13 Transformed Distribution Functions Probability and Statistics . . . 96

Q14 Mixtures and Bayes' Theorem Probability and Statistics . . . 99

STEP III

Q1 Improper Integrals and Volumes Pure . . . 102
Q2 Inverse Tangent Series Pure . . . 105
Q3 Curve Sketching Pure . . . 108
Q4 Diophantine Inequalities Pure . . . 112
Q5 Factorising Quartic Equations Pure . . . 115
Q6 Nonlinear Differential Equations Pure . . . 118
Q7 Coordinate Geometry Identities Pure . . . 122
Q8 Ptolemy's Theorem Pure . . . 126
Q9 Centre of Mass and Toppling Mechanics . . . 130
Q10 Rotation with Friction Mechanics . . . 134
Q11 Oblique Impact with Walls Mechanics . . . 138
Q12 Expectation by Conditioning Probability and Statistics . . . 142
Q13 Minimum of Exponentials Probability and Statistics . . . 146
Q14 Variance and Covariance Probability and Statistics . . . 149

WORKED SOLUTION

STEP I 2002, Question 1

Pure • Circles and Ellipses

> **STEP I 2002, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/02-s1-q1

The key idea

Find where the two ellipses actually meet. Both equations involve y only through y^2, so eliminating y^2 leaves a quadratic in x, and it turns out that the ellipses cross at just two real points. A circle through two given points has its centre on their perpendicular bisector, and that pins the equation down to a one-parameter family.

Solution

Finding the points of intersection

Expanding, the two ellipses are

$$x^2 + 4x + 2y^2 = 14,$$
$$9x^2 - 18x + 16y^2 = 16.$$

Multiply the first by 8 and subtract the second, which removes y^2:

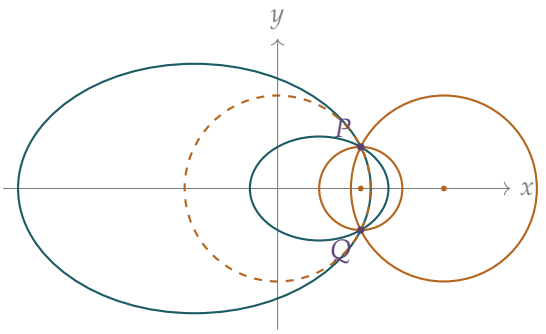
$$8x^2 + 32x - 9x^2 + 18x = 112 - 16 \implies x^2 - 50x + 96 = 0 \implies (x-2)(x-48) = 0.$$

> **Ask yourself** Why can't we just keep both roots?

Each root still has to give a real y. From the first ellipse, $2y^2 = 14 - x^2 - 4x$.

- $x = 48$: $2y^2 = 14 - 2304 - 192 < 0$, so there is no real point. (In any case the first ellipse only covers $-2 - 3\sqrt{2} \le x \le -2 + 3\sqrt{2}$.)
- $x = 2$: $2y^2 = 14 - 4 - 8 = 2$, so $y = \pm 1$.

So the ellipses meet in exactly two points, $P = (2, 1)$ and $Q = (2, -1)$. (Both satisfy the second equation as well: $9 + 16 = 25$.)



The figure shows the two ellipses (teal) and three of the circles through P and Q (amber), with centres $(0,0)$, $(2,0)$ and $(4,0)$.

The circles through P and Q

The centre of any circle through P and Q is equidistant from them, so it lies on the perpendicular bisector of PQ. The segment PQ is vertical with midpoint $(2,0)$, so its perpendicular bisector is the x-axis. Hence the centre is $(a,0)$ for some real a, and the radius is

$$r^2 = (2-a)^2 + 1^2.$$

The circle is therefore

$$\begin{aligned}(x-a)^2 + y^2 &= (2-a)^2 + 1\\ x^2 - 2ax + a^2 + y^2 &= 4 - 4a + a^2 + 1\\ x^2 - 2ax + y^2 &= 5 - 4a,\end{aligned}$$

which is the required form.

Conversely, for every real a this equation is a genuine circle, centre $(a,0)$ and radius $\sqrt{(a-2)^2+1} > 0$, and it passes through $(2,\pm 1)$ since $4 - 4a + 1 = 5 - 4a$. So the family is exactly the set of circles through the points of intersection.

Ask yourself What about the usual trick of taking a combination of the two equations?

Combining $7\times$(first) $-$ (second) makes the x^2 and y^2 coefficients equal and gives $x^2 - 23x + y^2 + 41 = 0$, which is the member $a = \frac{23}{2}$ of our family. But it is only one circle. Combinations of the two equations are the conics through all four (complex) common points, including the two non-real ones with $x = 48$, while a circle only has to pass through the two real points. That is why the direct approach is needed here.

Checks

- $a = 2$ gives $x^2 - 4x + y^2 = -3$, that is $(x-2)^2 + y^2 = 1$: the circle with PQ as diameter, as it should be.

- $a = 0$ gives $x^2 + y^2 = 5$, and indeed $2^2 + 1^2 = 5$.
- A numerical scan of $-6 \leq x \leq 6$ for points lying on both ellipses finds only $x = 2$. At $(2, 1)$ the normals to the ellipses are along $(8, 4)$ and $(18, 32)$, which are not parallel, so the ellipses genuinely cross there rather than touching.

Common slips

- Keeping $x = 48$ without checking that y is real.
- Producing the single circle from a combination of the ellipse equations and stopping there: the question asks about *any* circle through the points.

Beyond STEP

The combination $\lambda E_1 + \mu E_2 = 0$ of two conics is called a *pencil*: it consists of the conics through their four common points, counted in the complex projective plane. This question is a neat reminder that some of those points may be non-real, and that circles are special (they all pass through two fixed complex "circular points at infinity"). These ideas belong to projective and algebraic geometry; a friendly first look at plane curves and conics in this spirit is Reid, *Undergraduate Algebraic Geometry*, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

WORKED SOLUTION

STEP I 2002, Question 2

Pure • Curve Sketching

STEP I 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q2

The key idea

The polynomial $x^m(x-1)^n$ vanishes at $x=0$ and $x=1$, so something must turn round in between. Differentiate with the product rule and *take out every common factor*: the derivative then splits into simple pieces whose signs we can read off. The whole sketch is controlled by parity: whether a power is even or odd decides whether the curve touches the axis or crosses it, and which side of the axis it is on.

Solution

The stationary point

By the product rule,

$$\begin{aligned} \mathrm{f}'(x) &= mx^{m-1}(x-1)^n + nx^m(x-1)^{n-1} \\ &= x^{m-1}(x-1)^{n-1}\big[m(x-1)+nx\big] \\ &= x^{m-1}(x-1)^{n-1}\big[(m+n)x-m\big]. \end{aligned}$$

So $\mathrm{f}'(x)=0$ exactly when $x=0$, $x=1$ or

$$x = x_0 = \frac{m}{m+n}.$$

Since $m, n > 0$ we have $0 < m < m+n$, so $0 < x_0 < 1$. That is the required stationary point. (Since $m, n \geq 2$, the powers $m-1$ and $n-1$ are at least 1, so $x=0$ and $x=1$ are stationary points too. We will need them for the sketches.)

Its nature

Write $\mathrm{f}'(x) = g(x)\,h(x)$ with $g(x) = x^{m-1}(x-1)^{n-1}$ and $h(x) = (m+n)x - m$. Then

$$\mathrm{f}''(x) = g'(x)h(x) + g(x)h'(x) = g'(x)h(x) + (m+n)g(x).$$

Ask yourself Do we really need to differentiate g in full?

No. At x_0 we have $h(x_0) = 0$, so the first term vanishes, whatever g' is:

$$\mathrm{f}''(x_0) = (m+n)\, x_0^{m-1}\,(x_0-1)^{n-1}.$$

Here $m+n > 0$ and $x_0^{m-1} > 0$, but $x_0 - 1 < 0$. So $\mathrm{f}''(x_0)$ is nonzero, with the sign of $(-1)^{n-1}$:

- if n is even, $n-1$ is odd, $\mathrm{f}''(x_0) < 0$ and x_0 is a **maximum**;
- if n is odd, $n-1$ is even, $\mathrm{f}''(x_0) > 0$ and x_0 is a **minimum**.

Because $\mathrm{f}''(x_0) \neq 0$, the second-derivative test is conclusive. That is what we had to show.

The sketches

The four cases are given by the parities of m and n. To sketch each one, we collect these facts.

Sign of f. For $x > 1$ both factors are positive, so $\mathrm{f} > 0$. For $0 < x < 1$, $x^m > 0$ and $(x-1)^n$ has the sign of $(-1)^n$. For $x < 0$, the sign is $(-1)^m(-1)^n = (-1)^{m+n}$.

Behaviour at the roots. Near $x = 0$ we have $\mathrm{f}(x) \approx (-1)^n x^m$. If m is even, f keeps its sign either side of 0, so the curve touches the axis there (a turning point). If m is odd, f changes sign, and because $\mathrm{f}'(0) = 0$ the curve crosses the axis horizontally (a stationary point of inflection). The same reasoning with n applies at $x = 1$.

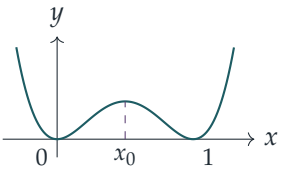
Ends. $\mathrm{f}(x)$ behaves like x^{m+n} for large $|x|$, so $\mathrm{f} \to +\infty$ as $x \to +\infty$. As $x \to -\infty$, $\mathrm{f} \to +\infty$ if $m+n$ is even and $\mathrm{f} \to -\infty$ if $m+n$ is odd.

No other stationary points. The only ones are $x = 0$, x_0 and $x = 1$.

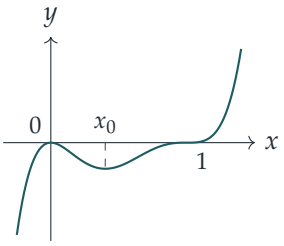
Putting these together:

- m **even,** n **even:** $\mathrm{f} \geq 0$ everywhere. There are minima on the axis at 0 and 1 and a maximum at x_0: a W shape.
- m **even,** n **odd:** the curve rises from $-\infty$ to touch the axis from below at 0 (a maximum), falls to the minimum at x_0, then rises through a stationary inflection at 1 and on to $+\infty$.
- m **odd,** n **even:** the curve rises from $-\infty$ through a stationary inflection at 0 to the maximum at x_0, falls to touch the axis at 1 (a minimum), then rises to $+\infty$.
- m **odd,** n **odd:** the curve comes down from $+\infty$, through a stationary inflection at 0 to the minimum at x_0, back up through a stationary inflection at 1, and on to $+\infty$.

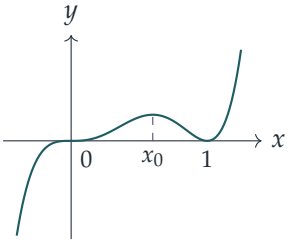
The sketches below use the smallest cases $(m,n) = (2,2), (2,3), (3,2), (3,3)$, each with its own vertical scale. The values at x_0 are small: $\frac{1}{16}$, $-\frac{108}{3125}$, $\frac{108}{3125}$ and $-\frac{1}{64}$.



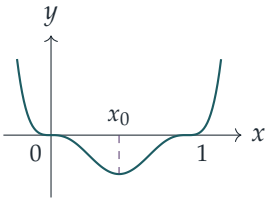
m even, n even



m even, n odd



m odd, n even



m odd, n odd

Checks

- Take $m = n = 2$: $\mathrm{f}(x) = \big(x(x-1)\big)^2$ is symmetric about $x = \frac{1}{2}$, and $x_0 = \frac{1}{2}$ is indeed a maximum, with value $\frac{1}{16}$.
- The substitution $x \mapsto 1 - x$ turns $x^m(x-1)^n$ into $(-1)^{m+n}x^n(x-1)^m$. So swapping m and n reflects the graph in $x = \frac{1}{2}$ (and also in the x-axis if $m + n$ is odd). This is why the two mixed-parity sketches are mirror images of each other.
- We checked f' against a numerical derivative, and the sign of $\mathrm{f}''(x_0)$ against a finite difference, for all $2 \le m, n \le 8$. For $(m, n) = (2, 3)$, for instance, $\mathrm{f}''(\frac{2}{5}) = 5 \cdot \frac{2}{5} \cdot \frac{9}{25} = 0.72 > 0$, a minimum, as predicted.

Common slips

- Expanding $\mathrm{f}''(x)$ in full. At x_0 the factor $(m+n)x - m$ is zero, so only one term survives.
- Forgetting that $x = 0$ and $x = 1$ are also stationary points (because $m, n \ge 2$), and drawing the curve crossing the axis at an angle there.
- Getting the sign for $x < 0$ wrong. It depends on $m + n$, not on m alone.

Beyond STEP

The local shape of a polynomial near a root of multiplicity k looks like $c(x-a)^k$, and the parity of k decides touching or crossing. This is the first step towards Taylor's theorem and the higher-derivative test for stationary points: if the first nonzero derivative at a stationary point has order k, the point is a turning point when k is even and an inflection when k is odd. Abbott's *Understanding Analysis* puts these ideas on a rigorous footing.

WORKED SOLUTION

STEP I 2002, Question 3

Pure • Inequalities by Calculus

STEP I 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q3

The key idea

Call the two square roots $\sqrt{P}$ and $\sqrt{Q}$. The quantities under them are nicely linked: $P + Q = a^2 + b^2$ is constant, and their derivatives are equal and opposite. That makes $\mathrm{d}y/\mathrm{d}\theta$ factorise cleanly, and its sign is controlled by a single trigonometric factor. Once we know the largest and smallest values of y, the double inequality is just "min $\leq y \leq$ max", and the first part tells us the two ends are in the right order.

Solution

The first inequality

Expanding,

$$2a^2 + 2b^2 - (a+b)^2 = a^2 - 2ab + b^2 = (a-b)^2 \geq 0,$$

so $(a+b)^2 \leq 2a^2 + 2b^2$, with equality exactly when $a = b$.

The stationary points

Write $c = \cos\theta$, $s = \sin\theta$ and

$$P = a^2c^2 + b^2s^2, \qquad Q = a^2s^2 + b^2c^2, \qquad y = \sqrt{P} + \sqrt{Q}.$$

Two special cases first. If $a^2 = b^2$, then $P = Q = a^2$ for every θ, so $y = 2|a|$ is constant and every point is (trivially) stationary; we set this aside. If $a = 0$ or $b = 0$ (but not both), P or Q vanishes at some θ and y is not differentiable there; we deal with this at the end. So for now assume $ab \neq 0$ and $a^2 \neq b^2$. Then $P \geq \min(a^2, b^2)(c^2 + s^2) > 0$, and likewise $Q > 0$, so y is differentiable everywhere.

Differentiating,

$$\frac{\mathrm{d}P}{\mathrm{d}\theta} = -2a^2cs + 2b^2sc = (b^2 - a^2)\sin 2\theta, \qquad \frac{\mathrm{d}Q}{\mathrm{d}\theta} = (a^2 - b^2)\sin 2\theta.$$

Hence

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{(b^2-a^2)\sin 2\theta}{2\sqrt{P}} - \frac{(b^2-a^2)\sin 2\theta}{2\sqrt{Q}} = \frac{(b^2-a^2)\sin 2\theta\left(\sqrt{Q}-\sqrt{P}\right)}{2\sqrt{PQ}}.$$

To see the sign of $\sqrt{Q}-\sqrt{P}$, rationalise:

$$\sqrt{Q}-\sqrt{P} = \frac{Q-P}{\sqrt{Q}+\sqrt{P}}, \qquad Q-P = (a^2-b^2)(s^2-c^2) = (b^2-a^2)\cos 2\theta.$$

Putting this in, and using $2\sin 2\theta\cos 2\theta = \sin 4\theta$,

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{(a^2-b^2)^2\sin 4\theta}{4\sqrt{PQ}\left(\sqrt{P}+\sqrt{Q}\right)}.$$

Ask yourself Why is it worth the trouble of rationalising $\sqrt{Q}-\sqrt{P}$?

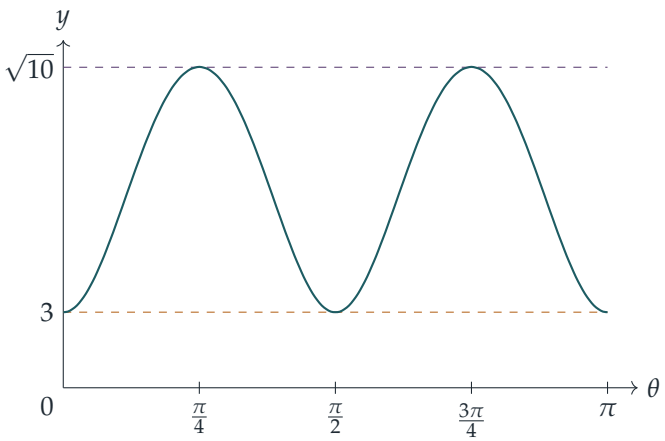
Because now everything except $\sin 4\theta$ is visibly positive: $(a^2-b^2)^2 > 0$, and P, Q are positive. So $\mathrm{d}y/\mathrm{d}\theta$ has exactly the sign of $\sin 4\theta$, which tells us both where the stationary points are and what kind they are.

The stationary points are where $\sin 4\theta = 0$, that is $\theta = \frac{1}{4}n\pi$ for integer n.

- When n is even, $\theta = \frac{1}{2}k\pi$, so $\{c^2, s^2\} = \{1, 0\}$ and $\{P, Q\} = \{a^2, b^2\}$. Here $y = |a| + |b|$ (remember $\sqrt{a^2} = |a|$).
- When n is odd, $\theta = \frac{1}{4}\pi + \frac{1}{2}k\pi$, so $c^2 = s^2 = \frac{1}{2}$ and $P = Q = \frac{1}{2}(a^2+b^2)$. Here $y = 2\sqrt{\frac{1}{2}(a^2+b^2)} = \sqrt{2a^2+2b^2}$.

Nature. As θ increases through $\frac{1}{2}k\pi$, $\sin 4\theta$ changes from negative to positive, so y goes from decreasing to increasing: these points are **minima**. As θ increases through $\frac{1}{4}\pi + \frac{1}{2}k\pi$, $\sin 4\theta$ changes from positive to negative, so these points are **maxima**.

The picture below shows $a = 2$, $b = 1$, with the vertical scale stretched: y oscillates between 3 and $\sqrt{10} \approx 3.162$, with period $\frac{1}{2}\pi$.



The final inequality

Replacing θ by $\theta + \frac{1}{2}\pi$ swaps c^2 and s^2, hence swaps P and Q, so y has period $\frac{1}{2}\pi$. On one period $[0, \frac{1}{2}\pi]$, y increases on $[0, \frac{1}{4}\pi]$ and decreases on $[\frac{1}{4}\pi, \frac{1}{2}\pi]$ (sign of $\sin 4\theta$), so its least value is $|a| + |b|$, at the ends, and its greatest is $\sqrt{2a^2 + 2b^2}$, at $\frac{1}{4}\pi$. By periodicity these are the least and greatest values for all θ, so

$$|a| + |b| \;\le\; y \;\le\; \sqrt{2a^2 + 2b^2}.$$

These two bounds are consistent: applying the first part to $|a|$ and $|b|$ gives $(|a| + |b|)^2 \le 2a^2 + 2b^2$, so the lower bound really does not exceed the upper one.

The cases set aside. If $a^2 = b^2$, then $y = 2|a|$, while $|a| + |b| = 2|a|$ and $\sqrt{2a^2 + 2b^2} = \sqrt{4a^2} = 2|a|$, so the inequality holds with equality throughout. If $b = 0$, then $y = |a|(|\cos\theta| + |\sin\theta|)$. Now $|\cos\theta| + |\sin\theta| \ge \cos^2\theta + \sin^2\theta = 1$ since each term is at most 1, and by the first part $(|\cos\theta| + |\sin\theta|)^2 \le 2$; so $|a| \le y \le \sqrt{2}\,|a|$, which is the required inequality. The case $a = 0$ is the same with a and b swapped, and $a = b = 0$ is trivial. This proves the result for all a, b and θ. ■

Checks

- With $a = 2$, $b = 1$: $y(0) = 2 + 1 = 3$ and $y(\frac{1}{4}\pi) = 2\sqrt{5/2} = \sqrt{10} \approx 3.1623$; a numerical scan of y over many θ stays between these.
- The derivative formula agrees with a finite-difference estimate of $\mathrm{d}y/\mathrm{d}\theta$ to about 10^{-9} at random a, b, θ.
- With $a = b$ both bounds collapse to $2|a|$, matching the constant y.

Common slips

- Writing $\sqrt{a^2} = a$: the minimum value is $|a| + |b|$, which is why moduli appear in the answer.
- Stopping at $\sin 2\theta = 0$ and missing the stationary points at odd multiples of $\frac{1}{4}\pi$, where $\sqrt{P} = \sqrt{Q}$.
- Calling points maxima or minima without a reason; the sign of $\sin 4\theta$ gives one in a line.

Beyond STEP

Geometrically, $\sqrt{P}$ is the distance from the centre of the ellipse $x = a\cos\phi$, $y = b\sin\phi$ to a point on it, and $\sqrt{Q}$ is the distance to the point a quarter-turn further round in parameter, so the question is about how these two "conjugate" radii share out a fixed total $a^2 + b^2$. The two bounds are also the ℓ^1 and (scaled) ℓ^2 norms of the vector (a, b),

and comparing norms like this is a routine tool in first-year analysis and linear algebra. For more on using derivatives to prove inequalities, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2002, Question 4

Pure • Tangents and Inequalities

STEP I 2002, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q4

The key idea

A line through $(0,1)$ that touches the curve further along must lie *below* the curve between those two points, because the curve is concave there and then flattens out. A region under a straight line is a trapezium, and rotating it about the y-axis gives a cylinder and a cone, so its area and volume are easy to find without calculus. Comparing them with the exact area and volume under the curve gives the two inequalities.

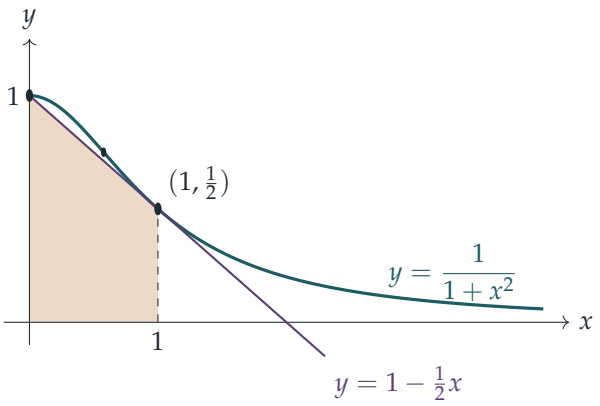
Solution

The sketch

Write $f(x) = \dfrac{1}{1+x^2}$. Then $f(0) = 1$, $f(x) > 0$, and $f(x) \to 0$ as $x \to \infty$, so the x-axis is an asymptote. Differentiating,

$$f'(x) = -\frac{2x}{(1+x^2)^2}, \qquad f''(x) = \frac{6x^2-2}{(1+x^2)^3}.$$

So $f'(0) = 0$ (the curve starts horizontally at $(0,1)$) and $f'(x) < 0$ for $x > 0$, so the curve decreases throughout. Also $f''(x) < 0$ for $0 \le x < 1/\sqrt{3}$ and $f''(x) > 0$ for $x > 1/\sqrt{3}$, so there is a point of inflection at $\left(1/\sqrt{3}, \frac{3}{4}\right)$: the curve is concave (bending down) near the y-axis and convex further out.



The shaded trapezium is the region under the line for $0 \le x \le 1$; it lies inside the region under the curve.

The line

The line passes through $(0, f(0)) = (0,1)$, so it is $y = 1 + mx$ for some gradient m. Suppose it is tangent at $x = a > 0$. Two conditions must hold there.

It meets the curve: $1 + ma = \dfrac{1}{1+a^2}$, so $ma = -\dfrac{a^2}{1+a^2}$, and since $a \neq 0$,

$$m = -\frac{a}{1+a^2}.$$

It has the curve's gradient: $m = f'(a) = -\dfrac{2a}{(1+a^2)^2}$.

Equating these and dividing by $-a/(1+a^2)$ (allowed, as $a > 0$) gives $1 = \dfrac{2}{1+a^2}$, so $a^2 = 1$ and, since $a > 0$, $a = 1$. Then $m = -\frac{1}{2}$, and the line is

$$y = 1 - \tfrac{1}{2}x,$$

touching the curve at $(1, \frac{1}{2})$.

No further intersections

Subtract the curve from the line:

$$\left(1 - \frac{x}{2}\right) - \frac{1}{1+x^2} = \frac{(2-x)(1+x^2) - 2}{2(1+x^2)} = \frac{-x^3 + 2x^2 - x}{2(1+x^2)} = -\frac{x(x-1)^2}{2(1+x^2)}.$$

This is zero only when $x = 0$ or $x = 1$. So the line meets the curve at $x = 0$ and at the point of tangency $x = 1$ (the repeated factor $(x-1)^2$ is the algebraic sign of tangency), and nowhere else: not for $x \ge 0$, and in fact not for negative x either.

We also read off something we need next. For $x > 0$ and $x \neq 1$ the expression is strictly negative, so

$$0 < 1 - \tfrac{1}{2}x < \frac{1}{1+x^2} \qquad \text{for } 0 < x < 1. \tag{$*$}$$

(The left inequality holds because $1 - \frac{1}{2}x \geq \frac{1}{2}$ on $[0,1]$.) So on $[0,1]$ the line lies strictly below the curve except at the endpoints, as in the sketch.

$\pi > 3$

Integrate $(*)$ over $[0,1]$. The two functions are continuous and the curve is strictly above the line on the open interval, so the inequality survives integration as a strict one:

$$\int_0^1 \frac{1}{1+x^2}\,\mathrm{d}x > \int_0^1 \Big(1 - \frac{x}{2}\Big)\mathrm{d}x.$$

The left side is $\frac{\pi}{4}$ (it is $\big[\arctan x\big]_0^1$). The right side is the area of the shaded trapezium, with parallel sides 1 and $\frac{1}{2}$ and width 1, so it is $\frac{1}{2}\big(1 + \frac{1}{2}\big) = \frac{3}{4}$. Hence $\frac{\pi}{4} > \frac{3}{4}$, that is,

$$\pi > 3.$$

$\ln 2 > 2/3$

Now rotate the same two regions (for $0 \leq x \leq 1$) through 2π about the y-axis. By $(*)$, the trapezium sits inside the region under the curve, so its solid of revolution sits inside the other one, and has smaller volume.

> **Ask yourself** How do we find a volume of revolution about the y-axis when the curve is given as y in terms of x?

Use cylindrical shells: the thin strip at x of width δx and height y sweeps out a shell of volume about $2\pi x\, y\, \delta x$, so the volume is $\int_0^1 2\pi x\, y\,\mathrm{d}x$. (Alternatively, slice horizontally; see the Checks.)

For the curve,

$$V_{\text{curve}} = \int_0^1 \frac{2\pi x}{1+x^2}\,\mathrm{d}x = \pi\big[\ln(1+x^2)\big]_0^1 = \pi \ln 2,$$

since the numerator $2x$ is the derivative of the denominator.

For the line, the solid is a cylinder of radius 1 and height $\frac{1}{2}$ (from the part of the trapezium below $y = \frac{1}{2}$) topped by a cone of radius 1 and height $\frac{1}{2}$ (from the triangle above it). So

$$V_{\text{line}} = \pi \cdot 1^2 \cdot \tfrac{1}{2} + \tfrac{1}{3}\pi \cdot 1^2 \cdot \tfrac{1}{2} = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}.$$

As a check by shells: $\int_0^1 2\pi x\big(1 - \frac{1}{2}x\big)\mathrm{d}x = 2\pi\big(\frac{1}{2} - \frac{1}{6}\big) = \frac{2\pi}{3}$.

The integrand $2\pi x\big(\frac{1}{1+x^2} - 1 + \frac{x}{2}\big)$ is continuous, non-negative on $[0,1]$ and strictly positive on $(0,1)$ by $(*)$, so $V_{\text{curve}} > V_{\text{line}}$:

$$\pi \ln 2 > \frac{2\pi}{3} \quad\Longrightarrow\quad \ln 2 > \frac{2}{3},$$

as required.

Checks

- Numerically, $\pi/4 \approx 0.7854 > 0.75$ and $\ln 2 \approx 0.6931 > 0.6667$, so both inequalities are true and neither is very slack.
- Slicing horizontally instead: for $\frac{1}{2} \le y \le 1$ the curve has $x^2 = \frac{1}{y} - 1$, so the volume is the cylinder $\pi \cdot 1^2 \cdot \frac{1}{2}$ plus $\int_{1/2}^{1} \pi\left(\frac{1}{y} - 1\right) \mathrm{d}y = \pi\left(\ln 2 - \frac{1}{2}\right)$, giving $\pi \ln 2$ again.
- Quick test of the factorisation: at $x = 2$ the line gives 0 and the curve $\frac{1}{5}$, a difference of $-\frac{1}{5}$; the formula gives $-\frac{2 \cdot 1}{2 \cdot 5} = -\frac{1}{5}$.

Common slips

- Finding the tangent at a general point and forgetting the condition that it passes through $(0, 1)$ (or vice versa): you need both conditions to pin down a.
- Stopping at "$\ge$": the strict inequalities need the line to be strictly below the curve on an interval, which is what the factor $x(x-1)^2$ gives you.
- Rotating about the x-axis by mistake, which gives $\int \pi y^2 \, \mathrm{d}x$ and different numbers.

Beyond STEP

The argument is a comparison principle: if $g \le h$ on an interval, with strict inequality somewhere and both continuous, then $\int g < \int h$. Making "strictly somewhere" precise is a typical exercise in a first analysis course, where the integral is defined via upper and lower sums. Bounding π by areas of simple shapes goes back to Archimedes, and much sharper versions of this question (such as $\int_0^1 x^4(1-x)^4/(1+x^2)\,\mathrm{d}x = \frac{22}{7} - \pi$) are worth looking up. For the analysis, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2002, Question 5

Pure • Integer Roots of Polynomials

STEP I 2002, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q5

The key idea

A polynomial written as a product of linear factors is easy to evaluate at special points: putting in $x = 0$, $x = 1$ or $x = -1$ turns the factorised form into a product of the k_i, $k_i + 1$ or $k_i - 1$, while the expanded form gives a sum of coefficients. For the quartic, those values are fixed integers, and integer roots mean every $k_i \pm 1$ must be a divisor of a known number. Factorising those numbers leaves very little room.

Solution

The three identities

Since the two expressions for $f(x)$ are the same polynomial, they agree at every value of x.

Putting $x = 0$: the expanded form gives $f(0) = a_n$, since every other term contains a factor of x. The factorised form gives $f(0) = k_1 k_2 \cdots k_n$. Hence

$$k_1 k_2 \cdots k_n = a_n.$$

Putting $x = 1$: every power of x becomes 1, so the expanded form gives $f(1) = 1 + a_1 + a_2 + \cdots + a_n$, while the factorised form gives $(1 + k_1)(1 + k_2) \cdots (1 + k_n)$. Hence

$$(k_1 + 1)(k_2 + 1) \cdots (k_n + 1) = 1 + a_1 + a_2 + \cdots + a_n,$$

as required.

Putting $x = -1$: the factorised form gives $(k_1 - 1)(k_2 - 1) \cdots (k_n - 1)$. In the expanded form, the term $a_j x^{n-j}$ becomes $(-1)^{n-j} a_j = (-1)^n (-1)^j a_j$ (using $(-1)^{-j} = (-1)^j$), so

$$(k_1 - 1)(k_2 - 1) \cdots (k_n - 1) = (-1)^n \left(1 - a_1 + a_2 - a_3 + \cdots + (-1)^n a_n\right).$$

For even n this is simply $1 - a_1 + a_2 - \cdots + a_n$; for odd n every sign is reversed.

The quartic

Write the quartic as $f(x) = (x+k_1)(x+k_2)(x+k_3)(x+k_4)$, so its roots are $-k_1, \ldots, -k_4$, and these are integers. Using the identities with $n = 4$:

$$k_1k_2k_3k_4 = 576 = 2^6 \cdot 3^2,$$
$$(k_1+1)(k_2+1)(k_3+1)(k_4+1) = 1 + 22 + 172 + 552 + 576 = 1323 = 3^3 \cdot 7^2,$$
$$(k_1-1)(k_2-1)(k_3-1)(k_4-1) = 1 - 22 + 172 - 552 + 576 = 175 = 5^2 \cdot 7.$$

Every k_i is at least 2. All the coefficients are positive, so $f(x) > 0$ for every $x \geq 0$; hence every root is negative and every k_i is a positive integer. Moreover no k_i equals 1, since otherwise the third product would be 0, not 175. So $k_i \geq 2$ for each i, and the numbers $k_i - 1$ and $k_i + 1$ are all positive integers.

? **Ask yourself** How do we turn three products into a short list of possible k_i?

Each $k_i - 1$ is a positive divisor of 175, and each $k_i + 1$ is a positive divisor of 1323. The divisors of 175 are $1, 5, 7, 25, 35, 175$, giving

$$k_i \in \{2, 6, 8, 26, 36, 176\}, \qquad k_i + 1 \in \{3, 7, 9, 27, 37, 177\}.$$

Of these values of $k_i + 1$, only $3, 7, 9, 27$ divide $1323 = 3^3 \cdot 7^2$ (since 37 is prime and $177 = 3 \cdot 59$). So each k_i is one of $2, 6, 8, 26$, and each $k_i + 1$ is one of $3, 7, 9, 27$.

Now look at the product $\prod(k_i + 1) = 3^3 \cdot 7^2$.

- The factor 7^2 can only come from factors equal to 7, so exactly two of the $k_i + 1$ equal 7: two of the k_i are 6.
- The other two factors are powers of 3 from $\{3, 9, 27\}$ with product $3^3 = 27$. Each is at least 3, so 27 is impossible (the partner would be 1), and the only option is 3×9. So the remaining k_i are 2 and 8.

Thus $\{k_1, k_2, k_3, k_4\} = \{2, 6, 6, 8\}$. As a check against the first product, $2 \cdot 6 \cdot 6 \cdot 8 = 576$, as required. Expanding confirms it:

$$(x+2)(x+8) = x^2 + 10x + 16, \qquad (x+6)^2 = x^2 + 12x + 36,$$

and

$$(x^2 + 10x + 16)(x^2 + 12x + 36) = x^4 + 22x^3 + 172x^2 + 552x + 576.$$

So the roots are

$$x = -2, \quad x = -6 \text{ (a repeated root)}, \quad x = -8.$$

Checks

- The sum of the k_i is $2 + 6 + 6 + 8 = 22$, matching the coefficient of x^3.

- $f(-6) = 1296 - 4752 + 6192 - 3312 + 576 = 0$, and $f'(x) = 4x^3 + 66x^2 + 344x + 552$ also vanishes at $x = -6$, confirming that -6 is a double root.
- A quick numerical test of the identities: for $f(x) = (x+2)(x-3) = x^2 - x - 6$ we get $(2-1)(-3-1) = -4$ and $1 - (-1) + (-6) = -4$.

Common slips

- Getting the sign wrong in the $x = -1$ result: the factor $(-1)^n$ matters when n is odd.
- Forgetting that a repeated root is allowed: once 7^2 is seen in 1323, two of the k_i must be equal.
- Listing only positive divisors without first showing that every $k_i \geq 2$; that step is what makes all the $k_i \pm 1$ positive.

Beyond STEP

The identities here are special cases of Vieta's formulae, which express the coefficients of a polynomial as the elementary symmetric functions of its roots. The divisor argument is a cousin of the rational root theorem, and of Gauss's lemma on factorising integer polynomials, which you meet in a first course on rings. Allenby's *Rings, Fields and Groups* develops these ideas carefully.

WORKED SOLUTION

STEP I 2002, Question 6

Pure • Pyramid Geometry

STEP I 2002, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

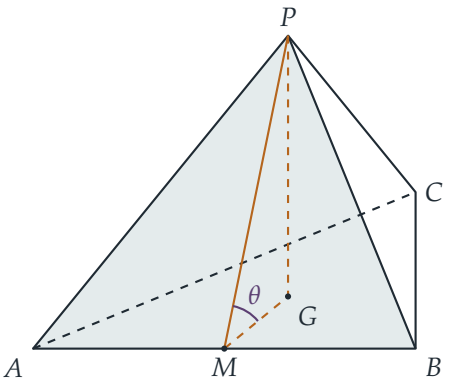
step.maths.org/questions/02-s1-q6

The key idea

By symmetry the apex sits directly above the centre of the equilateral base, so the height comes from one application of Pythagoras. Once we know the volume, we never need to find the new height directly: the volume is the same whichever face is on the ground, so $V = \frac{1}{3} \times (\text{area of the new base}) \times h$. The angle in the final part is the angle between two faces along their common edge, and the cleanest way to see it is in the cross-section through the apex that is perpendicular to that edge.

Solution

Label the base ABC, with sides a, and the apex P, with $PA = PB = PC = b$. Let G be the centre of ABC and M the midpoint of AB.



The volume

Let H be the height of P above the base and F the foot of the perpendicular from P. The right-angled triangles PFA, PFB, PFC share the side $PF = H$ and have hypotenuse

b, so

$$FA = FB = FC = \sqrt{b^2 - H^2}.$$

So F is equidistant from the three vertices: it is the circumcentre of ABC, which for an equilateral triangle is its centre G. The median of ABC has length $\frac{\sqrt{3}}{2}a$, and G lies two-thirds of the way along it from a vertex, so $GA = \frac{2}{3} \cdot \frac{\sqrt{3}}{2}a = \frac{a}{\sqrt{3}}$. By Pythagoras in triangle PGA,

$$H^2 = b^2 - \frac{a^2}{3} = \frac{3b^2 - a^2}{3}.$$

The base has area $\frac{1}{2}a^2 \sin 60^\circ = \frac{\sqrt{3}}{4}a^2$, so

$$V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4}a^2 \cdot \frac{\sqrt{3b^2 - a^2}}{\sqrt{3}} = \frac{1}{12}a^2 \left(3b^2 - a^2\right)^{\frac{1}{2}},$$

as required. (A pyramid of this shape exists only when $3b^2 > a^2$, which is exactly when this is real and positive.)

The new height

Now face PAB lies on the ground. It is isosceles with $PA = PB = b$ and $AB = a$, so PM is perpendicular to AB and, by Pythagoras in triangle PMA,

$$PM^2 = b^2 - \frac{a^2}{4} = \frac{4b^2 - a^2}{4}, \qquad \text{area of } PAB = \frac{1}{2}\, a \cdot PM = \frac{a}{4}\sqrt{4b^2 - a^2}.$$

Ask yourself Do we really need to locate the new top vertex to find h?

No. The solid hasn't changed, so its volume is still V, and the volume formula holds for any face taken as the base, with the height measured perpendicular to that face. So

$$V = \frac{1}{3} \cdot \frac{a}{4}\sqrt{4b^2 - a^2} \cdot h.$$

Equating with the first part,

$$\frac{a}{12}\sqrt{4b^2 - a^2}\, h = \frac{a^2}{12}\sqrt{3b^2 - a^2} \quad \Longrightarrow \quad h = \frac{a\sqrt{3b^2 - a^2}}{\sqrt{4b^2 - a^2}},$$

and squaring gives

$$h^2 = \frac{a^2(3b^2 - a^2)}{4b^2 - a^2},$$

as required. Here $4b^2 - a^2 > 0$ because $4b^2 > 3b^2 > a^2$.

The angle of the equilateral face

In the new position the ground is the plane of PAB, and the equilateral face ABC meets it along the edge AB. The angle between two planes is the angle between lines, one

in each plane, drawn perpendicular to their common edge at the same point. At M, the line MP in face PAB is perpendicular to AB (as PAB is isosceles), and the line MC in face ABC is perpendicular to AB (as ABC is equilateral), and G lies on MC. So the required angle is $\theta = \angle PMC = \angle PMG$.

Triangle PMG has a right angle at G, since PG is perpendicular to the whole base. Also GM is a third of the median, $GM = \frac{1}{3} \cdot \frac{\sqrt{3}}{2}a = \frac{a}{2\sqrt{3}}$. Therefore

$$\tan\theta = \frac{PG}{GM} = \frac{\sqrt{3b^2 - a^2}}{\sqrt{3}} \cdot \frac{2\sqrt{3}}{a} = \frac{2\sqrt{3b^2 - a^2}}{a},$$

so the angle between the equilateral triangle and the horizontal is

$$\theta = \tan^{-1}\left(\frac{2\sqrt{3b^2 - a^2}}{a}\right).$$

This angle is acute, because G lies inside the base, so it really is the angle between the planes and not its supplement.

Equivalently, using the new height: the vertex C is at height h above the ground and at distance $MC = \frac{\sqrt{3}}{2}a$ from the edge AB along the face, so $\sin\theta = h / \frac{\sqrt{3}}{2}a$, giving

$$\sin\theta = \frac{2\sqrt{3b^2 - a^2}}{\sqrt{3(4b^2 - a^2)}}, \qquad \cos\theta = \frac{a}{\sqrt{3(4b^2 - a^2)}},$$

which agree with the value of $\tan\theta$ above.

Checks

For a regular tetrahedron, $a = b = 1$: the formulas give $V = \frac{\sqrt{2}}{12}$, $h^2 = \frac{2}{3}$ (the same as the original height, as it must be by symmetry), and $\cos\theta = \frac{1}{3}$, the familiar dihedral angle $\theta \approx 70.53°$ of the regular tetrahedron. As $b \to a/\sqrt{3}$ the pyramid flattens and $\theta \to 0$; as $b \to \infty$ it becomes a tall spike and $\theta \to 90°$. We also built the pyramid in coordinates for 20,000 random pairs (a, b) and computed V from a triple product, h as the distance from C to the plane PAB, and θ from the normals; all three agreed with the formulas to about 10^{-13}. For instance $a = 2$, $b = 3$ gives $h \approx 1.6956$ and $\theta \approx 78.22°$.

Common slips

- Taking GA to be half the median, or using the inradius $\frac{a}{2\sqrt{3}}$ where the circumradius $\frac{a}{\sqrt{3}}$ is needed (and the other way round in the last part).
- Measuring the angle between the face and the ground with a line that isn't perpendicular to the common edge, such as AC or PC.

Beyond STEP

Computing the same volume with different faces as base is a first glimpse of the idea that volume does not depend on how we slice or orient a solid. In a first-year course the volume of a tetrahedron becomes $\frac{1}{6}|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|$, a scalar triple product (a 3×3 determinant), and the angle between faces is the angle between their normal vectors, which is how our numerical check worked. Strang's *Introduction to Linear Algebra* develops determinants as volumes.

WORKED SOLUTION

STEP I 2002, Question 7

Pure • Trigonometric Integrals

STEP I 2002, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q7

The key idea

Neither I nor J looks friendly alone, but their sum and difference both are. Adding them puts the denominator on top, so the integrand is 1. Subtracting them puts the *derivative* of the denominator on top, so the integral is a logarithm. The two later integrals use the same trick in a more general form: split the numerator into a multiple of the denominator plus a multiple of the denominator's derivative.

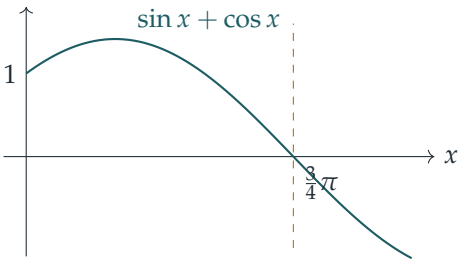
Solution

The main result

First, why the restriction on a? We can write

$$\sin x + \cos x = \sqrt{2}\,\sin\left(x + \tfrac{\pi}{4}\right),$$

which is positive for $-\frac{\pi}{4} < x < \frac{3\pi}{4}$ and first vanishes at $x = \frac{3}{4}\pi$. So for $0 \le a < \frac{3}{4}\pi$ the denominator is positive on all of $[0, a]$. Both integrals are then perfectly ordinary, and the logarithm below is of a positive number.



Adding the two integrals,

$$I + J = \int_0^a \frac{\cos x + \sin x}{\sin x + \cos x}\,\mathrm{d}x = \int_0^a 1\,\mathrm{d}x = a.$$

Subtracting them,

$$I - J = \int_0^a \frac{\cos x - \sin x}{\sin x + \cos x}\,\mathrm{d}x.$$

The numerator $\cos x - \sin x$ is exactly the derivative of the denominator $\sin x + \cos x$, so the integrand has the form $f'(x)/f(x)$ with $f > 0$ on $[0, a]$. Hence

$$I - J = \Big[\ln(\sin x + \cos x)\Big]_0^a = \ln(\sin a + \cos a) - \ln 1 = \ln(\sin a + \cos a).$$

Adding the two results gives

$$2I = a + \ln(\sin a + \cos a),$$

as required. (Subtracting them gives $2J = a - \ln(\sin a + \cos a)$ as a bonus.)

Part (i)

Since $p, q > 0$ and both $\sin x$ and $\cos x$ are non-negative on $[0, \frac{\pi}{2}]$ and never zero together, the denominator $D(x) = p\sin x + q\cos x$ is positive there. Its derivative is $D'(x) = p\cos x - q\sin x$.

> **Ask yourself** How do we generalise "consider $I + J$ and $I - J$" when the denominator has coefficients p and q?

Before, the sum and difference worked because the numerators added up to the denominator and subtracted to its derivative. So now we look for constants A and B with

$$\cos x = A\,(p\sin x + q\cos x) + B\,(p\cos x - q\sin x) \qquad \text{for all } x.$$

Comparing coefficients of $\sin x$ and $\cos x$:

$$pA - qB = 0, \qquad qA + pB = 1.$$

The first gives $B = pA/q$; then the second gives $qA + p^2A/q = 1$, so

$$A = \frac{q}{p^2 + q^2}, \qquad B = \frac{p}{p^2 + q^2}.$$

Now the integral splits into an easy piece and a logarithmic piece:

$$\begin{aligned}\int_0^{\pi/2} \frac{\cos x}{p\sin x + q\cos x}\,\mathrm{d}x &= \int_0^{\pi/2} \left(A + B\,\frac{D'(x)}{D(x)}\right)\mathrm{d}x \\ &= \frac{\pi}{2}A + B\Big[\ln D(x)\Big]_0^{\pi/2} = \frac{\pi}{2}A + B\ln\frac{p}{q},\end{aligned}$$

using $D(\frac{\pi}{2}) = p$ and $D(0) = q$, both positive. Therefore

$$\int_0^{\pi/2} \frac{\cos x}{p\sin x + q\cos x}\,\mathrm{d}x = \frac{\frac{1}{2}\pi q + p\ln(p/q)}{p^2 + q^2}.$$

Part (ii)

Now the denominator is $D(x) = 3\sin x + 4\cos x + 25$, with $D'(x) = 3\cos x - 4\sin x$. The denominator is positive, since $|3\sin x + 4\cos x| \leq 5 < 25$. The constant 25 in D means we must also match the constant term, so we seek A and B with

$$\cos x + 4 = A\,(3\sin x + 4\cos x + 25) + B\,(3\cos x - 4\sin x).$$

Comparing coefficients:

$$\sin x:\ 3A - 4B = 0, \qquad \cos x:\ 4A + 3B = 1, \qquad \text{constant}:\ 25A = 4.$$

The first two give $A = \frac{4}{25}$ and $B = \frac{3}{25}$, and then the constant equation $25 \cdot \frac{4}{25} = 4$ is satisfied too. With three equations and only two unknowns, this is not automatic: the "+4" in the numerator was chosen to make it work. So

$$\begin{aligned}\int_0^{\pi/2} \frac{\cos x + 4}{3\sin x + 4\cos x + 25}\,\mathrm{d}x &= \int_0^{\pi/2} \left(\frac{4}{25} + \frac{3}{25}\,\frac{D'(x)}{D(x)}\right)\mathrm{d}x \\ &= \frac{4}{25}\cdot\frac{\pi}{2} + \frac{3}{25}\Big[\ln D(x)\Big]_0^{\pi/2} \\ &= \frac{2\pi}{25} + \frac{3}{25}\ln\frac{28}{29},\end{aligned}$$

since $D(\frac{\pi}{2}) = 3 + 25 = 28$ and $D(0) = 4 + 25 = 29$.

Checks

- With $p = q = 1$, part (i) should agree with the main result at $a = \frac{\pi}{2}$, which gives $2I = \frac{\pi}{2} + \ln 1$, so $I = \frac{\pi}{4}$. The formula gives $\frac{\pi/2+0}{2} = \frac{\pi}{4}$. It matches.
- As $a \to \frac{3}{4}\pi$, $\ln(\sin a + \cos a) \to -\infty$. That fits the integrand of I being large and negative just before the denominator vanishes, since $\cos x < 0$ there.
- Numerically, part (ii) is $\frac{2\pi}{25} + \frac{3}{25}\ln\frac{28}{29} \approx 0.247116$, and Simpson's rule on the integral agrees to twelve decimal places. The main result and part (i) also agree with quadrature for 200 random values of a, and of p and q, respectively.

Common slips

- Getting $\ln(q/p)$ instead of $\ln(p/q)$ in part (i). At the upper limit $\frac{\pi}{2}$ it is the sin term that survives, giving p.
- In part (ii), forgetting the constant term and matching only $\sin x$ and $\cos x$. It happens to work, but you should check it rather than assume it.

Beyond STEP

Writing the numerator as "A times the denominator plus B times its derivative" is a small piece of linear algebra. The functions $\sin x$ and $\cos x$ span a two-dimensional

space, and D and D' form another basis of it whenever $p^2 + q^2 \neq 0$. The same idea, finding the right basis to expand in, drives partial fractions and the method of undetermined coefficients for differential equations, which you will meet in a first-year methods course. Siklos's *Advanced Problems in Mathematics* has more integrals that give way to this kind of symmetry trick.

WORKED SOLUTION

STEP I 2002, Question 8

Pure • Compound Interest Loans

> **STEP I 2002, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/02-s1-q8

The key idea

Write down what happens in a single year: the debt grows by a factor $1+\alpha$ and then the repayment is subtracted. Unrolling this one-step recurrence shows that each repayment R_k is "worth" $(1+\alpha)^{n-k}R_k$ by the end of year n, because it has stopped that much interest from building up. Everything else is a geometric series. For the final approximation, expand the powers of $1+\alpha$ by the binomial theorem, keeping just enough terms.

Solution

The general formula for C_{n+1}

At the end of year k the debt C_k becomes $(1+\alpha)C_k$ when interest is added, and then R_k is repaid. What is left is the debt at the start of year $k+1$:

$$C_{k+1} = (1+\alpha)C_k - R_k.$$

Applying this repeatedly,

$$\begin{aligned} C_{n+1} &= (1+\alpha)C_n - R_n \\ &= (1+\alpha)^2 C_{n-1} - (1+\alpha)R_{n-1} - R_n \\ &\vdots \\ &= (1+\alpha)^n C_1 - (1+\alpha)^{n-1}R_1 - (1+\alpha)^{n-2}R_2 - \cdots - R_n, \end{aligned}$$

so, since $C_1 = C$,

$$\boxed{C_{n+1} = (1+\alpha)^n C - \sum_{k=1}^{n}(1+\alpha)^{n-k}R_k}.$$

(To make this rigorous, check it holds for $n=1$, where it says $C_2 = (1+\alpha)C - R_1$, and that if it holds for n then multiplying by $1+\alpha$ and subtracting R_{n+1} gives it for $n+1$.)

Repayments $R_k = (1+\alpha)^k r$

Paying off the loan in N years means $C_{N+1} = 0$. With $R_k = (1+\alpha)^k r$, every term of the sum is the same:

$$(1+\alpha)^{N-k} R_k = (1+\alpha)^{N-k}(1+\alpha)^k r = (1+\alpha)^N r.$$

Hence

$$0 = C_{N+1} = (1+\alpha)^N C - N(1+\alpha)^N r = (1+\alpha)^N (C - Nr).$$

Since $(1+\alpha)^N \neq 0$, we get $C = Nr$, that is, $r = C/N$, as required.

This makes sense: the kth repayment is C/N "in today's money", grown by k years of interest, so altogether you repay the principal C plus exactly the interest on it.

Equal repayments R

Now $R_k = R$ for every k, and $C_{N+1} = 0$ gives

$$(1+\alpha)^N C = R \sum_{k=1}^{N} (1+\alpha)^{N-k} = R\big(1 + (1+\alpha) + \cdots + (1+\alpha)^{N-1}\big).$$

The bracket is a geometric series with first term 1, ratio $1+\alpha$ and N terms, so (as $\alpha > 0$, the ratio is not 1) it equals

$$\frac{(1+\alpha)^N - 1}{(1+\alpha) - 1} = \frac{(1+\alpha)^N - 1}{\alpha}.$$

Therefore $(1+\alpha)^N C = R\,\dfrac{(1+\alpha)^N - 1}{\alpha}$, and rearranging,

$$\frac{R}{C} = \frac{\alpha(1+\alpha)^N}{(1+\alpha)^N - 1},$$

which is the required result.

The case $\alpha = \frac{1}{50}$, $N = 4$

> **Ask yourself** How many terms of the binomial expansion should we keep, and why does the numerator need fewer than the denominator?

The denominator $(1+\alpha)^4 - 1$ starts at order α, while the numerator $\alpha(1+\alpha)^4$ is α times something starting at 1. So after cancelling a factor of α the numerator is $(1+\alpha)^4$ and the denominator is $\big((1+\alpha)^4 - 1\big)/\alpha$. To get both correct to first order in α we need $(1+\alpha)^4 \approx 1 + 4\alpha$ on top, but $(1+\alpha)^4 \approx 1 + 4\alpha + 6\alpha^2$ underneath (the $6\alpha^2$ becomes 6α after dividing by α). Thus

$$\frac{R}{C} = \frac{(1+\alpha)^4}{\big((1+\alpha)^4 - 1\big)/\alpha} = \frac{1 + 4\alpha + 6\alpha^2 + \ldots}{4 + 6\alpha + 4\alpha^2 + \alpha^3} \approx \frac{1 + 4\alpha}{4 + 6\alpha}.$$

With $\alpha = \frac{1}{50}$,

$$\frac{R}{C} \approx \frac{1 + \frac{4}{50}}{4 + \frac{6}{50}} = \frac{54}{206} = \frac{27}{103},$$

as required. The terms we dropped are of relative size about α^2, a fraction of a per cent here.

Checks

- With $N = 1$ the formula gives $R/C = 1 + \alpha$: borrow C, repay it with one year's interest. Correct.
- As $\alpha \to 0$, $(1+\alpha)^N - 1 \approx N\alpha$, so $R/C \to 1/N$: with no interest you simply repay C/N each year. This also agrees with $r = C/N$.
- The exact value is $R/C = \dfrac{6765201}{25760050} \approx 0.26262$, while $27/103 \approx 0.26214$; they differ by about 0.2%. Stepping the debt forward year by year with $R = 0.26262\,C$ leaves (to rounding) nothing owing after four years.

Common slips

- Getting the exponent wrong in the sum: R_k is paid at the end of year k, so it misses $n - k$ more years of interest, not $n - k + 1$.
- Approximating $(1+\alpha)^4 \approx 1 + 4\alpha$ in the denominator too: this gives $R/C \approx 27/100$, a cruder answer that doesn't match.

Beyond STEP

The quantity $\sum (1+\alpha)^{-k} R_k$ is the *present value* of the repayments, and the loan is cleared exactly when the present value of the repayments equals the amount borrowed; the formula for R is the standard annuity (mortgage) formula. Recurrences like $C_{k+1} = (1+\alpha)C_k - R_k$ are first-order linear difference equations, discrete cousins of the differential equation $y' = \alpha y - R(t)$, solved by the same "integrating factor" idea. For more problems that mix algebra with modelling in this way, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2002, Question 9

Mechanics • Toppling on an Incline

STEP I 2002, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

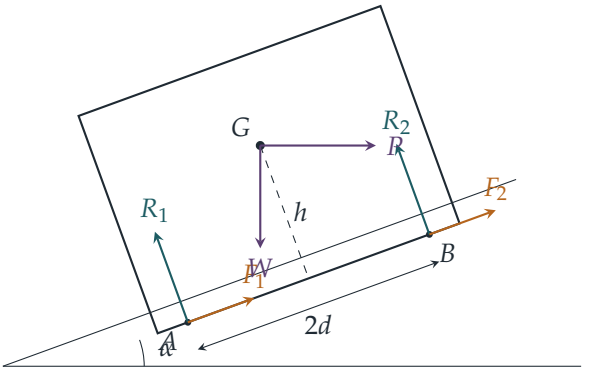
step.maths.org/questions/02-s1-q9

The key idea

The lorry is a rigid body in equilibrium, so we have three equations: resolve along the plane, resolve perpendicular to it, and take moments. Taking moments about G is the clever choice, because both the weight and P pass through G and drop out, leaving only the wheel forces. That one equation links the difference of the normal reactions to the total friction, and it does almost all the work. "Not tipping" then simply means that neither normal reaction is negative.

Solution

We model the lorry as a rigid body whose wheels touch the plane at two points, A (lower) and B (higher), a distance $2d$ apart. Let the normal reactions be R_1 at A and R_2 at B, and let the frictional forces be F_1 and F_2, both taken as positive *up* the plane (either may turn out negative). We assume friction is large enough to prevent sliding; the question is only about tipping. As in the diagram, P is horizontal, pointing towards the slope, so it has a component $P\cos\alpha$ up the plane and a component $P\sin\alpha$ into the plane.



Resolving perpendicular to the plane and along it (up the slope positive):

$$R_1 + R_2 = W\cos\alpha + P\sin\alpha, \tag{1}$$
$$F_1 + F_2 = W\sin\alpha - P\cos\alpha. \tag{2}$$

Now take moments about G. The weight and P act through G, so they contribute nothing. Each normal reaction acts at a perpendicular distance d from G, and each frictional force acts along the plane at a perpendicular distance h from G. Taking anticlockwise as positive (with the plane rising to the right, as in the diagram), R_2 turns the lorry anticlockwise about G, R_1 turns it clockwise, and a friction force up the plane, acting below G, turns it anticlockwise. So

$$d(R_2 - R_1) + h(F_1 + F_2) = 0. \tag{3}$$

Part (i)

If $R_1 = R_2$, equation (3) becomes $h(F_1 + F_2) = 0$. Since $h > 0$, this gives $F_1 + F_2 = 0$: the frictional forces sum to zero, as required.

(From (2), this happens exactly when $P = W\tan\alpha$: the up-slope part of P exactly balances the down-slope part of the weight, so the plane needn't supply any net friction.)

Part (ii)

Substituting (2) into (3),

$$R_2 - R_1 = \frac{h}{d}(P\cos\alpha - W\sin\alpha),$$

and combining this with (1),

$$2R_1 = W\cos\alpha + P\sin\alpha - \frac{h}{d}(P\cos\alpha - W\sin\alpha),$$
$$2R_2 = W\cos\alpha + P\sin\alpha + \frac{h}{d}(P\cos\alpha - W\sin\alpha).$$

Ask yourself What does "does not tip over" mean in terms of these equations?

The ground can push on a wheel but can't pull it, so each normal reaction must satisfy $R \geq 0$. If the formula for R_2 came out negative, the higher wheel would have to be held down; in reality it lifts and the lorry rotates about A. Similarly $R_1 < 0$ means rotation about B. So the lorry stays upright exactly when $R_1 \geq 0$ and $R_2 \geq 0$.

Throughout we use $\frac{d}{h} = \tan\beta$, with $0 < \beta < \frac{\pi}{2}$, and $0 < \alpha < \frac{\pi}{2}$.

The higher wheel stays down ($R_2 \geq 0$). Multiplying $2R_2 \geq 0$ by $d/h > 0$,

$$\tan\beta\,(W\cos\alpha + P\sin\alpha) + P\cos\alpha - W\sin\alpha \geq 0.$$

Multiplying by $\cos\beta > 0$ and collecting terms,

$$P(\cos\alpha\cos\beta + \sin\alpha\sin\beta) \geq W(\sin\alpha\cos\beta - \cos\alpha\sin\beta),$$

that is,

$$P\cos(\alpha-\beta) \geq W\sin(\alpha-\beta).$$

Since $-\frac{\pi}{2} < \alpha - \beta < \frac{\pi}{2}$, we have $\cos(\alpha-\beta) > 0$, so we may divide by it without reversing the inequality:

$$P \geq W\tan(\alpha-\beta).$$

The lower wheel stays down ($R_1 \geq 0$). In the same way, $2R_1 \geq 0$ gives

$$\tan\beta\,(W\cos\alpha + P\sin\alpha) - P\cos\alpha + W\sin\alpha \geq 0,$$

and multiplying by $\cos\beta > 0$,

$$P(\cos\alpha\cos\beta - \sin\alpha\sin\beta) \leq W(\sin\alpha\cos\beta + \cos\alpha\sin\beta),$$

that is,

$$P\cos(\alpha+\beta) \leq W\sin(\alpha+\beta).$$

Provided $\alpha + \beta < \frac{\pi}{2}$ (which the stated answer implicitly assumes, since otherwise $\tan(\alpha+\beta)$ is negative or undefined), $\cos(\alpha+\beta) > 0$, and dividing gives

$$P \leq W\tan(\alpha+\beta).$$

If instead $\alpha + \beta \geq \frac{\pi}{2}$, the left-hand side is ≤ 0 for $P \geq 0$ while the right-hand side is positive, so no horizontal push towards the slope, however large, can tip the lorry over its higher wheel.

Putting the two conditions together, the lorry does not tip if and only if

$$W\tan(\alpha-\beta) \leq P \leq W\tan(\alpha+\beta),$$

which is the required result. Too small a push and it topples down the slope about A; too large and it is pushed over uphill about B.

Checks

- *Geometric picture.* The weight and P combine into a single force through G, at angle θ to the vertical with $\tan\theta = P/W$. The lorry is about to tip when this resultant passes through a wheel contact. The line GA makes angle β with the normal to the plane, and the normal makes angle α with the vertical, so the resultant passes through A when $\theta = \alpha - \beta$, and through B when $\theta = \alpha + \beta$. These are exactly the two end-points.
- *No slope.* With $\alpha = 0$ the condition becomes $|P| \leq Wd/h$: the familiar condition for a box on level ground not to topple under a sideways push at height h.
- *Numbers.* With $W = 3$, $\alpha = 0.4$, $d = 1$, $h = 1.5$, the interval is $-0.571 \leq P \leq 4.551$. Checking $R_1, R_2 \geq 0$ directly against the interval for 20 000 random configurations gave complete agreement, with $R_2 = 0$ at the lower end and $R_1 = 0$ at the upper end.

Common slips

- Taking moments about a wheel instead of G. That works too, but then the weight and P have awkward moment arms, and sign errors creep in.
- Forgetting that the frictional forces have a moment about G. They act at the plane, a distance h below G, and this is where h enters the answer.
- Dividing by $\cos(\alpha \pm \beta)$ without checking its sign, which is what decides the direction of the inequality.

Beyond STEP

Friction between the wheels and the ground is *statically indeterminate*: equilibrium fixes only $F_1 + F_2$, never F_1 and F_2 separately, and finding the split needs a model of how the tyres deform. Engineers meet this all the time in structures with more supports than they strictly need. The geometric check above is the idea of the *line of action of the resultant* and the support polygon, which also underlies the stability of walking robots and stacked blocks. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers the statics of rigid bodies carefully.

WORKED SOLUTION

STEP I 2002, Question 10

Mechanics • Elastic Collisions

STEP I 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q10

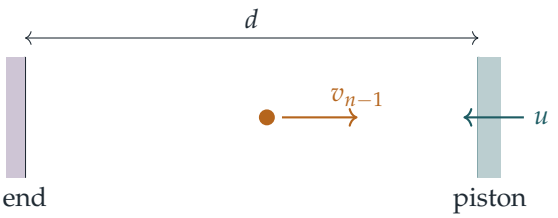
This question has a full guided treatment in *Beyond STEP*, Section 10.4, *A Molecule and a Piston*: https://beyondstep.keithmcnulty.org/ch10/piston.html

The key idea

A perfectly elastic collision with a wall that is far heavier than the particle reverses the *relative* velocity, so it is easiest to think in the frame of the piston: there the particle simply bounces off a stationary wall. Each bounce off the piston therefore adds $2u$ to the speed, while bounces off the fixed end change nothing. For the distances, notice that between two collisions with the piston the particle and the piston together cover a fixed pattern of lengths, which gives one linear equation for d_{n+1} in terms of d_n.

Solution

We model the air as a particle, the piston as a wall moving at constant speed u that is unaffected by collisions, and the end of the cylinder as a fixed wall. All motion is along the axis, so we take the direction from the end of the cylinder towards the piston as positive.



The speeds

Suppose the particle approaches the piston with speed w. The piston has velocity $-u$, so the particle's velocity relative to the piston is $w+u$ (towards it). In a perfectly elastic collision with a wall that is not deflected, Newton's experimental law with $e=1$ says the relative velocity is reversed, so afterwards the particle's velocity relative to the piston is $-(w+u)$. Its actual velocity is then

$$-u-(w+u)=-(w+2u),$$

so it leaves the piston with speed $w+2u$.

At the fixed end of the cylinder the same law, with the wall at rest, just reverses the velocity: the speed is unchanged. So between consecutive collisions with the piston the particle's speed stays the same.

Before the first collision the particle is moving towards the piston with speed v, so $v_1=v+2u$. Just after the nth collision it has speed v_n; it bounces off the end without change of speed and meets the piston again at speed v_n, leaving with $v_{n+1}=v_n+2u$. By induction,

$$v_n=v+2nu,$$

as required.

The distances

Between the nth and $(n+1)$th collisions (a time t_n) the piston moves inwards at speed u, so

$$d_n-d_{n+1}=u\,t_n.$$

In the same time the particle, moving at the constant speed v_n, travels from the piston to the end (a distance d_n, since the end is d_n from where the piston was) and back to the piston (a distance d_{n+1}, where the piston now is). So

$$d_n+d_{n+1}=v_nt_n.$$

Ask yourself How do we get rid of t_n?

Divide one equation by the other: $t_n>0$, so

$$\frac{d_n+d_{n+1}}{d_n-d_{n+1}}=\frac{v_n}{u}.$$

Cross-multiplying, $ud_n+ud_{n+1}=v_nd_n-v_nd_{n+1}$, so

$$(v_n+u)\,d_{n+1}=(v_n-u)\,d_n.$$

With $v_n=v+2nu$, we have $v_n-u=v+(2n-1)u$ and $v_n+u=v+(2n+1)u$, so

$$d_{n+1}=\frac{v+(2n-1)u}{v+(2n+1)u}\,d_n,$$

which is the required result.

A formula for d_n

Applying the recurrence repeatedly,

$$d_n = d_1 \prod_{k=1}^{n-1} \frac{v+(2k-1)u}{v+(2k+1)u} = d_1 \cdot \frac{v+u}{v+3u} \cdot \frac{v+3u}{v+5u} \cdots \frac{v+(2n-3)u}{v+(2n-1)u}.$$

The product telescopes: each denominator cancels with the next numerator, leaving

$$d_n = \frac{v+u}{v+(2n-1)u}\, d_1.$$

This also holds for $n = 1$, where it reads $d_1 = d_1$.

The case $v = u$

Putting $v = u$ gives

$$d_n = \frac{2u}{2nu}\, d_1 = \frac{d_1}{n}.$$

Then, from $ut_n = d_n - d_{n+1}$,

$$ut_n = \frac{d_1}{n} - \frac{d_1}{n+1} = \frac{d_1}{n(n+1)},$$

as required.

Checks

- With $u = 0$ the piston is a fixed wall: the formulae give $v_n = v$ and $d_n = d_1$ for all n, as they should.
- Take $u = 1$, $v = 3$, $d_1 = 75$. The formula predicts $d_2 = 50$, $d_3 = 37.5$, $d_4 = 30$ and speeds $5, 7, 9, 11$. An exact step-by-step simulation of the bouncing (in rational arithmetic) gives precisely these, and the same agreement holds for 200 random choices of u, v and starting position.
- When $v = u$, $\sum ut_n = d_1 \sum\left(\frac{1}{n} - \frac{1}{n+1}\right) = d_1$: the piston travels exactly d_1 in total, reaching the end of the cylinder, which is consistent with $d_n = d_1/n \to 0$.

Common slips

- Using $d_n - d_{n+1}$ wrongly, for example taking the particle's round trip as $2d_n$: the piston has moved in the meantime, so the return leg is only d_{n+1}.
- Adding u rather than $2u$ per collision. The relative velocity is reversed, so the change in the particle's velocity is twice the approach speed added by the piston.

Beyond STEP

Each collision changes the particle's speed by a fixed amount, but the collisions get more frequent as the gap closes, and the product $v_n d_n$ stays roughly constant for large n (here $v_n d_n \approx (v+u)d_1$ when n is large). Such a slowly-varying conserved quantity is an *adiabatic invariant*, and averaging over many molecules it explains why gas warms up when compressed quickly: for slow compression in one dimension, $T \propto 1/L^2$. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for collisions and reference frames.

WORKED SOLUTION

STEP I 2002, Question 11

Mechanics • Oblique Elastic Collisions

STEP I 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

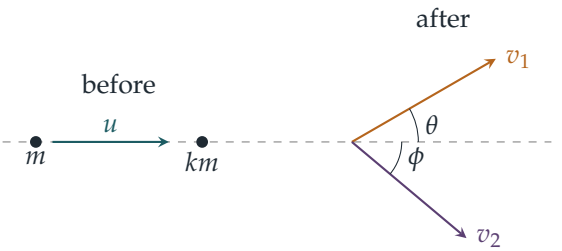
step.maths.org/questions/02-s1-q11

The key idea

A collision gives us two conservation laws: momentum (a vector, so two component equations) and, because no energy is lost, kinetic energy. The two momentum equations are linear in the unknown speeds after impact, so we can solve them for those speeds in terms of the angles. Substituting into the energy equation then removes every speed and leaves a relation between the angles and k alone.

Solution

We model P_1 and P_2 as particles moving in a plane, with no external forces acting during the collision. Let P_1 approach with speed $u > 0$ along the x-axis, and let the speeds after the collision be v_1 (for P_1, at angle θ above the axis) and v_2 (for P_2, at angle ϕ below it), with $0 < \theta < \pi$ and $0 < \phi < \pi$.



The identity

Conservation of momentum along the original line of motion, and perpendicular to it, gives (after cancelling m)

$$v_1 \cos\theta + kv_2 \cos\phi = u,$$
$$v_1 \sin\theta - kv_2 \sin\phi = 0,$$

where the minus sign is there because P_2 moves off on the other side of the axis. Conservation of kinetic energy gives, after cancelling $\frac{1}{2}m$,

$$v_1^2 + kv_2^2 = u^2. \qquad (*)$$

The momentum equations are two linear equations in the two unknowns v_1 and kv_2. Multiply the first by $\sin\phi$ and the second by $\cos\phi$ and add: the kv_2 terms cancel and

$$v_1(\sin\theta\cos\phi + \cos\theta\sin\phi) = u\sin\phi, \qquad \text{so} \qquad v_1 = \frac{u\sin\phi}{\sin(\theta+\phi)}.$$

Similarly, multiplying the first by $\sin\theta$ and the second by $\cos\theta$ and subtracting gives

$$kv_2 = \frac{u\sin\theta}{\sin(\theta+\phi)}.$$

Ask yourself Are we allowed to divide by $\sin(\theta+\phi)$?

We need $\theta + \phi \neq \pi$ (it lies strictly between 0 and 2π). If $\theta + \phi = \pi$, then $\sin\phi = \sin\theta$ and $\cos\phi = -\cos\theta$. The second momentum equation becomes $(v_1 - kv_2)\sin\theta = 0$, so $v_1 = kv_2$ because $\sin\theta \neq 0$. The first then reads $(v_1 - kv_2)\cos\theta = u$, giving $0 = u$, which is impossible. So $\sin(\theta+\phi) \neq 0$ and the division is fine.

Now we substitute into $(*)$, writing kv_2^2 as $(kv_2)^2/k$:

$$\frac{u^2\sin^2\phi}{\sin^2(\theta+\phi)} + \frac{u^2\sin^2\theta}{k\sin^2(\theta+\phi)} = u^2.$$

Multiplying through by $k\sin^2(\theta+\phi)/u^2$ (which is non-zero) gives

$$\sin^2\theta + k\sin^2\phi = k\sin^2(\theta+\phi),$$

as required.

(Geometrically, the momentum equations say that the momenta mv_1, kmv_2 and mu form a triangle with angles ϕ, θ and $\pi - (\theta+\phi)$, and the two expressions above are just the sine rule in that triangle.)

Right angle between the paths

If the paths are at right angles, then $\theta + \phi = \frac{\pi}{2}$, so $\sin^2(\theta+\phi) = 1$ and $\sin\theta = \cos\phi$. The identity becomes

$$\cos^2\phi + k\sin^2\phi = k, \qquad \text{that is,} \qquad \cos^2\phi = k(1-\sin^2\phi) = k\cos^2\phi.$$

Here $\cos\phi \neq 0$: otherwise $\phi = \frac{\pi}{2}$ and then $\theta = 0$, but the question tells us θ is non-zero. Dividing by $\cos^2\phi$ gives $k = 1$, as required.

Checks

- The familiar fact from snooker: when equal masses collide elastically with one at rest, they move off at right angles. Our result is the converse.
- A numerical example: take $k = 2$ and $\phi = 30°$. A smooth elastic impact along that line sends P_2 off at speed $\frac{2}{3}u\cos 30°$, and P_1 then moves at $\theta = 90°$. Both sides of the identity equal $\frac{3}{2}$.
- We also simulated 2×10^5 random elastic collisions (random k, u and line of impact). The identity and both formulas for v_1 and kv_2 held to within 10^{-12}.

Common slips

- Writing the perpendicular momentum equation as $v_1 \sin\theta + kv_2 \sin\phi = 0$: the particles move off on opposite sides, so the components have opposite signs.
- Forgetting the k in the energy equation, or failing to say why $\cos\phi \neq 0$ in the final step.

Beyond STEP

In the frame moving with the centre of mass, an elastic collision just rotates both velocities, keeping their sizes. All of these angle relations come from mapping that circle of possible outcomes back to the laboratory frame. This viewpoint is the standard way to treat scattering, from billiard balls to particle physics. In particle physics, measured scattering angles are used to work out the mass ratio of the colliding particles. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the centre-of-mass treatment of collisions carefully.

WORKED SOLUTION

STEP I 2002, Question 12

Probability and Statistics • Geometric Probability

STEP I 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q12

The key idea

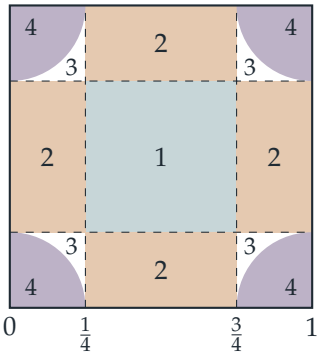
The grid repeats itself, so "placing the hoof at random" just means the centre of the hoof is uniformly distributed over one unit square. How many squares the hoof touches depends only on how close the centre is to the edges and corners of that square, so each answer corresponds to a region of the square, and its probability is the area of that region. Once we have the four probabilities, the rest is comparing them, and their mean, with what Harry actually does.

Solution

The distribution of Harry's answers

Let the centre C of the hoof lie in the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$. If Harry places his hoof at random, then by the periodicity of the grid C is uniformly distributed over this square, so the probability that C lies in a region is the region's area. Let N be the number of squares partly covered.

The hoof has radius $\frac{1}{4}$, which is less than $\frac{1}{2}$, so it can cross at most one vertical grid line and at most one horizontal one. It crosses a vertical line exactly when $x < \frac{1}{4}$ or $x > \frac{3}{4}$, and a horizontal line exactly when $y < \frac{1}{4}$ or $y > \frac{3}{4}$. (Positions exactly at distance $\frac{1}{4}$ from a line have zero area, so we ignore them.)



The events "crosses a vertical line" and "crosses a horizontal line" each have probability $\frac{1}{2}$, and they are independent because x and y are independent uniform coordinates. So:

- $N = 1$ when the hoof crosses no line: C is in the central square of side $\frac{1}{2}$, so $\mathrm{P}(N = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.
- $N = 2$ when it crosses exactly one line: $\mathrm{P}(N = 2) = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$.
- When it crosses one line of each kind, C is in one of the four corner squares of side $\frac{1}{4}$, with total probability $\frac{1}{4}$. The two lines meet at a corner of the grid, and the hoof covers part of all four squares round that corner if and only if it covers the corner point itself.

Ask yourself In a corner square, why does the answer depend on the distance to the grid corner?

Take the corner square nearest the origin, with $0 < x < \frac{1}{4}$ and $0 < y < \frac{1}{4}$. The hoof certainly reaches the square containing C, the square to its left and the square below it. The diagonal square $x < 0$, $y < 0$ has nearest point the origin, at distance $\sqrt{x^2 + y^2}$ from C, so the hoof enters it exactly when $\sqrt{x^2 + y^2} < \frac{1}{4}$. So $N = 4$ when C is within $\frac{1}{4}$ of a grid corner, and $N = 3$ otherwise.

The points of the unit square within $\frac{1}{4}$ of one of its corners form four quarter-discs of radius $\frac{1}{4}$, of total area $\pi\left(\frac{1}{4}\right)^2 = \frac{\pi}{16}$. Hence

$$\mathrm{P}(N = 1) = \tfrac{1}{4}, \qquad \mathrm{P}(N = 2) = \tfrac{1}{2}, \qquad \mathrm{P}(N = 3) = \tfrac{1}{4} - \tfrac{\pi}{16}, \qquad \mathrm{P}(N = 4) = \tfrac{\pi}{16}.$$

Is the evidence so far consistent with a random horse?

When the correct answer is $4 \div 4 = 1$, a random horse is right with probability $\mathrm{P}(N = 1) = \frac{1}{4}$. When the correct answer is $2 + 2 = 4$, it is right with probability $\mathrm{P}(N = 4) = \frac{\pi}{16} \approx 0.196$. These are exactly the success rates observed, so yes, the evidence is entirely consistent with my suspicion that Harry is just placing his hoof at random.

The expected answer, and the new horse

If Harry places his hoof at random, his answer doesn't depend on the question, so the expected value of his answers is just $\mathrm{E}(N)$ (the equal frequency of the true answers plays no part):

$$\begin{aligned}\mathrm{E}(N) &= 1\cdot\tfrac{1}{4} + 2\cdot\tfrac{1}{2} + 3\left(\tfrac{1}{4} - \tfrac{\pi}{16}\right) + 4\cdot\tfrac{\pi}{16} \\ &= \tfrac{1}{4} + 1 + \tfrac{3}{4} + \tfrac{\pi}{16} \\ &= 2 + \tfrac{\pi}{16} \approx 2.20.\end{aligned}$$

Compare the observed average of 2 with two hypotheses.

- *Harry calculates correctly.* The true answers are $1, 2, 3, 4$ equally often, so the average should be $\frac{5}{2}$. An average of 2 is well below this, so Harry is certainly getting many answers wrong. In fact each wrong answer is at most 3 below the truth, and the average shortfall is $\frac{1}{2}$, so at least $\frac{1}{6}$ of his answers must be wrong.
- *Harry is random.* The average should be close to $2 + \frac{\pi}{16} \approx 2.20$. A single answer has variance $\mathrm{E}(N^2) - \mathrm{E}(N)^2 = \frac{9}{2} + \frac{7\pi}{16} - \left(2 + \frac{\pi}{16}\right)^2 \approx 1.05$, so over n problems the average has standard deviation about $1.02/\sqrt{n}$. With "many" problems (say $n = 400$, standard deviation about 0.05) an average of 2 is about four standard deviations too low, which is very unlikely by chance.

So over many problems, an average of 2 is too low even for a horse placing its hoof at random: Harry seems to favour small answers, as if he presses lightly or aims for the middle of a square. Either way, he is not doing the calculations, and he does worse than a random horse would. Yes: it's time for a new horse (though perhaps one that can do arithmetic).

Checks

- The four probabilities add up: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} - \frac{\pi}{16} + \frac{\pi}{16} = 1$, and $\mathrm{P}(N = 3) = \frac{1}{4} - \frac{\pi}{16} \approx 0.054$ is positive, as it must be.
- A simulation of 400 000 random hoof placements, counting the squares within distance $\frac{1}{4}$ of the centre, gave relative frequencies 0.2495, 0.4990, 0.0547, 0.1968 and mean 2.199, against $2 + \frac{\pi}{16} \approx 2.196$.
- Sanity check: with a tiny hoof the answer would almost always be 1, and the mean would tend to 1. A larger hoof gives larger answers, and ours sits in between.

Common slips

- Assuming the hoof covers four squares whenever it crosses two grid lines. It also has to reach the corner where they meet, which is what makes π appear.
- Treating the average of 2 as evidence of randomness because it is "below 2.5".

Randomness predicts about 2.2, so the comparison has to be made with that value, taking account of the spread of the average.

Beyond STEP

This is *geometric probability*: when a point is uniform over a region, probabilities are areas, and the same idea gives Buffon's needle, whose answer famously also involves π. The final part is a first taste of hypothesis testing, using the central limit theorem to judge whether an observed mean is plausible under a model. Grimmett and Welsh, *Probability: An Introduction*, covers both continuous uniform distributions and the limit theorems behind this kind of argument.

WORKED SOLUTION

STEP I 2002, Question 13

Probability and Statistics • Conditional Expectation

STEP I 2002, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q13

The key idea

There are only six possible pairs (U, V), so everything comes down to a small table of joint probabilities and a little algebra on each case. For the quadratic, the sign of V decides everything, because U^2 is at most 1. For the cubic, the coefficients are suspiciously tangled up with U and V; that is a strong hint that it factorises, and once it does, the roots can be read off.

Solution

First we build the joint distribution. Since $\mathrm{P}(U = u,\, V = v) = \mathrm{P}(U = u)\,\mathrm{P}(V = v \mid U = u)$ and $\mathrm{P}(U = u) = \frac{1}{3}$ for each u,

	$U = 1$	$U = 0$	$U = -1$
$V = 1$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{2}{9}$
$V = -1$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{1}{9}$

The six entries add to 1, as they should.

Part (i)

The roots of $x^2 + Ux + V = 0$ are both real exactly when the discriminant is non-negative: $U^2 - 4V \geq 0$.

- If $V = -1$, the discriminant is $U^2 + 4 > 0$, so the roots are real (and distinct) whatever U is.
- If $V = 1$, the discriminant is $U^2 - 4 \leq 1 - 4 < 0$, since $U^2 \leq 1$. So the roots are never real.

Hence both roots are real if and only if $V = -1$, and

$$\mathrm{P}(\text{both roots real}) = \mathrm{P}(V = -1) = \tfrac{2}{9} + \tfrac{1}{6} + \tfrac{1}{9} = \tfrac{4+3+2}{18} = \tfrac{1}{2},$$

as required.

Part (ii)

By part (i), conditioning on "both roots real" is the same as conditioning on $V = -1$. Dividing the bottom row of the table by $\mathrm{P}(V = -1) = \frac{1}{2}$ gives

$$\mathrm{P}(U = 1 \mid V = -1) = \tfrac{4}{9}, \qquad \mathrm{P}(U = 0 \mid V = -1) = \tfrac{1}{3}, \qquad \mathrm{P}(U = -1 \mid V = -1) = \tfrac{2}{9}.$$

Ask yourself Why can't we just use $\mathrm{P}(U = u) = \frac{1}{3}$ here?

Because U and V are not independent: knowing that $V = -1$ makes $U = 1$ more likely (since $V = -1$ has probability $\frac{2}{3}$ when $U = 1$). The expectation we want is conditional, so the weights must be the conditional probabilities.

With $V = -1$ the equation is $x^2 + Ux - 1 = 0$, whose larger root is

$$R = \frac{-U + \sqrt{U^2 + 4}}{2}.$$

So $R = \frac{1}{2}(\sqrt{5} - 1)$ when $U = 1$, $R = 1$ when $U = 0$, and $R = \frac{1}{2}(\sqrt{5} + 1)$ when $U = -1$. Therefore

$$\begin{aligned} \mathrm{E}(R \mid \text{roots real}) &= \tfrac{4}{9} \cdot \frac{\sqrt{5} - 1}{2} + \tfrac{1}{3} \cdot 1 + \tfrac{2}{9} \cdot \frac{\sqrt{5} + 1}{2} \\ &= \frac{4\sqrt{5} - 4 + 6 + 2\sqrt{5} + 2}{18} \\ &= \frac{6\sqrt{5} + 4}{18} = \frac{3\sqrt{5} + 2}{9}. \end{aligned}$$

Part (iii)

The constant term is U, which suggests a factor $(x + U)$. Indeed

$$\begin{aligned} (x + U)(x^2 - 2Vx + 1) &= x^3 - 2Vx^2 + x + Ux^2 - 2UVx + U \\ &= x^3 + (U - 2V)x^2 + (1 - 2UV)x + U, \end{aligned}$$

which is exactly the given cubic. So the roots are $x = -U$ together with the roots of $x^2 - 2Vx + 1 = 0$. That quadratic has discriminant $4V^2 - 4 = 0$, because $V = \pm 1$ gives $V^2 = 1$. So it has a repeated root $x = V$, and the cubic is

$$(x + U)(x - V)^2 = 0,$$

with roots $-U$, V, V.

All three roots are positive if and only if $-U > 0$ and $V > 0$, that is, $U = -1$ and $V = 1$. (When $U = 0$ one root is 0, which is not positive.) From the table,

$$\mathrm{P}(\text{all roots positive}) = \mathrm{P}(U = -1,\ V = 1) = \tfrac{1}{3} \cdot \tfrac{2}{3} = \tfrac{2}{9}.$$

Checks

- The conditional probabilities $\frac{4}{9}, \frac{1}{3}, \frac{2}{9}$ add to 1.
- The answer to (ii) is about 0.967. It must lie between the smallest possible larger root, $\frac{1}{2}(\sqrt{5}-1) \approx 0.618$, and the largest, $\frac{1}{2}(\sqrt{5}+1) \approx 1.618$, and it should sit below the middle value 1, because the smallest root carries the biggest weight. It does.
- In (iii), try $U = -1$, $V = 1$: the cubic becomes $x^3 - 3x^2 + 3x - 1 = (x-1)^3$, with roots $1, 1, 1$, all positive. Try $U = 1$, $V = 1$: it becomes $x^3 - x^2 - x + 1 = (x+1)(x-1)^2$, which has the negative root -1.
- We confirmed all three answers by exact enumeration over the six cases with fractions, and by finding the cubic's roots numerically in each case.

Common slips

- Using the unconditional weights $\frac{1}{3}$ in (ii), which gives $\frac{1}{3}(\sqrt{5}+1) \approx 1.079$ instead.
- Counting $U = 0$ in (iii): a root of 0 is not positive.
- Trying to analyse the cubic with discriminants or sign changes instead of looking for the factorisation.

Beyond STEP

The heart of this question is conditional expectation, $\mathrm{E}(X \mid A)$, computed by reweighting with conditional probabilities. At university this idea grows into the conditional expectation $\mathrm{E}(X \mid Y)$ as a random variable in its own right, and the tower law $\mathrm{E}\big(\mathrm{E}(X \mid Y)\big) = \mathrm{E}(X)$, which makes many hard expectations easy. Grimmett and Welsh, *Probability: An Introduction*, develops this clearly.

WORKED SOLUTION

STEP I 2002, Question 14

Probability and Statistics • Geometric Series in Probability

STEP I 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q14

The key idea

Condition on how many numbers I punch in altogether. Once that number is fixed, the time between the alarm going off and my getting the money is fixed too, so the police probability is just $1 - \mathrm{e}^{-\lambda t}$ for that time. Averaging over the number of attempts gives the answer. In both plans the averaging turns into a geometric series in $q = \mathrm{e}^{-\lambda/r}$, so it's worth naming q early.

Solution

Punching in order

Let K be the position of my number in the list $1, 2, \ldots, n$. Then K is the number of attempts I make, and $P(K = k) = \frac{1}{n}$ for $k = 1, \ldots, n$.

Suppose $K = k$. I punch numbers at a rate of r per minute, so successive numbers are $1/r$ minutes apart. Measure time from the first number I punch.

- If $k = 1$, the first number is right, no wrong number is ever entered and the police are never called. The probability that they arrive first is $0 = 1 - \mathrm{e}^{0}$.
- If $k \geq 2$, the first number is wrong, so the alarm is raised at time 0, and the correct number is the kth, entered at time $(k-1)/r$. The police arrive before I get my money if they arrive within $t = (k-1)/r$ minutes, which has probability $1 - \mathrm{e}^{-\lambda(k-1)/r}$.

The formula $1 - \mathrm{e}^{-\lambda(k-1)/r}$ therefore covers every k, including $k = 1$. Let A be the event that the police arrive before I get my money. By the law of total probability,

$$P(A) = \sum_{k=1}^{n} P(K = k)\, P(A \mid K = k) = \sum_{k=1}^{n} \frac{1 - \mathrm{e}^{-\lambda(k-1)/r}}{n},$$

as required.

To simplify, put $q = \mathrm{e}^{-\lambda/r}$. Since $\lambda, r > 0$ we have $0 < q < 1$, and $\mathrm{e}^{-\lambda(k-1)/r} = q^{k-1}$. So

$$P(A) = 1 - \frac{1}{n}\sum_{k=1}^{n} q^{k-1} = 1 - \frac{1-q^n}{n(1-q)}$$
$$= 1 - \frac{1-\mathrm{e}^{-\lambda n/r}}{n\left(1-\mathrm{e}^{-\lambda/r}\right)},$$

summing the geometric series, which is valid because $q \neq 1$.

Punching at random

Now each attempt is a fresh number chosen uniformly from $1, \ldots, n$, independently of earlier attempts (I don't remember what I've tried). So each attempt is correct with probability $\frac{1}{n}$, independently. If N is the number of attempts up to and including the correct one, then N is geometric:

$$P(N = k) = \frac{1}{n}\left(\frac{n-1}{n}\right)^{k-1}, \qquad k = 1, 2, 3, \ldots$$

Ask yourself Does the timing argument still work when numbers can be repeated?

Yes. The police are alerted at the first wrong number, and if $N = k \geq 2$ that is the very first attempt, at time 0. The money comes at the kth attempt, at time $(k-1)/r$. Repeats change only the distribution of N, not the timing. So exactly as before, $P(A \mid N = k) = 1 - q^{k-1}$, and this is also right for $k = 1$.

Write $p = \frac{n-1}{n}$. Then

$$P(A) = \sum_{k=1}^{\infty} \frac{1}{n} p^{k-1}\left(1 - q^{k-1}\right) = \frac{1}{n}\sum_{k=1}^{\infty} p^{k-1} - \frac{1}{n}\sum_{k=1}^{\infty} (pq)^{k-1}$$
$$= \frac{1}{n} \cdot \frac{1}{1-p} - \frac{1}{n} \cdot \frac{1}{1-pq}.$$

Splitting the sum is fine because both series converge: $0 \leq p < 1$ and $0 < pq < 1$. The first term is $\frac{1}{n} \cdot n = 1$, which is just the statement that N is certainly finite. For the second,

$$n(1-pq) = n - (n-1)q,$$

so

$$P(A) = 1 - \frac{1}{n - (n-1)\mathrm{e}^{-\lambda/r}},$$

as required.

Checks

- With $n = 1$ there is never a wrong number, and both formulae give 0: the first because $1 - q^1 = 1 - q$, the second because the denominator is 1.

- As $\lambda \to \infty$ (the police arrive at once), $q \to 0$ and both answers tend to $1 - \frac{1}{n}$. That's the probability that the first attempt is wrong, as it should be.
- Random punching should be worse. For $n = 5$, $\lambda = 0.7$, $r = 3$ we get 0.3382 in order and 0.4543 at random. A simulation with 2×10^5 trials of each gave 0.339 and 0.456.

Common slips

- Using time k/r instead of $(k-1)/r$. The clock starts at the first (wrong) number, so the kth number comes $k - 1$ gaps later.
- In the random case, giving each k a probability of $\frac{1}{n}$, or stopping the sum at n. With repeats allowed, N is geometric and unbounded.

Beyond STEP

The police arrival time here is an exponential random variable, and the calculation is $P(T < X)$ for a random X, found by conditioning on X. When X is geometric the answer has a neat closed form because $E\left(q^X\right)$ is the probability generating function of X evaluated at q. Indeed both answers are $1 - E\left(q^{N-1}\right)$. Generating functions and the exponential distribution are central to first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers both well.

WORKED SOLUTION

STEP II 2002, Question 1

Pure • Integration by Substitution

STEP II 2002, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q1

The key idea

The double-angle formula turns $1-\cos 2\theta$ into $2\sin^2\theta$, and $\operatorname{cosec}^2\theta$ is something we can integrate on sight. The suggested substitution $x=\sin 2\theta$ is designed so that $\sqrt{1-x^2}$ becomes $\cos 2\theta$, which pulls the second integral back to the first. For the last integral, the limits 1 and $2/\sqrt{3}$ are the reciprocals of the limits 1 and $\sqrt{3}/2$, which strongly suggests trying $y=1/x$.

Solution

The first integral

Since $\cos 2\theta = 1-2\sin^2\theta$, we have $1-\cos 2\theta = 2\sin^2\theta$, which is positive on $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, so the integrand is well behaved there. Using $\frac{\mathrm{d}}{\mathrm{d}\theta}(\cot\theta) = -\operatorname{cosec}^2\theta$,

$$\begin{aligned}\int_{\pi/6}^{\pi/4} \frac{1}{1-\cos 2\theta}\,\mathrm{d}\theta &= \frac{1}{2}\int_{\pi/6}^{\pi/4} \operatorname{cosec}^2\theta\,\mathrm{d}\theta = \frac{1}{2}\Big[-\cot\theta\Big]_{\pi/6}^{\pi/4} \\ &= \frac{1}{2}\left(\cot\tfrac{\pi}{6} - \cot\tfrac{\pi}{4}\right) = \frac{1}{2}\left(\sqrt{3}-1\right) = \frac{\sqrt{3}}{2} - \frac{1}{2},\end{aligned}$$

as required.

The second integral

Put $x=\sin 2\theta$, so $\mathrm{d}x = 2\cos 2\theta\,\mathrm{d}\theta$. When $\theta = \frac{1}{6}\pi$, $x = \sin\frac{1}{3}\pi = \frac{\sqrt{3}}{2}$, and when $\theta = \frac{1}{4}\pi$, $x = \sin\frac{1}{2}\pi = 1$; as θ runs over $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, x increases steadily from $\frac{\sqrt{3}}{2}$ to 1.

Ask yourself Is $\sqrt{1-x^2}$ equal to $\cos 2\theta$, or to $-\cos 2\theta$?

We have $\sqrt{1-x^2} = \sqrt{\cos^2 2\theta} = |\cos 2\theta|$. On our range $2\theta \in [\frac{1}{3}\pi, \frac{1}{2}\pi]$, where $\cos 2\theta \geq 0$,

so $\sqrt{1-x^2} = \cos 2\theta$. Therefore

$$\int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x = \int_{\pi/6}^{\pi/4} \frac{2\cos 2\theta}{1-\cos 2\theta}\,\mathrm{d}\theta.$$

To bring in the first integral, write the numerator in terms of the denominator:

$$\frac{\cos 2\theta}{1-\cos 2\theta} = \frac{1-(1-\cos 2\theta)}{1-\cos 2\theta} = \frac{1}{1-\cos 2\theta} - 1.$$

Hence, using the first result,

$$\begin{aligned}\int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x &= 2\int_{\pi/6}^{\pi/4} \frac{1}{1-\cos 2\theta}\,\mathrm{d}\theta - 2\int_{\pi/6}^{\pi/4} 1\,\mathrm{d}\theta \\ &= 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) - 2\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \sqrt{3} - 1 - \frac{\pi}{6},\end{aligned}$$

as required.

The final integral

Put $y = 1/x$, so $\mathrm{d}y = -\mathrm{d}x/x^2$. When $y = 1$, $x = 1$, and when $y = 2/\sqrt{3}$, $x = \sqrt{3}/2$. On this range $x > 0$, so

$$\sqrt{y^2-1} = \sqrt{\frac{1-x^2}{x^2}} = \frac{\sqrt{1-x^2}}{x},$$
$$y\left(y - \sqrt{y^2-1}\right) = \frac{1}{x}\cdot\frac{1-\sqrt{1-x^2}}{x} = \frac{1-\sqrt{1-x^2}}{x^2}.$$

Ask yourself Where does the minus sign from $\mathrm{d}y = -\mathrm{d}x/x^2$ go?

The substitution reverses the order of the limits, and swapping them back absorbs the minus sign:

$$\begin{aligned}\int_{1}^{2/\sqrt{3}} \frac{1}{y\left(y-\sqrt{y^2-1}\right)}\,\mathrm{d}y &= \int_{1}^{\sqrt{3}/2} \frac{x^2}{1-\sqrt{1-x^2}}\cdot\left(-\frac{1}{x^2}\right)\mathrm{d}x \\ &= \int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{1-x^2}}\,\mathrm{d}x \\ &= \sqrt{3} - 1 - \frac{\pi}{6}.\end{aligned}$$

The factors of x^2 cancel exactly, which is why the extra factor of y was put in the denominator.

Checks

- The integrands of the second and third integrals are both positive, so the answer must be positive. Indeed $\sqrt{3} - 1 - \frac{\pi}{6} \approx 0.73205 - 0.52360 = 0.20845$.
- Simpson's rule on each of the three integrals gives $0.366025\ldots$ for the first (matching $\frac{\sqrt{3}}{2} - \frac{1}{2}$) and $0.208452\ldots$ for the second and third (matching $\sqrt{3} - 1 - \frac{\pi}{6}$).
- A rough estimate: in the second integral the interval has length $1 - \frac{\sqrt{3}}{2} \approx 0.134$, and the integrand runs from 2 (at $x = \frac{\sqrt{3}}{2}$) down to 1 (at $x = 1$), so the integral should lie between 0.134 and 0.268. It does.

Common slips

- Writing $\sqrt{1 - \sin^2 2\theta} = \cos 2\theta$ without checking the sign of $\cos 2\theta$ on the range.
- Forgetting to change the limits, or losing the minus sign when the limits reverse under $y = 1/x$.
- Evaluating $\left[-\cot\theta \right]$ as $\cot\frac{\pi}{4} - \cot\frac{\pi}{6}$, which gives the right size but the wrong sign.

Beyond STEP

The final part is a small example of a general principle: a substitution is a change of variables that can map one integral exactly onto another, and spotting reciprocal limits is a reliable hint that $y = 1/x$ will do it. In a first analysis course, substitution is justified rigorously from the chain rule and the fundamental theorem of calculus, with care about where the substitution is monotonic and the integrand continuous, exactly the conditions we checked here. Abbott's *Understanding Analysis* gives a clear account of the theory behind these manipulations.

WORKED SOLUTION

STEP II 2002, Question 2

Pure • Reciprocal Polynomial Equations

STEP II 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q2

The key idea

Look at the coefficients from the outside in: 1 and 1, then 5 and -5, then 4 in the middle. Coefficients that match or flip sign in pairs mean that dividing by the middle power of z leaves only combinations of $z^k + z^{-k}$ and $z^k - z^{-k}$, and each of these is a polynomial in $w = z - z^{-1}$. That halves the degree. The degree-8 equation has the same structure, and once reduced it turns out to have it *again*, so the trick is used twice.

Solution

The quartic

Since $z = 0$ is not a root (the constant term is 1), we may divide by z^2:

$$z^2 + 5z + 4 - 5z^{-1} + z^{-2} = 0, \qquad \text{that is,} \qquad \left(z^2 + z^{-2}\right) + 5\left(z - z^{-1}\right) + 4 = 0.$$

With $w = z - z^{-1}$ we have $w^2 = z^2 - 2 + z^{-2}$, so $z^2 + z^{-2} = w^2 + 2$. Substituting,

$$(w^2 + 2) + 5w + 4 = 0 \quad \Longleftrightarrow \quad w^2 + 5w + 6 = 0,$$

as required. This factorises as $(w + 2)(w + 3) = 0$, so $w = -2$ or $w = -3$.

To get back to z, multiply $z - z^{-1} = w$ by z:

$$z^2 - wz - 1 = 0.$$

- $w = -2$: $z^2 + 2z - 1 = 0$, so $z = -1 \pm \sqrt{2}$.
- $w = -3$: $z^2 + 3z - 1 = 0$, so $z = \frac{1}{2}\left(-3 \pm \sqrt{13}\right)$.

These four distinct values are the roots of the quartic:

$$z = -1 \pm \sqrt{2}, \qquad z = \frac{-3 \pm \sqrt{13}}{2}.$$

The equation of degree 8

Again $z = 0$ is not a root, so we divide by z^4 and pair the coefficients symmetrically about the middle:

$$2\left(z^4 + z^{-4}\right) - 3\left(z^3 - z^{-3}\right) - 12\left(z^2 + z^{-2}\right) + 12\left(z - z^{-1}\right) + 22 = 0.$$

(Check the pairs: z^8 and z^0 both have 2; z^7 has -3 and z^1 has $+3$; z^6 and z^2 both have -12; z^5 has 12 and z^3 has -12.)

We need each bracket in terms of $w = z - z^{-1}$. We already have $z^2 + z^{-2} = w^2 + 2$. Cubing,

$$w^3 = z^3 - 3z + 3z^{-1} - z^{-3} \quad \Longrightarrow \quad z^3 - z^{-3} = w^3 + 3w,$$

and squaring $z^2 + z^{-2}$,

$$z^4 + z^{-4} = \left(z^2 + z^{-2}\right)^2 - 2 = (w^2 + 2)^2 - 2 = w^4 + 4w^2 + 2.$$

Substituting,

$$\begin{aligned} 0 &= 2(w^4 + 4w^2 + 2) - 3(w^3 + 3w) - 12(w^2 + 2) + 12w + 22 \\ &= 2w^4 - 3w^3 - 4w^2 + 3w + 2. \end{aligned}$$

Ask yourself We are left with a quartic in w. Have we gained anything?

Yes: look at its coefficients, 2, -3, -4, 3, 2. They have exactly the same pattern as before (the outer pair equal, the next pair opposite in sign), so the same trick works one level down. Since $w = 0$ is not a root, divide by w^2:

$$2\left(w^2 + w^{-2}\right) - 3\left(w - w^{-1}\right) - 4 = 0.$$

Put $v = w - w^{-1}$, so that $w^2 + w^{-2} = v^2 + 2$:

$$2(v^2 + 2) - 3v - 4 = 0 \quad \Longleftrightarrow \quad 2v^2 - 3v = 0 \quad \Longleftrightarrow \quad v(2v - 3) = 0.$$

So $v = 0$ or $v = \frac{3}{2}$. Now unwind, using $w^2 - vw - 1 = 0$:

- $v = 0$: $w^2 = 1$, so $w = \pm 1$.
- $v = \frac{3}{2}$: $2w^2 - 3w - 2 = 0$, that is $(2w + 1)(w - 2) = 0$, so $w = 2$ or $w = -\frac{1}{2}$.

Finally unwind once more with $z^2 - wz - 1 = 0$:

- $w = 1$: $z^2 - z - 1 = 0$, so $z = \frac{1}{2}\left(1 \pm \sqrt{5}\right)$.
- $w = -1$: $z^2 + z - 1 = 0$, so $z = \frac{1}{2}\left(-1 \pm \sqrt{5}\right)$.
- $w = 2$: $z^2 - 2z - 1 = 0$, so $z = 1 \pm \sqrt{2}$.
- $w = -\frac{1}{2}$: $2z^2 + z - 2 = 0$, so $z = \frac{1}{4}\left(-1 \pm \sqrt{17}\right)$.

Each quadratic $z^2 - wz - 1 = 0$ has discriminant $w^2 + 4 > 0$, so all these roots are real, and the eight values are distinct. Since a polynomial of degree 8 has at most eight roots, these are all of them:

$$z = \frac{1 \pm \sqrt{5}}{2}, \qquad z = \frac{-1 \pm \sqrt{5}}{2}, \qquad z = 1 \pm \sqrt{2}, \qquad z = \frac{-1 \pm \sqrt{17}}{4}.$$

Checks

- Each pair of roots from $z^2 - wz - 1 = 0$ has product -1, which is a quick check on the arithmetic: for example $(1 + \sqrt{2})(1 - \sqrt{2}) = -1$.
- For the degree-8 equation the sum of the roots should be $\frac{3}{2}$ (minus the z^7 coefficient over the z^8 coefficient). The four pairs contribute $1 - 1 + 2 - \frac{1}{2} = \frac{3}{2}$. Good.
- Numerically, $z = \frac{1}{4}(-1 + \sqrt{17}) \approx 0.7808$ makes the degree-8 polynomial vanish to within 10^{-9}, as do the other seven roots and the four roots of the quartic.

Common slips

- Writing $z^3 - z^{-3} = w^3$: the cross terms give the extra $3w$.
- Stopping at the quartic in w and trying to factorise it by trial, instead of noticing that its coefficients have the same symmetry.
- Giving only the values of w (or v) and forgetting to return to z.

Beyond STEP

Polynomials whose coefficients read the same (or the same up to alternating sign) backwards are called *reciprocal* or *palindromic*; their roots come in pairs related by $z \mapsto 1/z$ or $z \mapsto -1/z$, and the substitution here is exactly the reduction this symmetry allows. In Galois theory this is the observation that the equation is invariant under a group of transformations of the roots, so it can be solved in stages through a tower of smaller equations, which is what happened here: degree 8, then 4, then 2. Allenby's *Rings, Fields and Groups* is a good place to see these ideas develop.

WORKED SOLUTION

STEP II 2002, Question 3

Pure • Fermat Numbers and Primes

STEP II 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q3

This question has a full guided treatment in *Beyond STEP*, Section 1.1, *A Prime Factory*: https://beyondstep.keithmcnulty.org/ch01/fermat-numbers.html

The key idea

Write $F_k - 2 = 2^{2^k} - 1$. Multiplying this by $F_k = 2^{2^k} + 1$ gives a difference of two squares, and that is exactly the step the induction needs. Once the product formula is in place, any common factor of two Fermat numbers has to divide 2. Since every Fermat number is odd, that factor can only be 1.

Solution

The first four Fermat numbers

Since $2^0 = 1$, $2^1 = 2$, $2^2 = 4$ and $2^3 = 8$,

$$F_0 = 2^1 + 1 = 3, \qquad F_1 = 2^2 + 1 = 5, \qquad F_2 = 2^4 + 1 = 17, \qquad F_3 = 2^8 + 1 = 257.$$

Now we check $(*)$ for the first three values of k:

$$\begin{aligned} k = 1: &\quad F_0 = 3 = 5 - 2 = F_1 - 2, \\ k = 2: &\quad F_0F_1 = 15 = 17 - 2 = F_2 - 2, \\ k = 3: &\quad F_0F_1F_2 = 15 \times 17 = 255 = 257 - 2 = F_3 - 2. \end{aligned}$$

Proof of $(*)$ for all $k \geq 1$

We use induction on k. The case $k = 1$ is done above.

Suppose $(*)$ holds for some $k \geq 1$, so that $F_0F_1 \cdots F_{k-1} = F_k - 2$. Then

$$\begin{aligned} F_0F_1 \cdots F_{k-1}F_k &= (F_k - 2)\, F_k \\ &= \left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right) \\ &= \left(2^{2^k}\right)^2 - 1 \\ &= 2^{2 \cdot 2^k} - 1 = 2^{2^{k+1}} - 1 \\ &= F_{k+1} - 2. \end{aligned}$$

So $(*)$ holds for $k+1$ as well, and by induction it holds for every $k \geq 1$.

Ask yourself Why is $\left(2^{2^k}\right)^2$ equal to $2^{2^{k+1}}$, and not $2^{2^{2k}}$ or something else?

Squaring a power doubles the exponent: $\left(2^a\right)^2 = 2^{2a}$. Here $a = 2^k$, and $2a = 2 \cdot 2^k = 2^{k+1}$. Taking care over towers of exponents like this is most of what the induction step involves.

No two Fermat numbers share a factor

Take two different Fermat numbers F_m and F_n with $m < n$. Then $n \geq 1$, and F_m is one of the factors $F_0, \ldots, F_{n-1}$ in $(*)$ with $k = n$. So

$$F_n - 2 = F_m \times (\text{an integer}),$$

which means F_m divides $F_n - 2$.

Now suppose a positive integer d divides both F_m and F_n. Since $d \mid F_m$ and $F_m \mid F_n - 2$, we have $d \mid F_n - 2$. Since d divides F_n as well, it divides the difference:

$$d \mid F_n - (F_n - 2) = 2,$$

so $d = 1$ or $d = 2$. But $F_m = 2^{2^m} + 1$ is odd, being an even number plus 1 (note $2^m \geq 1$). So $d \neq 2$, and therefore $d = 1$. Hence no two Fermat numbers have a common factor greater than 1.

Infinitely many primes

Each F_n is at least 3, so it has at least one prime factor. Choose one and call it p_n. This gives a prime for each $n = 0, 1, 2, \ldots$.

Ask yourself Couldn't two different Fermat numbers give the same prime?

No. If $p_m = p_n$ with $m \neq n$, that prime would divide both F_m and F_n. It would then be a common factor greater than 1, which the previous part rules out. So the primes $p_0, p_1, p_2, \ldots$ are all different. Since there are infinitely many Fermat numbers, there are infinitely many distinct primes.

Notice that we never needed F_n itself to be prime, only that each one has *some* prime factor that no other Fermat number shares.

Checks

- $(*)$ with $k = 4$: $255 \times 257 = 65535 = 65537 - 2 = F_4 - 2$. We also checked $(*)$ exactly by computer for all k up to 12.
- The smallest prime factors of $F_0, \ldots, F_5$ are $3, 5, 17, 257, 65537, 641$, all different, just as the argument predicts. $F_5 = 4294967297 = 641 \times 6700417$ is the first Fermat number that isn't prime.
- We confirmed numerically that $\gcd(F_m, F_n) = 1$ for all $0 \leq m < n \leq 10$.

Common slips

- Reading 2^{2^n} as $(2^2)^n = 4^n$. The exponent is 2^n, so $F_3 = 2^8 + 1$, not $4^3 + 1$.
- Getting $d \mid 2$ and then forgetting to rule out $d = 2$. The fact that Fermat numbers are odd is what finishes the argument.
- Assuming every Fermat number is prime. The final part works without this, and it is false anyway ($641 \mid F_5$).

Beyond STEP

This proof is due to Pólya. It belongs to a family of proofs of the infinitude of primes. Each builds an infinite sequence of pairwise coprime integers, and each term then contributes its own new prime. Euclid's proof can be read the same way. Fermat numbers come up again in number theory courses: Gauss showed that a regular n-gon can be constructed with ruler and compass exactly when n is a power of 2 times a product of distinct Fermat primes, and no Fermat prime beyond F_4 is known. For more, read Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2002, Question 4

Pure • Inequalities from Integrals

STEP II 2002, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q4

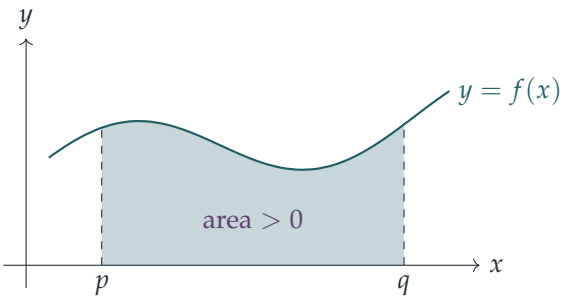
The key idea

A quadratic $as^2 - bs + c$ with $a > 0$ and $b^2 < 4ac$ has no real roots and a positive leading coefficient, so it is positive for *every* real s. Whatever we substitute for s (x, $\sin x$ or $1/x$), the resulting function is positive, so its integral over any interval is positive. Each part is then just a matter of choosing the substitution and the interval so that the integral produces exactly the numbers in the required inequality.

Solution

The opening sketch

The integral $\int_p^q f(x)\,\mathrm{d}x$ is the signed area between the curve $y = f(x)$ and the x-axis, from $x = p$ to $x = q$. Area above the axis counts as positive and area below counts as negative. If $f(x) > 0$ throughout $p < x < q$, the whole curve lies above the axis there, so the region has no negative part and has a genuine (non-zero) area. Hence the integral is strictly positive. (All the functions in this question are continuous, so the area is well defined.)



A fact used in every part

Let $g(s) = as^2 - bs + c$ with $a > 0$ and $b^2 < 4ac$. Completing the square,

$$g(s) = a\left(s - \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}.$$

The first term is at least 0 because $a > 0$, and the second is strictly positive because $4ac - b^2 > 0$ and $a > 0$. So $g(s) > 0$ for all real s.

Part (i)

Take $f(x) = ax^2 - bx + c = g(x)$, which is positive for all x. We may integrate over any interval we like, and $[0, 1]$ gives the simplest numbers:

$$0 < \int_0^1 (ax^2 - bx + c)\,\mathrm{d}x = \frac{a}{3} - \frac{b}{2} + c.$$

Multiplying by 6 gives $2a - 3b + 6c > 0$, that is, $3b < 2a + 6c$, as required.

Part (ii)

Now $f(x) = a\sin^2 x - b\sin x + c = g(\sin x)$. Since g is positive at every real number, it is certainly positive at $\sin x$, so $f(x) > 0$ for all x.

? **Ask yourself** Which interval should we integrate over?

The target has π multiplying a and c but not b. Over $[0, \frac{\pi}{2}]$, $\int \sin x\,\mathrm{d}x = 1$ has no π, while $\int \sin^2 x\,\mathrm{d}x$ and $\int 1\,\mathrm{d}x$ do. Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\int_0^{\pi/2} \sin^2 x\,\mathrm{d}x = \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\pi/2} = \frac{\pi}{4}, \qquad \int_0^{\pi/2} \sin x\,\mathrm{d}x = \Big[-\cos x\Big]_0^{\pi/2} = 1.$$

Hence

$$0 < \int_0^{\pi/2} f(x)\,\mathrm{d}x = \frac{a\pi}{4} - b + \frac{c\pi}{2}.$$

Multiplying by 4 gives $a\pi + 2c\pi - 4b > 0$, that is, $4b < (a + 2c)\pi$, as required.

Part (iii)

? **Ask yourself** What function has integrals giving $\ln(q/p)$, $\frac{1}{p} - \frac{1}{q}$ and $q - p$?

These are $\int_p^q \frac{1}{x}\,\mathrm{d}x$, $\int_p^q \frac{1}{x^2}\,\mathrm{d}x$ and $\int_p^q 1\,\mathrm{d}x$. So take

$$f(x) = \frac{a}{x^2} - \frac{b}{x} + c = g\left(\frac{1}{x}\right),$$

which is defined and continuous for $x > 0$, and positive there because g is positive everywhere. Since $q > p > 0$, the interval $[p, q]$ avoids $x = 0$, so

$$0 < \int_p^q \left(\frac{a}{x^2} - \frac{b}{x} + c\right) \mathrm{d}x = \left[-\frac{a}{x} - b\ln x + cx\right]_p^q$$
$$= a\left(\frac{1}{p} - \frac{1}{q}\right) - b\ln\frac{q}{p} + c(q-p).$$

Rearranging, $b\ln(q/p) < a\left(\frac{1}{p} - \frac{1}{q}\right) + c(q-p)$, as required. (The condition $p > 0$ also lets us write $\ln q - \ln p = \ln(q/p)$ with both logarithms defined.)

Checks

Take $a = b = c = 1$, so $b^2 = 1 < 4 = 4ac$. Then (i) says $3 < 8$, (ii) says $4 < 3\pi \approx 9.42$, and (iii) with $p = 1$, $q = 2$ says $\ln 2 \approx 0.693 < \frac{1}{2} + 1 = 1.5$. All hold. We also tested all three inequalities on 20 000 random triples (a, b, c) satisfying the conditions, including cases with b^2 just below $4ac$, and confirmed each integral against Simpson's rule. Note that the inequalities can fail without the conditions: $a = 1$, $b = 10$, $c = 1$ breaks (i), since $30 > 8$.

Common slips

- Forgetting to justify that $f > 0$: the conditions $a > 0$ and $b^2 < 4ac$ are exactly what makes the quadratic positive, and this has to be said.
- In (ii), taking $\int_0^{\pi/2} \sin^2 x \,\mathrm{d}x = \frac{\pi}{2}$ instead of $\frac{\pi}{4}$.
- In (iii), not mentioning that $p > 0$ keeps $x = 0$ (where f is undefined) out of the interval.

Beyond STEP

The principle "a positive continuous function has a positive integral" is one of the basic properties proved for the Riemann integral in a first analysis course. Together with linearity, it is the source of many classical inequalities, including the integral form of the Cauchy–Schwarz inequality, which comes from integrating the square $(f + tg)^2 \geq 0$ and using the same discriminant argument as here. Abbott's *Understanding Analysis* develops the integral carefully from this point of view.

WORKED SOLUTION

STEP II 2002, Question 5

Pure • Logistic Map Iteration

STEP II 2002, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q5

The key idea

Everything hangs on the parabola $y = kx(1-x)$, whose largest value on $[0,1]$ is $\frac{k}{4}$ at $x = \frac{1}{2}$. That alone settles part (i) by induction. Parts (ii) and (iii) ask for fixed points and then points of period two. The trick in (iii) is to write x_1 in terms of b: $x_1 = ka(1-a) = ab$. Then $x_2 = a$ turns into a cubic in b, and that cubic has an obvious root which is exactly the fixed point from (ii).

Solution

Part (i)

We use induction on n. The case $n = 0$ is given. Suppose $0 < x_n < 1$. Then $x_n > 0$ and $1 - x_n > 0$, so $x_n(1-x_n) > 0$. Completing the square,

$$x_n(1-x_n) = \tfrac{1}{4} - \left(x_n - \tfrac{1}{2}\right)^2 \leq \tfrac{1}{4}.$$

Since $0 < k < 4$, multiplying through by k keeps the inequalities, and

$$0 < x_{n+1} = kx_n(1-x_n) \leq \tfrac{k}{4} < 1.$$

So $0 < x_{n+1} < 1$, and by induction $0 < x_n < 1$ for all n. Notice where each hypothesis is used: $k > 0$ for the lower bound and $k < 4$ for the upper one.

Part (ii)

A constant sequence needs only $x_1 = x_0$, that is $a = ka(1-a)$. Since $a \neq 0$ we may divide by a to get $1 = k(1-a)$, and since $a \neq 1$ we may divide by $1-a$:

$$k = \frac{1}{1-a}.$$

Conversely, with this k we have $x_1 = a$, and so every later term is a too.

Part (iii)

With $b = k(1-a)$ we have

$$x_1 = ka(1-a) = ab.$$

If $x_2 = a$ the sequence repeats with period two, so the whole condition is just $x_2 = a$:

$$k\,(ab)(1-ab) = a.$$

Divide by $a \neq 0$ to get $kb(1-ab) = 1$. Since $a \neq 1$ we can write $k = \dfrac{b}{1-a}$, so

$$\frac{b^2(1-ab)}{1-a} = 1 \quad \Longrightarrow \quad b^2 - ab^3 = 1-a \quad \Longrightarrow \quad ab^3 - b^2 + (1-a) = 0,$$

which is the required result.

Ask yourself Which root of the cubic can we spot, and what does it mean?

Putting $b = 1$ gives $a - 1 + 1 - a = 0$, so $b = 1$ is a root. But $b = 1$ means $k(1-a) = 1$, which is the fixed point of part (ii): then $x_1 = ab = a$. So the extra condition $x_1 \neq a$, which says $ab \neq a$, is exactly $b \neq 1$. Dividing out the factor $b - 1$:

$$ab^3 - b^2 + (1-a) = (b-1)\big(ab^2 + (a-1)b + (a-1)\big),$$

which you can confirm by expanding. So b must be a root of

$$ab^2 - (1-a)b - (1-a) = 0$$

other than 1. Since $a \neq 0$, this really is a quadratic, with

$$b = \frac{(1-a) \pm \sqrt{(1-a)^2 + 4a(1-a)}}{2a} = \frac{(1-a) \pm \sqrt{(1-a)(1+3a)}}{2a}.$$

For real roots we need $(1-a)(1+3a) \geq 0$, which means $-\frac{1}{3} \leq a \leq 1$. With $a \neq 0, 1$ this leaves $-\frac{1}{3} \leq a < 1$, $a \neq 0$. For any other value of a no real k works.

In that range $1 - a > 0$, so dividing by $1 - a$ gives $k = b/(1-a)$ in a neat form, using $\sqrt{(1-a)(1+3a)}/(1-a) = \sqrt{(1+3a)/(1-a)}$:

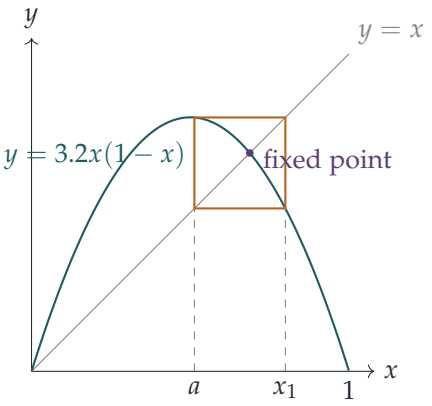
$$\boxed{k = \frac{1}{2a}\left(1 \pm \sqrt{\frac{1+3a}{1-a}}\right), \qquad -\tfrac{1}{3} \leq a < 1,\ a \neq 0.}$$

One last thing: we must make sure neither value is the excluded root $b = 1$. The quadratic has $b = 1$ as a root only when $a + (a-1) + (a-1) = 0$, i.e. $a = \frac{2}{3}$. Then $\sqrt{(1+3a)/(1-a)} = 3$ and the two values are $k = 3$ (which is $b = 1$, the fixed point, so it is excluded) and $k = -\frac{3}{2}$. So:

- for $a = \frac{2}{3}$ the only possibility is $k = -\frac{3}{2}$;
- for $a = -\frac{1}{3}$ the two signs coincide and $k = -\frac{3}{2}$;

- for every other allowed a both signs give a genuine 2-cycle.

The picture below shows one such cycle, for $k = 3.2$. The orbit bounces between $a \approx 0.513$ and $x_1 \approx 0.800$, tracing a square between the parabola and the line $y = x$.



Checks

- Take $a = 0.7$. The formula gives $k \approx 3.0104$ or $k \approx -1.5818$. With $k \approx 3.0104$ we get $x_1 \approx 0.6322$ and then $x_2 \approx 0.7000$, so the orbit really does have period two.
- At $a = \frac{2}{3}$, $k = 3$ gives $x_1 = 3 \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{3}$, the fixed point, as predicted. With $k = -\frac{3}{2}$ we get $x_1 = -\frac{1}{3}$ and $x_2 = -\frac{3}{2} \cdot (-\frac{1}{3}) \cdot \frac{4}{3} = \frac{2}{3}$, a genuine 2-cycle.
- Numerically, $x_2 = a$ held for thousands of random $a \in [-\frac{1}{3}, 1)$ with both signs. For a outside that interval, a scan over k found no solution except the fixed point $k = 1/(1-a)$.

Common slips

- Forgetting that the cubic must contain the fixed-point solution. Spotting the root $b = 1$ is what makes the cubic easy.
- Leaving out the reality condition $-\frac{1}{3} \leq a < 1$, or not noticing that at $a = \frac{2}{3}$ one of the two values of k is the fixed point and must be thrown away.
- In (i), proving only $x_{n+1} \leq \frac{1}{4}k$ without noting this is strictly less than 1 because $k < 4$.

Beyond STEP

This is the logistic map, the classic example of a simple rule with very rich behaviour. For k in the usual range $0 < k \leq 4$ the 2-cycle you found exists exactly when $k > 3$: at $k = 3$ the fixed point loses stability and splits into two points. This is the first

period-doubling bifurcation. Further doublings follow, at k-values that bunch together geometrically (at a rate set by Feigenbaum's constant), until the map becomes chaotic. Strogatz's *Nonlinear Dynamics and Chaos* tells the whole story at first-year level.

WORKED SOLUTION

STEP II 2002, Question 6

Pure • Lines in Inclined Planes

STEP II 2002, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q6

The key idea

Everything rests on one fact about an inclined plane: if a line in the plane makes angle ψ with a line of greatest slope, then its angle γ with the horizontal satisfies

$$\sin\gamma = \cos\psi \, \sin\beta,$$

where β is the inclination of the line of greatest slope. Once you have this, each line in the question gives one trigonometric equation, and the perpendicularity of l_1 and l_3 turns a cosine into a sine. After that it is just algebra.

Solution

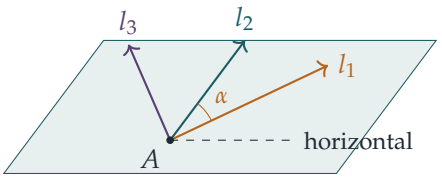
The slope of a line in an inclined plane

Set up axes at A with the z-axis vertical. Let $\mathbf{h}$ be a horizontal unit vector lying in P, and let $\mathbf{u}$ be the unit vector up the line of greatest slope l_2, which is perpendicular to $\mathbf{h}$ in P. Since l_2 makes angle β with the horizontal, we may take

$$\mathbf{h} = (1, 0, 0), \qquad \mathbf{u} = (0, \cos\beta, \sin\beta).$$

A unit vector in P making angle ψ with $\mathbf{u}$ is $\mathbf{d} = \cos\psi\,\mathbf{u} + \sin\psi\,\mathbf{h}$, because $\mathbf{u}$ and $\mathbf{h}$ are orthonormal and span P. Its vertical component is $\cos\psi\sin\beta$. The angle γ between a line with unit direction $\mathbf{d}$ and the horizontal has $\sin\gamma = |\text{vertical component of } \mathbf{d}|$, so

$$\sin\gamma = |\cos\psi| \, \sin\beta. \qquad (*)$$



The first configuration

In P, line l_1 makes acute angle α with l_2, and its angle with the horizontal is $\pi/6$. By $(*)$,

$$\cos\alpha \sin\beta = \sin\tfrac{\pi}{6} = \tfrac{1}{2},$$

as required (here $\cos\alpha > 0$ since α is acute).

Ask yourself What angle does l_3 make with l_2?

All three lines lie in the same plane P and pass through A. Since $l_3 \perp l_1$, the direction of l_3 is that of l_1 turned through a right angle within P, so l_3 makes angle $\alpha \pm \frac{\pi}{2}$ with l_2 (the sign depends on which way we turn, and doesn't matter). Either way $|\cos(\alpha \pm \frac{\pi}{2})| = \sin\alpha$, since α is acute. So $(*)$ for l_3 gives

$$\sin\alpha \sin\beta = \sin\tfrac{\pi}{4} = \tfrac{1}{\sqrt{2}}.$$

Squaring and adding the two results,

$$\sin^2\beta\,(\cos^2\alpha + \sin^2\alpha) = \tfrac{1}{4} + \tfrac{1}{2}, \qquad \text{so} \qquad \sin^2\beta = \tfrac{3}{4}.$$

The angle β between a line and the horizontal lies in $[0, \frac{\pi}{2}]$, so $\sin\beta = \frac{\sqrt{3}}{2}$ and $\beta = \pi/3$, as required. (For the record, dividing the two equations gives $\tan\alpha = \sqrt{2}$, which is indeed acute, so the configuration exists.)

The rotated configuration

The plane, and hence l_2 and $\beta = \pi/3$, are unchanged. Now l_1 makes acute angle θ with l_2 and l_3 makes angle $\theta \pm \frac{\pi}{2}$, exactly as before. Applying $(*)$ to each line, with $\sin\beta = \frac{\sqrt{3}}{2}$:

$$\sin\phi = \tfrac{\sqrt{3}}{2}\cos\theta,$$
$$\sin 2\phi = \tfrac{\sqrt{3}}{2}\sin\theta.$$

Since 2ϕ is an angle with the horizontal, $0 \le 2\phi \le \frac{\pi}{2}$, and $\phi \neq 0$ because $\cos\theta > 0$. So $\sin\phi > 0$ and we may divide the second equation by the first:

$$\tan\theta = \frac{\sin 2\phi}{\sin\phi} = \frac{2\sin\phi\cos\phi}{\sin\phi} = 2\cos\phi.$$

Write $t = \tan^2\theta$, so $\cos^2\phi = t/4$ and $\sin^2\phi = 1 - t/4$. To eliminate θ we use $\cos^2\theta = 1/(1+\tan^2\theta)$ in the square of the first equation:

$$1 - \frac{t}{4} = \frac{3}{4}\cdot\frac{1}{1+t} \quad\Longrightarrow\quad (4-t)(1+t) = 3 \quad\Longrightarrow\quad t^2 - 3t - 1 = 0.$$

Hence $t = \frac{1}{2}\left(3 \pm \sqrt{13}\right)$. Since $t = \tan^2\theta \ge 0$ and $\sqrt{13} > 3$, we must take the plus sign:

$$\tan^2\theta = \frac{3+\sqrt{13}}{2},$$

as required.

The root is genuinely attainable: it gives $\cos^2\phi = t/4 \approx 0.826$, so $\phi \approx 24.7^\circ$, $2\phi \approx 49.4^\circ < 90^\circ$, and $\theta \approx 61.2^\circ$ is acute.

Checks

- In the first configuration, $\tan\alpha = \sqrt{2}$ gives $\cos\alpha = 1/\sqrt{3}$, $\sin\alpha = \sqrt{2/3}$, and with $\sin\beta = \sqrt{3}/2$ the two products are $\frac{1}{2}$ and $1/\sqrt{2}$, as they should be.
- Numerically, $\tan^2\theta \approx 3.3028$, so $\tan\theta \approx 1.8174$, $\theta \approx 61.18^\circ$; then $\sin\phi = \frac{\sqrt{3}}{2}\cos\theta \approx 0.4175$ gives $\phi \approx 24.68^\circ$, and $\frac{\sqrt{3}}{2}\sin\theta \approx 0.7589 \approx \sin(49.35^\circ) = \sin 2\phi$.
- Limiting cases of $(*)$: $\psi = 0$ gives l_2 itself (angle β), and $\psi = \frac{\pi}{2}$ gives a horizontal line (angle 0).

Common slips

- Using $\tan\gamma = \cos\psi\tan\beta$ instead of $\sin\gamma = \cos\psi\sin\beta$. The tangent version is right only if ψ is measured in the horizontal plan view, but here α and θ are angles measured within P.
- Forgetting to reject the negative root $\frac{1}{2}(3 - \sqrt{13})$: a square can't be negative.

Beyond STEP

The formula $(*)$ is really a dot product: the sine of a line's angle with the horizontal is the component of its unit vector along the vertical, and writing that vector in an orthonormal basis of the plane does all the work. Thinking of planes as spans of orthonormal vectors, and of angles as dot products, is the starting point of first-year linear algebra and of the geometry of rotations. Strang's *Introduction to Linear Algebra* develops these ideas carefully.

WORKED SOLUTION

STEP II 2002, Question 7

Pure • Vector Geometry of Lines

STEP II 2002, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q7

The key idea

Everything here is a dot product. A line through the origin with unit direction $\mathbf{u}$ makes an angle of $\pi/4$ with m_1 exactly when $|\mathbf{u}\cdot\mathbf{d}_1| = |\mathbf{d}_1|\cos\frac{\pi}{4}$, which turns the first part into simultaneous equations. For the second half, "perpendicular" is a dot product equal to zero, and two lines in space meet only if they are coplanar and not parallel; coplanarity is a scalar triple product equal to zero. The twist is that the coplanar case turns out to be a parallel one.

Solution

The lines m_3 and m_4

Let $\mathbf{d}_1 = \mathbf{i}+\mathbf{j}$ and $\mathbf{d}_2 = \mathbf{i}+\mathbf{k}$, both of length $\sqrt{2}$. Let $\mathbf{u} = a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ be a unit vector along a line through O. The angle between two lines is the acute angle between them, so the condition is

$$|\mathbf{u}\cdot\mathbf{d}_1| = \sqrt{2}\cos\tfrac{\pi}{4} = 1, \qquad |\mathbf{u}\cdot\mathbf{d}_2| = 1,$$

that is, $a+b = \pm 1$ and $a+c = \pm 1$. Replacing $\mathbf{u}$ by $-\mathbf{u}$ gives the same line and flips both signs, so there are only two cases.

Same signs: $a+b=1$, $a+c=1$. Then $b=c=1-a$, and $a^2+b^2+c^2=1$ gives

$$a^2+2(1-a)^2 = 1 \iff 3a^2-4a+1=0 \iff (3a-1)(a-1)=0.$$

So $a=1$, giving $\mathbf{u}=\mathbf{i}$, or $a=\frac{1}{3}$, giving $\mathbf{u}=\frac{1}{3}(\mathbf{i}+2\mathbf{j}+2\mathbf{k})$.

Opposite signs: $a+b=1$, $a+c=-1$. Then $b=1-a$, $c=-1-a$, and $a^2+(1-a)^2+(1+a)^2=1$ gives $3a^2+2=1$, which has no real solution.

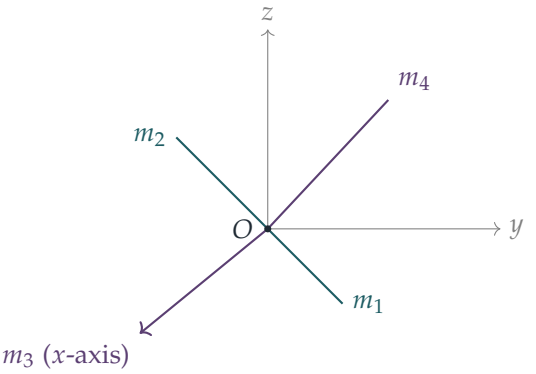
So the two lines have directions

$$m_3:\ \mathbf{i}, \qquad m_4:\ \mathbf{i}+2\mathbf{j}+2\mathbf{k}$$

(the labels could equally be swapped). With unit vectors $\mathbf{e}_3=\mathbf{i}$ and $\mathbf{e}_4=\frac{1}{3}(\mathbf{i}+2\mathbf{j}+2\mathbf{k})$,

$$\cos\theta = \mathbf{e}_3\cdot\mathbf{e}_4 = \tfrac{1}{3},$$

which is positive, so this is the acute angle: its cosine is $\frac{1}{3}$.



Setting up the points

Since A lies on m_1 at distance $\lambda\sqrt{2}$ from O, it is $\pm\lambda(1,1,0)$; non-negative coordinates force us to take $A = \lambda(1,1,0)$ with $\lambda \geq 0$ (we may take $\lambda \geq 0$ throughout, since $\lambda\sqrt{2}$ is a distance). Likewise $B = \lambda(1,0,1)$. The unit points on m_3, m_4 with non-negative coordinates are $(1,0,0)$ and $\frac{1}{3}(1,2,2)$.

Which of these is P and which is Q depends on the labelling of m_3 and m_4. It doesn't matter: the reflection $(x,y,z) \mapsto (x,z,y)$ swaps A and B and fixes both $(1,0,0)$ and $\frac{1}{3}(1,2,2)$, so it carries one labelling of the configuration onto the other (with the roles of the lines AQ and BP exchanged). Perpendicularity and intersection are preserved by a reflection, so it is enough to take

$$A = (\lambda,\lambda,0), \quad B = (\lambda,0,\lambda), \quad P = (1,0,0), \quad Q = \left(\tfrac{1}{3},\tfrac{2}{3},\tfrac{2}{3}\right).$$

Then

$$\overrightarrow{AQ} = \left(\tfrac{1}{3}-\lambda,\ \tfrac{2}{3}-\lambda,\ \tfrac{2}{3}\right), \qquad \overrightarrow{BP} = (1-\lambda,\ 0,\ -\lambda).$$

Part (i)

AQ is perpendicular to BP exactly when $\overrightarrow{AQ}\cdot\overrightarrow{BP} = 0$:

$$\begin{aligned}\overrightarrow{AQ}\cdot\overrightarrow{BP} &= \left(\tfrac{1}{3}-\lambda\right)(1-\lambda) + 0 - \tfrac{2}{3}\lambda \\ &= \lambda^2 - \tfrac{4}{3}\lambda + \tfrac{1}{3} - \tfrac{2}{3}\lambda = \lambda^2 - 2\lambda + \tfrac{1}{3}.\end{aligned}$$

This vanishes when $(\lambda-1)^2 = \frac{2}{3}$, that is,

$$\lambda = 1 \pm \sqrt{\tfrac{2}{3}} = 1 \pm \tfrac{\sqrt{6}}{3}.$$

Both values are positive (since $\sqrt{6} < 3$), so both are allowed by the non-negativity condition, and the quadratic has no other roots. Hence there are exactly two values of λ for which $AQ \perp BP$, as required.

Part (ii)

Ask yourself How do we test whether two lines in three dimensions meet?

Two distinct lines meet if and only if they lie in a common plane and are not parallel. The lines AQ and BP lie in a common plane exactly when the four points A, B, P, Q do, which happens exactly when the scalar triple product of $\overrightarrow{AB}$, $\overrightarrow{AP}$, $\overrightarrow{AQ}$ is zero.

Here $\overrightarrow{AB} = (0, -\lambda, \lambda)$ and $\overrightarrow{AP} = (1-\lambda, -\lambda, 0)$, so

$$\begin{vmatrix} 0 & -\lambda & \lambda \\ 1-\lambda & -\lambda & 0 \\ \frac{1}{3}-\lambda & \frac{2}{3}-\lambda & \frac{2}{3} \end{vmatrix} = \lambda \cdot \tfrac{2}{3}(1-\lambda) + \lambda\Big[(1-\lambda)\big(\tfrac{2}{3}-\lambda\big) + \lambda\big(\tfrac{1}{3}-\lambda\big)\Big]$$

$$= \lambda\Big[\tfrac{2}{3} - \tfrac{2}{3}\lambda + \tfrac{2}{3} - \tfrac{4}{3}\lambda\Big] = 2\lambda\big(\tfrac{2}{3}-\lambda\big),$$

expanding along the first row. For $\lambda \neq 0$, the lines can therefore only meet if $\lambda = \frac{2}{3}$.

But when $\lambda = \frac{2}{3}$,

$$\overrightarrow{AQ} = \big(-\tfrac{1}{3},\ 0,\ \tfrac{2}{3}\big), \qquad \overrightarrow{BP} = \big(\tfrac{1}{3},\ 0,\ -\tfrac{2}{3}\big) = -\overrightarrow{AQ},$$

so the lines are parallel. They are not the same line, because $\overrightarrow{BA} = (0, \frac{2}{3}, -\frac{2}{3})$ is not a multiple of $\overrightarrow{BP}$, so A is not on BP. Two distinct parallel lines never meet.

So for every non-zero λ the lines AQ and BP fail to intersect, as required. (When $\lambda = 0$, $A = B = O$ and the two lines do meet at O, which is why the question excludes it.)

Checks

- $\mathbf{i}$ makes angle $\pi/4$ with $\mathbf{i} + \mathbf{j}$ by inspection, which is a reassuring first root of the quadratic.
- At $\lambda = 1 + \sqrt{2/3} \approx 1.8165$ the dot product $\overrightarrow{AQ} \cdot \overrightarrow{BP}$ evaluates numerically to 0 (to 10^{-16}), and likewise at $\lambda \approx 0.1835$.
- Computing the shortest distance between the lines AQ and BP for $\lambda = 0.001, 0.002, \ldots, 5$ (and for the other labelling of P and Q) never gives zero; it only tends to zero as $\lambda \to 0$.

Common slips

- Forgetting the case $a + b = 1$, $a + c = -1$: it gives nothing, but it has to be ruled out.
- Stopping at "the triple product vanishes when $\lambda = \frac{2}{3}$, so they intersect". Coplanar lines can be parallel, and here they are.

Beyond STEP

The scalar triple product is a 3×3 determinant, and its vanishing says that three vectors are linearly dependent; the shortest distance between two skew lines, $|\overrightarrow{AB} \cdot (\mathbf{u} \times \mathbf{v})|/|\mathbf{u} \times \mathbf{v}|$, is built from the same pieces. In a first-year linear algebra course these ideas are unified through determinants, rank and the geometry of subspaces. Strang's *Introduction to Linear Algebra* develops this with plenty of three-dimensional pictures.

WORKED SOLUTION

STEP II 2002, Question 8

Pure • First-Order Differential Equations

STEP II 2002, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q8

The key idea

Each equation is the simplest first-order linear equation there is, $y' = p(x)\,y$, so on any interval where p is given by a single formula the solution is a constant times $\exp\big(\int p\,\mathrm{d}x\big)$. The catch is that the problem is split at $x = 0$: we get a separate arbitrary constant on each side, fixed on one side by the given value of y and on the other side only by insisting that the two pieces meet at $x = 0$. The first part is a rehearsal of exactly this, with an absolute value hidden in the second.

Solution

The first pair of equations

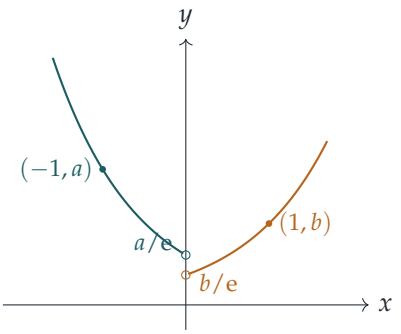
For $x < 0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -y$ gives $y = A\mathrm{e}^{-x}$ for some constant A. (To see that there are no other solutions, note that $\frac{\mathrm{d}}{\mathrm{d}x}\big(y\mathrm{e}^{x}\big) = \mathrm{e}^{x}(y' + y) = 0$, so $y\mathrm{e}^{x}$ is constant on the interval $x < 0$.) The condition $y = a$ at $x = -1$ gives $A\mathrm{e} = a$, so

$$y = a\,\mathrm{e}^{-x-1} \qquad (x < 0).$$

Similarly, for $x > 0$ we have $y = B\mathrm{e}^{x}$, and $y = b$ at $x = 1$ gives $B = b/\mathrm{e}$, so

$$y = b\,\mathrm{e}^{x-1} \qquad (x > 0).$$

As $x \to 0^{-}$, $y \to a/\mathrm{e}$; as $x \to 0^{+}$, $y \to b/\mathrm{e}$. In general these differ, and the curve jumps at the y-axis. The sketch shows a typical case with $a, b > 0$ and $a > b$: on the left a decaying exponential, falling as it approaches the axis, and on the right a growing exponential, with open circles marking the two (different) limiting values.



There is no jump exactly when the two limits agree, $a/\mathrm{e} = b/\mathrm{e}$, that is, when

$$a = b.$$

In that case the curve is $y = a\,\mathrm{e}^{|x|-1}$, a symmetric "V"-like curve with a corner at $x = 0$.

The equation with $|\mathrm{e}^x - 1|$

Since $\mathrm{e}^x > 1$ exactly when $x > 0$,

$$|\mathrm{e}^x - 1| = \begin{cases} \mathrm{e}^x - 1, & x \geq 0, \\ 1 - \mathrm{e}^x, & x < 0. \end{cases}$$

So the equation splits into two ordinary linear equations, just as before.

For $x > 0$: $y' = (\mathrm{e}^x - 1)y$. An antiderivative of $\mathrm{e}^x - 1$ is $\mathrm{e}^x - x$, so $\frac{\mathrm{d}}{\mathrm{d}x}\left(y\,\mathrm{e}^{-(\mathrm{e}^x - x)}\right) = 0$ and

$$y = C\exp(\mathrm{e}^x - x).$$

At $x = 1$, $y = \mathrm{e}^{\mathrm{e}}$, so $C\,\mathrm{e}^{\mathrm{e}-1} = \mathrm{e}^{\mathrm{e}}$, giving $C = \mathrm{e}$. Hence

$$y = \exp(\mathrm{e}^x - x + 1) \qquad (x \geq 0).$$

For $x < 0$: $y' = (1 - \mathrm{e}^x)y$. An antiderivative of $1 - \mathrm{e}^x$ is $x - \mathrm{e}^x$, so

$$y = D\exp(x - \mathrm{e}^x).$$

Ask yourself There is no data point with $x < 0$. What fixes D?

Continuity at $x = 0$ does. From the right, $y \to \exp(1 - 0 + 1) = \mathrm{e}^2$. From the left, $y \to D\exp(0 - 1) = D/\mathrm{e}$. Equating, $D = \mathrm{e}^3$, so

$$y = \exp(x - \mathrm{e}^x + 3) \qquad (x < 0).$$

Putting the two together,

$$y = \begin{cases} \exp(\mathrm{e}^x - x + 1), & x \geq 0, \\ \exp(x - \mathrm{e}^x + 3), & x < 0. \end{cases}$$

Notice that this time there is not even a corner at $x = 0$: both formulas give $y'(0) = 0$, matching $|e^0 - 1|\,y = 0$. Since $y > 0$ and $|e^x - 1| \geq 0$, the solution is increasing everywhere, with a stationary point of inflection at $x = 0$, where $y = e^2$.

The limits

(i) For $x > 0$,

$$y\exp(-e^x) = \exp(e^x - x + 1 - e^x) = e^{1-x} \to 0 \quad \text{as } x \to +\infty.$$

So the limit is 0: the factor $\exp(-e^x)$ cancels the double-exponential growth exactly, leaving a decaying e^{1-x}.

(ii) For $x < 0$,

$$y\,e^{-x} = \exp(x - e^x + 3 - x) = \exp(3 - e^x) \to e^3 \quad \text{as } x \to -\infty,$$

since $e^x \to 0$. So the limit is e^3. In other words, for large negative x the solution behaves like $e^3 e^x$, as you would expect: there $|e^x - 1| \approx 1$ and the equation is nearly $y' = y$.

Checks

- Differentiate the answer: for $x > 0$, $y' = (e^x - 1)y$, and for $x < 0$, $y' = (1 - e^x)y$. At $x = 1$, $\exp(e - 1 + 1) = e^e$.
- Numerically, stepping the equation from $y(1) = e^e \approx 15.154$ back to $x = 0$ with a Runge–Kutta method gives $y(0) \approx 7.38906 = e^2$, and continuing to $x = -5$ agrees with $\exp(x - e^x + 3)$ to six significant figures.
- For limit (ii), at $x = -10$ we get $ye^{-x} \approx 20.0846$, close to $e^3 \approx 20.0855$; for (i), at $x = 10$, $e^{1-x} \approx 1.2 \times 10^{-4}$.

Common slips

- Using the condition at $x = 1$ to fix the constant on *both* sides of $x = 0$. The two intervals are separate problems, and the left-hand constant comes only from continuity.
- Dropping the modulus and solving $y' = (e^x - 1)y$ for all x; this gives the wrong curve for $x < 0$ and the wrong limit in (ii).
- Writing $\int (e^x - 1)\,dx = e^x - x$ but then $\exp(e^x - x) = e^{e^x}/x$ (it is $e^{e^x}e^{-x}$).

Beyond STEP

The equation $y' = |e^x - 1|\,y$ has a coefficient that is continuous but not differentiable at $x = 0$. The existence and uniqueness theorem for $y' = f(x, y)$ (Picard–Lindelöf) only needs f continuous in x and Lipschitz in y, so here a single condition fixes the solution

on the whole line, and continuity at 0 is automatic. In the first part the right-hand side itself jumps at $x = 0$, which is why an extra condition was needed to join the pieces. Boyce and DiPrima, *Elementary Differential Equations*, discusses these existence and uniqueness questions for first-order equations.

WORKED SOLUTION

STEP II 2002, Question 9

Mechanics • Projectile Motion

STEP II 2002, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

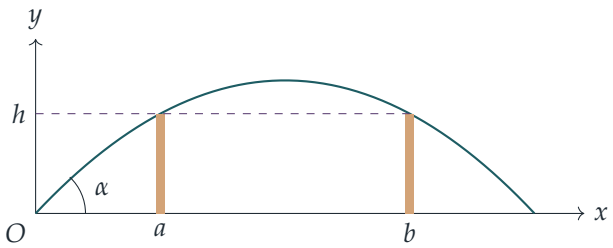
step.maths.org/questions/02-s2-q9

The key idea

Write down the equation of the path, y as a function of x. "Just passes over both walls" says the path goes through the two wall tops, so the equation $y = h$ has roots exactly $x = a$ and $x = b$. That is a quadratic with known roots, and comparing it with its sum and product of roots gives everything at once. The second part is then a matter of differentiating the two formulae with respect to h.

Solution

We model the particle as a point moving under gravity alone (no air resistance), and take O as origin with x horizontal and y vertically up. We assume, as the picture intends, $0 < a < b$, $h > 0$ and $0 < \alpha < \frac{\pi}{2}$.



Finding $\tan\alpha$ and V

With initial velocity components $V\cos\alpha$ and $V\sin\alpha$, after time t

$$x = Vt\cos\alpha, \qquad y = Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(V\cos\alpha)$ gives the path

$$y = x\tan\alpha - \frac{gx^2}{2V^2\cos^2\alpha} = x\tan\alpha - kx^2, \qquad \text{where } k = \frac{g\sec^2\alpha}{2V^2} = \frac{g(1+\tan^2\alpha)}{2V^2} > 0.$$

The particle just clears each wall, so the path passes through the tops (a,h) and (b,h). Hence a and b are roots of

$$x\tan\alpha - kx^2 = h, \qquad \text{that is,} \qquad kx^2 - x\tan\alpha + h = 0.$$

This is a quadratic in x (as $k \neq 0$), so it has at most two roots, and therefore its roots are exactly a and b. Comparing with the sum and product of the roots,

$$a + b = \frac{\tan\alpha}{k}, \qquad ab = \frac{h}{k}.$$

The second gives $k = h/(ab)$, and then the first gives

$$\boxed{\tan\alpha = \frac{h(a+b)}{ab}.}$$

Finally, from the definition of k,

$$\begin{aligned}\frac{2V^2}{g} = \frac{1+\tan^2\alpha}{k} &= \frac{ab}{h}\left(1 + \frac{h^2(a+b)^2}{a^2b^2}\right)\\ &= \frac{ab}{h} + \frac{(a+b)^2 h}{ab},\end{aligned}$$

as required.

(It is worth noticing that with these values the path is $y = \dfrac{h}{ab}\,x(a+b-x)$, which visibly equals h at $x = a$ and $x = b$.)

Raising the walls

The second particle also just clears both walls, now of height $h + \delta h$, at the same distances a and b. So the formulae above hold with h, α, V replaced by $h + \delta h$, $\alpha + \delta\alpha$, $V + \delta V$:

$$\tan\alpha = \frac{a+b}{ab}\,h, \qquad \frac{2V^2}{g} = \frac{ab}{h} + \frac{(a+b)^2}{ab}\,h.$$

Each is a function of h alone (with a, b fixed), so for small changes we may use $\delta(\text{quantity}) \approx \dfrac{\mathrm{d}(\text{quantity})}{\mathrm{d}h}\,\delta h$.

The angle. Differentiating $\tan\alpha = \frac{a+b}{ab}h$ with respect to h gives $\sec^2\alpha\,\dfrac{\mathrm{d}\alpha}{\mathrm{d}h} = \dfrac{a+b}{ab}$, so

$$\sec^2\alpha\,\delta\alpha \approx \frac{a+b}{ab}\,\delta h.$$

Since $\sec^2\alpha > 0$, $a, b > 0$ and $\delta h > 0$, the right-hand side is positive, so $\delta\alpha > 0$: the particle must be thrown more steeply.

Ask yourself The "$\approx$" is only first-order. Can we really conclude the sign of $\delta\alpha$ from it?

Yes, and in fact without approximating at all. The formula for $\tan\alpha$ is linear in h, so exactly $\tan(\alpha+\delta\alpha)-\tan\alpha = \frac{a+b}{ab}\,\delta h > 0$. As tan is strictly increasing on $(0, \frac{\pi}{2})$, this forces $\alpha+\delta\alpha > \alpha$, so $\delta\alpha > 0$.

The speed. Differentiating $\dfrac{2V^2}{g}$ with respect to h,

$$\frac{4V}{g}\frac{\mathrm{d}V}{\mathrm{d}h} = -\frac{ab}{h^2} + \frac{(a+b)^2}{ab} = \frac{(a+b)^2}{abh^2}\left(h^2 - \frac{a^2b^2}{(a+b)^2}\right),$$

so

$$\delta V \approx \frac{g(a+b)^2}{4Vabh^2}\left(h - \frac{ab}{a+b}\right)\left(h + \frac{ab}{a+b}\right)\delta h.$$

Every factor except $h - \frac{ab}{a+b}$ is positive (as $V, a, b, h, \delta h > 0$). Hence δV has the sign of $h - \frac{ab}{a+b}$: it is positive if $h > ab/(a+b)$ and negative if $h < ab/(a+b)$, as required. (Since this derivative is non-zero away from $h = ab/(a+b)$, the first-order term really does fix the sign for small enough δh.)

Interpretation. Comparing with $\tan\alpha = h(a+b)/(ab)$, the condition $h > ab/(a+b)$ is exactly $\tan\alpha > 1$, that is $\alpha > 45^\circ$. As a function of h, $2V^2/g$ has the form $p/h + qh$, which is smallest when $h = \sqrt{p/q} = ab/(a+b)$, where $\alpha = 45^\circ$ and $2V^2/g = 2(a+b)$. So: if the throw is already steeper than 45°, raising the walls needs more speed; if it is shallower, raising the walls (and steepening the throw towards 45°) actually needs less.

Checks

- Take $a = 3\,\mathrm{m}$, $b = 7\,\mathrm{m}$, $h = 2.5\,\mathrm{m}$. Then $\tan\alpha = 25/21$ and $2V^2/g = 8.4 + 11.90\ldots = 20.30\ldots\,\mathrm{m}$. Stepping the motion numerically with these values, the particle is at height 2.5 m (to 10^{-11}) at both $x = 3$ and $x = 7$.
- Symmetry: the answers are symmetric in a and b, as they must be, since the path is a parabola symmetric about $x = \frac{1}{2}(a+b)$.
- Units: ab/h and $(a+b)^2h/(ab)$ are lengths, matching $2V^2/g$; and $\tan\alpha$ is dimensionless.
- At $h = ab/(a+b)$ we get $\alpha = 45^\circ$ and $2V^2/g = 2(a+b)$; for $a = 3$, $b = 7$ this is 20 m, slightly less than the 20.30 m above, consistent with it being the minimum.

Common slips

- Writing two separate equations for the walls and solving them simultaneously by brute force: it works, but the sum and product of roots does it in two lines.

- Forgetting that $\sec^2\alpha = 1 + \tan^2\alpha$ when converting k back into V.
- Quoting the sign of δV without factorising the derivative of $2V^2/g$: the clean condition comes from $h^2 > a^2b^2/(a+b)^2$ with $h > 0$.

Beyond STEP

The second part is a small sensitivity analysis: we linearised the map $h \mapsto (\alpha, V)$ around a solution and read off how the outputs respond. Linearisation, and the related idea of envelopes of families of trajectories (the "parabola of safety" bounding everything reachable at a given speed), are standard in first-year mechanics and calculus courses. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and these approximation techniques with care.

WORKED SOLUTION

STEP II 2002, Question 10

Mechanics • Kinematics and Optimisation

STEP II 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q10

The key idea

The total distance is fixed, so the race time T and the switching time t are tied together by a single equation. Clearing the fraction turns it into a quadratic in t, and a time T is achievable only if that quadratic has a real root. The discriminant condition gives the minimum T directly, with no calculus. For the second part, once we know both runners' positions as functions of time, the distance between them is a piecewise quadratic, and we maximise it in the usual way.

Solution

Throughout, distances are in km and times in hours, and $42\frac{3}{8} = \frac{339}{8}$.

The minimum race time

The first competitor covers $13t$ in the first t hours and then $\left(14 + \frac{2t}{T}\right)(T - t)$ in the remaining $T - t$ hours. So

$$13t + \left(14 + \frac{2t}{T}\right)(T - t) = \frac{339}{8}.$$

Expanding, $13t + 14T - 14t + 2t - \dfrac{2t^2}{T} = \dfrac{339}{8}$, that is

$$t + 14T - \frac{2t^2}{T} = \frac{339}{8}.$$

Multiplying by $T > 0$ and collecting powers of t:

$$2t^2 - Tt + \left(\frac{339}{8}T - 14T^2\right) = 0. \qquad (*)$$

Ask yourself We want the smallest T, but T is not given as a function of t. How can we tell which values of T are possible at all?

A value of T can occur only if $(*)$ has a real solution t, so its discriminant must be non-negative:

$$T^2 - 8\Big(\frac{339}{8}T - 14T^2\Big) \geq 0 \iff 113T^2 - 339T \geq 0 \iff 113T(T-3) \geq 0.$$

Since $T > 0$, this forces $T \geq 3$.

The bound is attained. When $T = 3$ the discriminant is zero and $(*)$ becomes

$$2t^2 - 3t + \Big(\frac{1017}{8} - 126\Big) = 2t^2 - 3t + \frac{9}{8} = 2\Big(t - \frac{3}{4}\Big)^2 = 0,$$

so $t = \frac{3}{4}$. This lies between 0 and T, so it is a genuine race plan: 45 minutes at 13, then the remaining $2\frac{1}{4}$ hours at $14 + \frac{2(3/4)}{3} = 14.5$. Indeed $13 \cdot \frac{3}{4} + 14.5 \cdot \frac{9}{4} = 9.75 + 32.625 = 42.375$. Hence the minimum possible value of T is 3, as required.

(Equivalently, differentiating $(*)$ implicitly and setting $\frac{\mathrm{d}T}{\mathrm{d}t} = 0$ gives $T = 4t$, and substituting back gives $T = 3$. The discriminant argument has the advantage of proving that 3 is a global minimum, not just a stationary value.)

The maximum separation

Let τ be the time since the start. The second competitor's speed is $16 - k\tau$ for some constant k, so their distance run is

$$s_2(\tau) = 16\tau - \tfrac{1}{2}k\tau^2.$$

Finishing in 3 hours gives $48 - \frac{9}{2}k = \frac{339}{8}$, so $\frac{9}{2}k = \frac{45}{8}$ and $k = \frac{5}{4}$. Thus $s_2 = 16\tau - \frac{5}{8}\tau^2$, and the speed falls from 16 to 12.25, staying positive.

The first competitor also takes 3 hours, which by the first part means $t = \frac{3}{4}$. Their distance run is

$$s_1(\tau) = \begin{cases} 13\tau, & 0 \leq \tau \leq \frac{3}{4}, \\ \frac{39}{4} + \frac{29}{2}\big(\tau - \frac{3}{4}\big), & \frac{3}{4} \leq \tau \leq 3. \end{cases}$$

Let $d = s_2 - s_1$. The second runner starts faster, so we expect $d \geq 0$.

For $0 \leq \tau \leq \frac{3}{4}$*:* $d = 3\tau - \frac{5}{8}\tau^2$, with $\frac{\mathrm{d}d}{\mathrm{d}\tau} = 3 - \frac{5}{4}\tau > 0$. So d increases on this interval, reaching $d\big(\frac{3}{4}\big) = \frac{9}{4} - \frac{45}{128} = \frac{243}{128}$.

For $\frac{3}{4} \leq \tau \leq 3$*:*

$$d = 16\tau - \tfrac{5}{8}\tau^2 - \tfrac{39}{4} - \tfrac{29}{2}\tau + \tfrac{87}{8} = \tfrac{1}{8}\big(9 + 12\tau - 5\tau^2\big) = \tfrac{1}{8}(3-\tau)(3+5\tau).$$

This is a downward parabola with vertex at $\tau = \frac{12}{10} = \frac{6}{5}$, which lies inside $\big[\frac{3}{4}, 3\big]$. (Equivalently, the gap stops growing when the speeds are equal: $16 - \frac{5}{4}\tau = 14.5$.) There

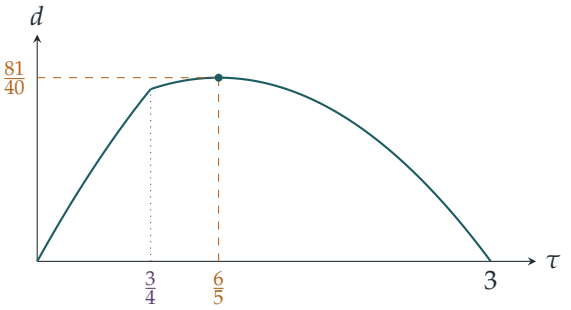
$$d\big(\tfrac{6}{5}\big) = \tfrac{1}{8} \cdot \tfrac{9}{5} \cdot 9 = \tfrac{81}{40}.$$

The factorised form shows $d \geq 0$ on the whole interval, falling to 0 at the finish, as it must.

Comparing the two pieces, $\frac{81}{40} = 2.025 > \frac{243}{128} \approx 1.898$. So the maximum distance between the runners is

$$\boxed{\tfrac{81}{40}\text{ km} = 2.025\text{ km}},$$

reached 1.2 hours (72 minutes) into the race, with the second competitor ahead.



Checks

- Both runners finish at $\tau = 3$, and the formula gives $d(3) = 0$, as it must.
- The maximum of the second piece occurs exactly when the speeds are equal (14.5 each), which is the natural condition for a gap to stop growing.
- Numerically: solving the quadratic for T over a fine grid of t gives the smallest T as 3.000 at $t = 0.75$, and evaluating d at 300 000 times in $[0, 3]$ gives a maximum of 2.025 at $\tau = 1.2$, with $d \geq 0$ throughout.

Common slips

- Using the first competitor's minimum-time plan without noticing that "run time of 3 hours" is precisely what fixes $t = \frac{3}{4}$.
- Maximising only at the kink $\tau = \frac{3}{4}$: the gap keeps growing after the switch, because the second runner is still faster than 14.5 until $\tau = \frac{6}{5}$.
- Writing the second runner's distance as $16\tau - k\tau^2$, forgetting the $\frac{1}{2}$ from integrating a linearly decreasing speed.

Beyond STEP

The first part is an optimisation over a constraint given implicitly by $F(t, T) = 0$. Asking "for which T does a solution t exist?" is the idea behind the implicit function theorem and, more generally, Lagrange multipliers: at the optimum the level curve of the

constraint has $\partial F/\partial t = 0$, which is exactly the double-root condition here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats kinematics with calculus throughout, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation.

WORKED SOLUTION

STEP II 2002, Question 11

Mechanics • Friction and Limiting Equilibrium

STEP II 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q11

The key idea

The first part is a direct centre-of-mass integral; a substitution $t = x/l$ makes it a one-liner. For the ladder, "limiting at both ends" turns all four contact forces into multiples of just one unknown, so resolving twice and taking moments about A gives $\tan\theta$ at once. For the final part we need one more idea: the beam is safer the steeper it stands, so "slides for every $\theta < \pi/4$" says the limiting angle is at least $\pi/4$.

Solution

The centre of mass

A small piece of length δx at distance x from B has weight $\alpha W l^{-1}(x/l)^{\alpha-1}\,\delta x$. With $t = x/l$ (so $\mathrm{d}x = l\,\mathrm{d}t$), the total weight is

$$\int_0^l \frac{\alpha W}{l}\left(\frac{x}{l}\right)^{\alpha-1}\mathrm{d}x = \alpha W\int_0^1 t^{\alpha-1}\,\mathrm{d}t = W,$$

as it should be (the integral converges since $\alpha > 0$). The moment of the weight about B is

$$\int_0^l x\,\frac{\alpha W}{l}\left(\frac{x}{l}\right)^{\alpha-1}\mathrm{d}x = \alpha W l\int_0^1 t^{\alpha}\,\mathrm{d}t = \frac{\alpha W l}{\alpha+1}.$$

Dividing by W, the centre of mass G_0 is at distance $\alpha l/(\alpha+1)$ from B, as required. So it is at distance

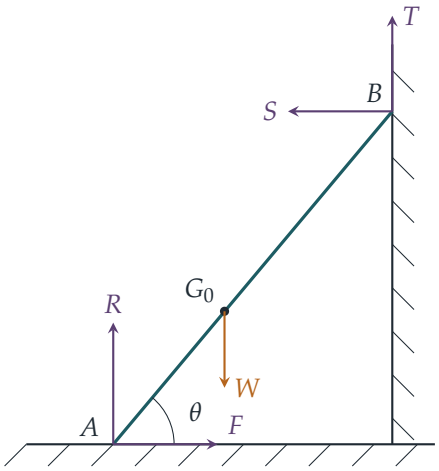
$$d = l - \frac{\alpha l}{\alpha+1} = \frac{l}{\alpha+1}$$

from A, which is what we will use for moments about A.

Limiting equilibrium at both ends

Model the beam as a rigid rod, with rough contacts at A and B. If it slips, A moves away from the wall and B moves down, so friction acts towards the wall at A and

upwards at B. Let the floor exert normal reaction R and friction F at A, and the wall exert normal reaction S and friction T at B. The weight acts at G_0, a distance $l/(\alpha+1)$ from A.



Resolving horizontally and vertically, and taking moments about A (the weight acts at horizontal distance $d\cos\theta$ from A; B is at height $l\sin\theta$ and horizontal distance $l\cos\theta$):

$$\begin{aligned} F &= S, \\ R + T &= W, \\ S\,l\sin\theta + T\,l\cos\theta &= W\,\frac{l}{\alpha+1}\cos\theta. \end{aligned}$$

Limiting at both ends means $F = \mu R$ and $T = \mu S$. Then $S = F = \mu R$ and $T = \mu S = \mu^2 R$, and the vertical equation gives

$$R(1+\mu^2) = W, \qquad \text{so} \qquad S = \frac{\mu W}{1+\mu^2}, \quad T = \frac{\mu^2 W}{1+\mu^2}.$$

Dividing the moment equation by $l\cos\theta$ and substituting,

$$\frac{\mu W}{1+\mu^2}\tan\theta + \frac{\mu^2 W}{1+\mu^2} = \frac{W}{1+\alpha} \quad\Longrightarrow\quad \mu\tan\theta = \frac{1+\mu^2}{1+\alpha} - \mu^2 = \frac{1-\alpha\mu^2}{1+\alpha},$$

that is,

$$\tan\theta = \frac{1-\alpha\mu^2}{(1+\alpha)\mu},$$

as required. Call this angle θ_0, and write $t_0 = \tan\theta_0$.

The greatest value of μ

> **Ask yourself** Why does "slides for every $\theta < \pi/4$" translate into a condition on θ_0?

We show that equilibrium is possible exactly when $\tan\theta \geq t_0$. Write $t = \tan\theta$ and $k = 1/(1+\alpha)$, so the moment equation reads $St + T = kW$. The friction laws are $F \leq \mu R$ and $|T| \leq \mu S$.

Necessity. From $T \leq \mu S$ and $S = F \leq \mu R = \mu(W - T)$ we get $T \leq \mu^2(W - T)$, so $T \leq \mu^2 W/(1+\mu^2)$. Also $S \leq \mu(W - T)$, so

$$kW = St + T \leq \mu t(W - T) + T = \mu t W + (1 - \mu t)T.$$

If $\mu t < 1$, using the bound on T gives $kW \leq W(\mu t + \mu^2)/(1+\mu^2)$, which rearranges to $t \geq t_0$. If $\mu t \geq 1$ then $t \geq 1/\mu > t_0$ anyway, since $\mu t_0 = (1 - \alpha\mu^2)/(1+\alpha) < 1$. Either way, equilibrium forces $t \geq t_0$.

Sufficiency. If $t \geq t_0$ (and $t > 0$), take the wall friction limiting, $T = \mu S$, with $S = kW/(t+\mu)$ from the moment equation, and $F = S$, $R = W - T$. The only condition left is $S \leq \mu R$, i.e. $S(1+\mu^2) \leq \mu W$, i.e. $k(1+\mu^2) \leq \mu(t+\mu)$, which is exactly $t \geq t_0$. (Then $R > 0$ too.)

So the beam slides at angle θ precisely when $\tan\theta < t_0$. "It slides for every $\theta < \pi/4$" means $t_0 \geq 1$: if $t_0 \geq 1$, every $\theta < \pi/4$ has $\tan\theta < 1 \leq t_0$; if $t_0 < 1$, any angle with $\max(t_0, 0) < \tan\theta < 1$ is below $\pi/4$ yet in equilibrium.

With $\alpha = \frac{3}{2}$,

$$t_0 = \frac{1 - \frac{3}{2}\mu^2}{\frac{5}{2}\mu} = \frac{2 - 3\mu^2}{5\mu} \geq 1 \iff 3\mu^2 + 5\mu - 2 \leq 0 \iff (3\mu - 1)(\mu + 2) \leq 0,$$

multiplying by $5\mu > 0$. Since $\mu + 2 > 0$, this holds exactly when $\mu \leq \frac{1}{3}$. The greatest possible value is

$$\mu = \tfrac{1}{3}.$$

Checks

- With $\alpha = 1$ the beam is uniform and the formula becomes $\tan\theta = (1 - \mu^2)/(2\mu)$, the familiar result for a uniform ladder with equal coefficients at both ends.
- At $\mu = \frac{1}{3}$, $\alpha = \frac{3}{2}$: $t_0 = (2 - \frac{1}{3})/\frac{5}{3} = 1$ exactly, so the limiting angle is $\pi/4$. Numerically, $\mu = 0.332$ gives $t_0 \approx 1.004$ and $\mu = 0.334$ gives $t_0 \approx 0.996$.
- As $\alpha \to \infty$ the mass concentrates at B ($d \to 0$), and $\tan\theta$ becomes negative for any $\mu > 0$: the beam can stand at any angle, sensibly, since the weight then acts almost at the wall.

Common slips

- Using $\alpha l/(\alpha+1)$ as the distance from A in the moment equation; it is measured from B.
- Putting the wall friction downwards: when the beam slips, B slides down, so friction on the beam at B acts up.
- Reversing the final inequality: a *steeper* beam is safer, so "slides below $\pi/4$" needs $\tan\theta_0 \geq 1$, and the largest μ makes θ_0 exactly $\pi/4$.

Beyond STEP

The necessity-and-sufficiency argument above is really a small *linear programming* problem: the friction laws define a convex region of admissible forces, and equilibrium is possible when the moment line meets it. Statically indeterminate problems like this one, where the equations alone don't fix the forces, are a standard theme in first-year mechanics and engineering. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, torque and rigid-body equilibrium with this kind of care.

WORKED SOLUTION

STEP II 2002, Question 12

Probability and Statistics • Normal and Poisson Approximations

STEP II 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q12

The key idea

There are three layers of randomness stacked inside each other: heads within a coin, coins within a day, and days within the week. Each layer is a binomial count whose "success" probability comes from the layer below. The formula is built by approximating the bottom layer with a normal distribution (with a continuity correction), the middle layer with a Poisson distribution and then a normal one, and leaving the top layer as an exact binomial. Once you see the three layers, the job is to identify which approximation produced each expression in the formula, and then to ask whether each approximation is any good for the given numbers.

Solution

One coin on one day

Fix a day and a coin. Let X be the number of heads in its M throws. The throws are independent, each with probability p of a head, so

$$X \sim \mathrm{B}(M, p), \qquad \mathrm{E}(X) = Mp, \qquad \mathrm{Var}(X) = Mp(1-p).$$

For large M we approximate X by a normal variable $X' \sim \mathrm{N}\big(Mp,\, Mp(1-p)\big)$. Since X is integer-valued, "fewer than m heads" means $X \leq m-1$, and with a continuity correction this becomes $X' < m - \frac{1}{2}$. So

$$r = \mathrm{P}(X < m) = \mathrm{P}(X \leq m-1) \approx \Phi\left(\frac{m - \frac{1}{2} - Mp}{\sqrt{Mp(1-p)}}\right) = \Phi\left(\frac{2m - 1 - 2Mp}{2\sqrt{Mp(1-p)}}\right).$$

Comparing with the question, this says $h \approx Lr$.

All the coins on one day

Let Y be the number of the L coins that produce fewer than m heads on that day. The coins are thrown independently and each behaves in the same way, so each coin

independently "succeeds" with probability r, and

$$Y \sim \mathrm{B}(L, r), \qquad \mathrm{E}(Y) = Lr = h, \qquad \mathrm{Var}(Y) = Lr(1-r) = h(1-r).$$

Ask yourself The formula for q has $\sqrt{h}$ in the denominator. Why is the standard deviation $\sqrt{h}$ and not $\sqrt{h(1-r)}$?

The formula is designed for the case where r is small: fewer than m heads is a rare event for a single coin, while L is large. In that regime $1 - r \approx 1$, so the variance is approximately the mean, which is exactly the situation of the Poisson approximation $Y \approx \mathrm{Po}(h)$. If h is also reasonably large, $\mathrm{Po}(h)$ is in turn approximately $\mathrm{N}(h, h)$. (Equivalently, approximate $\mathrm{B}(L, r)$ directly by a normal and replace $1 - r$ by 1.)

So take $Y' \sim \mathrm{N}(h, h)$. "More than l coins" means $Y \geq l + 1$, which with a continuity correction becomes $Y' > l + \frac{1}{2}$. Hence

$$q = \mathrm{P}(Y > l) \approx 1 - \Phi\left(\frac{l + \frac{1}{2} - h}{\sqrt{h}}\right) = \Phi\left(\frac{h - l - \frac{1}{2}}{\sqrt{h}}\right) = \Phi\left(\frac{2h - 2l - 1}{2\sqrt{h}}\right),$$

using the symmetry $1 - \Phi(z) = \Phi(-z)$ of the standard normal distribution. This is the given q.

All the days

Call a day *good* if more than l of its coins produce fewer than m heads. Each day has the same probability q of being good, and different days involve different throws, so the days are independent. The number of good days among the K is therefore $\mathrm{B}(K, q)$, and

$$\mathrm{P}(\text{exactly } k \text{ good days}) = \binom{K}{k} q^k (1-q)^{K-k},$$

with q approximated as above. That is the required result.

The numerical case

Here $Mp = 60$ and $Mp(1-p) = 24$, so

$$\frac{2m - 1 - 2Mp}{2\sqrt{Mp(1-p)}} = \frac{96 - 1 - 120}{2\sqrt{24}} = \frac{-25}{9.798} \approx -2.55,$$

giving $r \approx \Phi(-2.55) \approx 0.0054$ and $h = 500r \approx 2.68$. Then

$$\frac{2h - 2l - 1}{2\sqrt{h}} \approx \frac{5.36 - 9}{3.27} \approx -1.11, \qquad q \approx \Phi(-1.11) \approx 0.133,$$

and the approximation gives $\binom{7}{2} q^2 (1-q)^5 \approx 21 \times 0.0178 \times 0.490 \approx 0.18$.

Ask yourself Each approximation is standard. Which of them are we using outside the conditions where it is reliable?

Take the three layers in turn.

- **Top layer.** The $\mathrm{B}(7, q)$ step is exact (given q), so it introduces no error of its own. But it squares the error in q, as $k = 2$.
- **Bottom layer.** $M = 100$ with $Mp = 60$ and $M(1-p) = 40$ is comfortably in the range where the normal approximation to the binomial is usually used. But we are using it 2.55 standard deviations into the tail, to find a probability of about 0.005. The normal approximation is good for the bulk of the distribution in *absolute* terms; in the tails its *relative* error can be large, especially as $p \neq \frac{1}{2}$ makes the binomial skewed. Every relative error in r passes straight into $h = Lr$, and q is sensitive to h.
- **Middle layer.** Replacing $\mathrm{B}(500, r)$ by $\mathrm{Po}(h)$ is excellent, since r is tiny and L large. But then approximating $\mathrm{Po}(h)$ by a normal with $h \approx 2.7$ is not justified: a Poisson distribution with mean under 3 is markedly skewed, and the usual rule of thumb asks for a mean of at least about 10 (some say 15 or 20).

So no, we would not expect the approximation to be accurate. The doubtful steps are the normal approximation to the tail of $\mathrm{B}(100, 0.6)$ and the normal approximation to a Poisson distribution with small mean.

In fact the exact value of r is $\mathrm{P}(X \leq 47) \approx 0.00576$, about 7% more than the normal estimate, so h should be about 2.88; and then the exact $\mathrm{P}(Y > 4)$ is about 0.165, not 0.133. The final probability is about 0.23 rather than 0.18. (For these particular numbers the normal approximation to $\mathrm{Po}(2.68)$ happens to give almost the same tail as the Poisson itself, 0.133 against 0.134, so most of the damage is done in the bottom layer. You could not have counted on that luck in advance.)

Checks

- The continuity corrections point the right way: "fewer than m" is $X \leq m - 1$, so we stop at $m - \frac{1}{2}$; "more than l" is $Y \geq l + 1$, so we start at $l + \frac{1}{2}$. The given formulae contain exactly $2m - 1$ and $2l + 1$.
- Numerically, $\Phi(-2.5516) = 0.005362$, $h = 2.681$ and $q = \Phi(-1.1108) = 0.1333$, giving 0.1825. Exact binomial sums give $r = 0.005761$, $q = 0.1645$ (confirmed by a simulation of 2×10^4 days, which gave 0.164) and a final answer of 0.2314.

Common slips

- Using m instead of $m - \frac{1}{2}$ (or l instead of $l + \frac{1}{2}$): then the formulae don't match.
- Using $\sqrt{h(1-r)}$ and then being unable to reach $\sqrt{h}$. The question's formula drops the factor $1 - r$ because it is effectively using the Poisson approximation, and you should say so.

- Judging accuracy only by "$Mp > 5$ and $M(1-p) > 5$". That rule is about the centre of the distribution, not about a tail probability of 0.005.

Beyond STEP

The question is about how errors propagate through a chain of approximations, and about the difference between absolute and relative error in the tails. At university, the central limit theorem is sharpened by the Berry–Esseen theorem, which bounds the absolute error of the normal approximation. Tail probabilities need different tools, such as Chernoff bounds and large-deviation theory, which describe how fast tail probabilities shrink. Blitzstein and Hwang, *Introduction to Probability*, covers the Poisson and normal approximations and Chernoff-type inequalities clearly; Grimmett and Welsh, *Probability: An Introduction*, is a good companion.

WORKED SOLUTION

STEP II 2002, Question 13

Probability and Statistics • Transformed Distribution Functions

STEP II 2002, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q13

The key idea

Although it sits in the mechanics section, this is a probability question about transforming a distribution function. Everything rests on one formula: differentiating G gives $G'(y) = f(y) \cdot \dfrac{2}{(2-F(y))^2}$, where $f = F'$ is the density of X. So the density of Y is the density of X multiplied by a weight that is trapped between $\frac{1}{2}$ and 2. Once you have that, each expectation of Y is an integral of X's density times a weight in $[\frac{1}{2}, 2]$, and the inequalities follow by comparing integrands. The condition $a > 0$ is there to make the integrands non-negative.

Solution

Properties of G

We assume, as the question intends, that X has a density $f(y) = F'(y) \geq 0$ on $[a, b]$, so that all of X's probability lies in $[a, b]$.

Since $F(a) = 0$ and $F(b) = 1$,

$$G(a) = \frac{0}{2-0} = 0, \qquad G(b) = \frac{1}{2-1} = 1.$$

By the quotient rule,

$$G'(y) = \frac{F'(y)\big(2-F(y)\big) - F(y)\big(-F'(y)\big)}{(2-F(y))^2} = \frac{2F'(y)}{(2-F(y))^2}.$$

The numerator is non-negative, because a distribution function is non-decreasing, so $F'(y) \geq 0$. The denominator is a square of $2 - F(y) \geq 1$, hence positive. So $G'(y) \geq 0$.

For the bound, $0 \leq F(y) \leq 1$ because F is a distribution function. So

$$1 \leq 2 - F(y) \leq 2 \quad \Longrightarrow \quad 1 \leq (2-F(y))^2 \leq 4 \quad \Longrightarrow \quad \tfrac{1}{2} \leq \frac{2}{(2-F(y))^2} \leq 2,$$

where squaring keeps the inequalities because everything is positive, and taking reciprocals and multiplying by 2 reverses and then rescales them. This is the required double inequality.

Together, $G(a) = 0$, $G(b) = 1$ and G non-decreasing mean that G is a genuine distribution function on $[a, b]$, so the random variable Y makes sense. Its density is

$$g(y) = G'(y) = w(y)\, f(y), \qquad \text{where } w(y) = \frac{2}{(2 - F(y))^2} \in \left[\tfrac{1}{2}, 2\right].$$

Bounds on $\mathrm{E}(Y)$

Ask yourself Why can we multiply the inequality $\frac{1}{2} \le w(y) \le 2$ by $yf(y)$ and keep its direction?

Because $yf(y) \ge 0$ on $[a, b]$: we have $f(y) \ge 0$, and $y \ge a > 0$. This is exactly where the hypothesis $a > 0$ is used. Hence

$$\tfrac{1}{2}\, yf(y) \le y\, w(y) f(y) \le 2\, yf(y) \qquad \text{for } a \le y \le b,$$

and integrating over $[a, b]$ preserves the inequalities:

$$\tfrac{1}{2} \int_a^b yf(y)\, \mathrm{d}y \;\le\; \int_a^b y\, g(y)\, \mathrm{d}y \;\le\; 2 \int_a^b yf(y)\, \mathrm{d}y,$$

that is,

$$\tfrac{1}{2}\, \mathrm{E}(X) \le \mathrm{E}(Y) \le 2\, \mathrm{E}(X),$$

as required.

Bound on $\mathrm{Var}(Y)$

In the same way, $y^2 f(y) \ge 0$ (this needs only $f \ge 0$), so multiplying $w(y) \le 2$ by it and integrating gives

$$\mathrm{E}(Y^2) = \int_a^b y^2\, w(y) f(y)\, \mathrm{d}y \le 2 \int_a^b y^2 f(y)\, \mathrm{d}y = 2\, \mathrm{E}(X^2).$$

For the subtracted term we need a *lower* bound on $(\mathrm{E}(Y))^2$. From the previous part, $\mathrm{E}(Y) \ge \frac{1}{2}\mathrm{E}(X)$, and $\mathrm{E}(X) > 0$ since $X \ge a > 0$. Both sides are positive, so squaring is safe:

$$(\mathrm{E}(Y))^2 \ge \tfrac{1}{4}(\mathrm{E}(X))^2.$$

Therefore

$$\begin{aligned} \mathrm{Var}(Y) &= \mathrm{E}(Y^2) - (\mathrm{E}(Y))^2 \\ &\le 2\, \mathrm{E}(X^2) - \tfrac{1}{4}(\mathrm{E}(X))^2 \\ &= 2\big(\mathrm{Var}(X) + (\mathrm{E}(X))^2\big) - \tfrac{1}{4}(\mathrm{E}(X))^2 \\ &= 2\, \mathrm{Var}(X) + \tfrac{7}{4}(\mathrm{E}(X))^2, \end{aligned}$$

using $\mathrm{E}(X^2) = \mathrm{Var}(X) + (\mathrm{E}(X))^2$ in the third line. This is the required bound.

Checks

Take X uniform on $[1,3]$, so $F(y) = \frac{1}{2}(y-1)$ and $\mathrm{E}(X) = 2$. Since $Y \geq 1$, $\mathrm{E}(Y) = 1 + \int_1^3 (1 - G(y))\,\mathrm{d}y$; substituting $u = F(y)$ gives

$$\mathrm{E}(Y) = 1 + 2\int_0^1 \frac{2(1-u)}{2-u}\,\mathrm{d}u = 1 + 4(1 - \ln 2) \approx 2.227,$$

comfortably between 1 and 4. Intuitively, w is small where F is small and large where F is near 1, so Y leans towards the top of the interval and $\mathrm{E}(Y) > \mathrm{E}(X)$ here. We also checked numerically, for 300 random power-law distributions on random intervals, that $\frac{1}{2}\mathrm{E}(X) \leq \mathrm{E}(Y) \leq 2\mathrm{E}(X)$ and the variance bound both hold; the observed ratios $\mathrm{E}(Y)/\mathrm{E}(X)$ lay between about 0.92 and 1.31.

Common slips

- Forgetting why $a > 0$ matters: if X could be negative, $yf(y)$ could be negative and multiplying by it would flip the inequality.
- Bounding $\mathrm{Var}(Y)$ using $\mathrm{E}(Y) \leq 2\mathrm{E}(X)$: the term $(\mathrm{E}(Y))^2$ is subtracted, so you need its lower bound, not its upper one.
- Trying to compare $\mathrm{Var}(Y)$ with $\mathrm{Var}(X)$ directly by integrating $(y-\mu)^2$ against g: the two variances use different means, so work with $\mathrm{E}(Y^2)$ instead.

Beyond STEP

Writing $g = wf$ is a simple instance of a change of measure: the new distribution has a density (a *likelihood ratio*, or Radon–Nikodym derivative) with respect to the old one, and bounds on that ratio transfer directly to bounds on expectations. This idea underlies importance sampling in statistics and the change of probability measure used in financial mathematics. Grimmett and Welsh, *Probability: An Introduction*, covers distribution functions, densities and expectations at the right level to take this further.

WORKED SOLUTION

STEP II 2002, Question 14

Probability and Statistics • Mixtures and Bayes' Theorem

STEP II 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q14

The key idea

This is a two-stage experiment: first a region is chosen, then accidents happen in it. So condition on the region. The law of total expectation gives the mean, the law of total probability gives the distribution, and Bayes' theorem turns the question round at the end. The number of accidents in a region is modelled as Poisson, because the question gives only a mean, and the target formula, with its $e^{-\lambda}\lambda^k/k!$ pattern, confirms the model.

Solution

Setting up

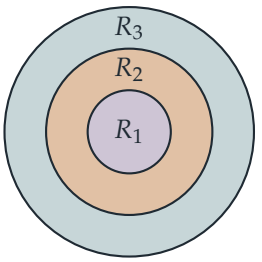
Let S be the index of the selected region and $\lambda_n = 2 - \frac{n}{N}$ the mean number of accidents in R_n. Region R_n is an annulus with radii $n-1$ and n, so its area is

$$\pi n^2 - \pi(n-1)^2 = \pi(2n-1).$$

The whole island has area πN^2. (As a check, $\sum_{n=1}^{N}(2n-1) = N^2$.) So

$$\mathrm{P}(S = n) = \frac{2n-1}{N^2}, \qquad n = 1, \ldots, N.$$

Given $S = n$, we model X as Poisson with mean λ_n. Since $\lambda_n \geq 1 > 0$ for every n, each of these Poisson distributions makes sense.



area of $R_n = \pi(2n-1)$

The long-term average

In the long run the average number recorded per day is $\mathrm{E}(X)$, by the law of large numbers. Conditioning on the region gives

$$\begin{aligned}\mathrm{E}(X) &= \sum_{n=1}^{N} \mathrm{P}(S=n)\,\mathrm{E}(X \mid S=n) = \sum_{n=1}^{N} \frac{2n-1}{N^2}\Big(2-\frac{n}{N}\Big)\\ &= 2\sum_{n=1}^{N}\frac{2n-1}{N^2} \;-\; \frac{1}{N^3}\sum_{n=1}^{N}(2n^2-n)\\ &= 2-\frac{1}{N^3}\Big(\frac{N(N+1)(2N+1)}{3}-\frac{N(N+1)}{2}\Big).\end{aligned}$$

The first sum is 1 because the selection probabilities add up to 1. In the bracket we used the given formula for $\sum n^2$ and $\sum n = \frac{1}{2}N(N+1)$. Taking out $\frac{1}{6}N(N+1)$ leaves $2(2N+1)-3 = 4N-1$, so

$$\mathrm{E}(X) = 2-\frac{(N+1)(4N-1)}{6N^2} = 2-\frac{1}{6}\Big(1+\frac{1}{N}\Big)\Big(4-\frac{1}{N}\Big),$$

splitting N^2 as $N\cdot N$ between the two factors. This is the required result.

The distribution of X

By the law of total probability, for $k = 0, 1, 2, \ldots$,

$$\mathrm{P}(X=k) = \sum_{n=1}^{N}\mathrm{P}(S=n)\,\mathrm{P}(X=k \mid S=n) = \sum_{n=1}^{N}\frac{2n-1}{N^2}\cdot\frac{\mathrm{e}^{-\lambda_n}\lambda_n^{k}}{k!}.$$

Now write each piece in terms of N:

$$\lambda_n^{k} = \Big(\frac{2N-n}{N}\Big)^{k} = N^{-k}(2N-n)^{k}, \qquad \mathrm{e}^{-\lambda_n} = \mathrm{e}^{-2+n/N} = \mathrm{e}^{-2}\,\mathrm{e}^{n/N}.$$

The factors e^{-2}, N^{-2}, N^{-k} and $1/k!$ do not depend on n, so they come out of the sum:

$$\mathrm{P}(X=k) = \frac{\mathrm{e}^{-2}N^{-k-2}}{k!}\sum_{n=1}^{N}(2n-1)(2N-n)^{k}\,\mathrm{e}^{n/N},$$

as required.

The case $N=3$, two accidents

We want $\mathrm{P}(S=2 \mid X=2)$. By Bayes' theorem,

$$\mathrm{P}(S=2 \mid X=2) = \frac{\mathrm{P}(S=2)\,\mathrm{P}(X=2 \mid S=2)}{\mathrm{P}(X=2)}.$$

Ask yourself Do we need to work out $\mathrm{P}(X=2)$ in full?

No. The numerator is just the $n = 2$ term of the sum for $\mathrm{P}(X = 2)$. So the common factor $\mathrm{e}^{-2}N^{-k-2}/k!$ cancels, and the probability is the $n = 2$ term of $\sum(2n-1)(2N-n)^k\mathrm{e}^{n/N}$ divided by the whole sum.

With $N = 3$ and $k = 2$, the terms $(2n-1)(6-n)^2\mathrm{e}^{n/3}$ are

$$n = 1:\ 1 \cdot 25\,\mathrm{e}^{1/3}, \qquad n = 2:\ 3 \cdot 16\,\mathrm{e}^{2/3} = 48\,\mathrm{e}^{2/3}, \qquad n = 3:\ 5 \cdot 9\,\mathrm{e} = 45\,\mathrm{e}.$$

Hence, dividing top and bottom by $\mathrm{e}^{2/3}$,

$$\mathrm{P}(S = 2 \mid X = 2) = \frac{48\,\mathrm{e}^{2/3}}{25\,\mathrm{e}^{1/3} + 48\,\mathrm{e}^{2/3} + 45\,\mathrm{e}} = \frac{48}{48 + 45\,\mathrm{e}^{1/3} + 25\,\mathrm{e}^{-1/3}},$$

which is the required result.

Checks

- For $N = 1$ there is one region with mean 1, and the formula gives $2 - \frac{1}{6} \cdot 2 \cdot 3 = 1$. As $N \to \infty$ the mean tends to $2 - \frac{4}{6} = \frac{4}{3}$. That agrees with $\int_0^1 2x(2-x)\,\mathrm{d}x = \frac{4}{3}$, where a uniform point at relative radius x has density $2x$.
- For $N = 3$ the mean is $\frac{32}{27} \approx 1.185$. A simulation of 4×10^5 days (pick a uniform point in the disc, then draw a Poisson count) gave a mean of 1.182. It also gave $\mathrm{P}(X = 2) \approx 0.209$ (exact 0.2094), and the fraction of two-accident days spent in R_2 came out at 0.372. The final answer evaluates to 0.3729.
- The distribution formula, summed over k, gives total probability 1 for every N we tried.

Common slips

- Choosing regions with equal probability $1/N$ instead of in proportion to area $2n-1$.
- Averaging the Poisson *means* and then claiming X is Poisson with that mean. A mixture of Poissons is not Poisson. Only the expectation passes through the averaging so simply.

Beyond STEP

The variable X has a *mixture distribution*: the region acts as a hidden variable, and the last part does Bayesian inference about it. The second slip above is explained by the law of total variance, $\mathrm{Var}(X) = \mathrm{E}(\mathrm{Var}(X \mid S)) + \mathrm{Var}(\mathrm{E}(X \mid S))$, which shows that a mixture is always more spread out than a single Poisson with the same mean. Grimmett and Welsh, *Probability: An Introduction*, covers conditional expectation and these mixtures well.

WORKED SOLUTION

STEP III 2002, Question 1

Pure • Improper Integrals and Volumes

STEP III 2002, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

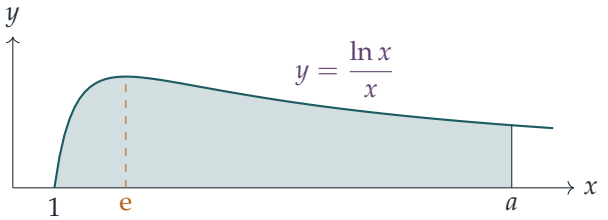
step.maths.org/questions/02-s3-q1

The key idea

Both integrals become easy once you notice that $\frac{1}{x}$ is the derivative of $\ln x$. The area integrand is "a function times its own derivative", so it integrates at sight. The volume integrand $(\ln x)^2/x^2$ has a leftover power of x, so we integrate by parts twice, each time differentiating a power of $\ln x$ to lower it. The interesting part is the contrast between the two limits: the curve decays just slowly enough for the area to grow without bound, but squaring it makes it decay fast enough for the volume to settle down.

Solution

For $x \geq 1$ we have $\ln x \geq 0$, so the curve lies on or above the x-axis and the area is simply the integral. It starts at $(1,0)$, rises to a maximum at $x = \mathrm{e}$ (where $y' = (1-\ln x)/x^2 = 0$), and then decays slowly towards the axis.



The area

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(\ln x)^2\right) = \dfrac{\ln x}{x}$,

$$A(a) = \int_1^a \frac{\ln x}{x}\,\mathrm{d}x = \left[\tfrac{1}{2}(\ln x)^2\right]_1^a = \tfrac{1}{2}(\ln a)^2,$$

because $\ln 1 = 0$. (Equivalently, substitute $u = \ln x$, $\mathrm{d}u = \mathrm{d}x/x$, to get $\int_0^{\ln a} u\,\mathrm{d}u$.)

As $a \to \infty$, $\ln a \to \infty$, so $A(a) \to \infty$: the area is unbounded, although it grows very slowly.

The volume

Rotating about the x-axis, the volume is $V(a) = \pi \int_1^a y^2 \,\mathrm{d}x = \pi \int_1^a \dfrac{(\ln x)^2}{x^2} \,\mathrm{d}x$.

Integrate by parts with $u = (\ln x)^2$ and $\mathrm{d}v = x^{-2}\,\mathrm{d}x$, so $\mathrm{d}u = \frac{2\ln x}{x}\,\mathrm{d}x$ and $v = -\frac{1}{x}$:

$$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2}{x} + 2\int \frac{\ln x}{x^2}\,\mathrm{d}x.$$

Doing the same again with $u = \ln x$, $\mathrm{d}v = x^{-2}\,\mathrm{d}x$:

$$\int \frac{\ln x}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} - \frac{1}{x}.$$

Putting these together,

$$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2 + 2\ln x + 2}{x} + C.$$

At $x = 1$ the bracket is $0 + 0 + 2$, so the lower limit contributes $-(-2) = +2$, and

$$V(a) = \pi\left(2 - \frac{(\ln a)^2 + 2\ln a + 2}{a}\right).$$

Ask yourself How do we know $(\ln a)^2/a$ tends to 0, when its top and bottom both tend to infinity?

Logarithms grow more slowly than any positive power. To make this precise, start from $\ln t < t$ for all $t > 0$ (the line $y = t$ lies above $y = \ln t$; the gap $t - \ln t$ has its minimum value 1 at $t = 1$). Put $t = a^{1/4}$: then $\frac{1}{4}\ln a < a^{1/4}$, that is $\ln a < 4a^{1/4}$. So for $a > 1$,

$$0 < \frac{(\ln a)^2}{a} < \frac{16\,a^{1/2}}{a} = \frac{16}{\sqrt{a}} \to 0, \qquad 0 < \frac{\ln a}{a} < \frac{4}{a^{3/4}} \to 0,$$

and of course $2/a \to 0$. Hence the fraction in $V(a)$ tends to 0, and

$$V(a) \to 2\pi \quad \text{as } a \to \infty.$$

Since the integrand is positive, $V(a)$ increases with a, so the volume rises steadily towards 2π but never reaches it: the solid has finite volume even though the region being rotated has infinite area.

Checks

- At $a = 1$ both formulae give 0, as they must: $A(1) = 0$ and $V(1) = \pi(2 - 2) = 0$.

- Differentiating $-\left((\ln x)^2 + 2\ln x + 2\right)/x$ gives back $(\ln x)^2/x^2$ (the $2\ln x$ and 2 terms cancel against the derivative of the bracket).
- Numerically, Simpson's rule on $\pi\int_1^{10}(\ln x)^2/x^2\,\mathrm{d}x$ gives 2.54247, matching the formula; and with $x = \mathrm{e}^t$ the infinite volume is $\pi\int_0^{\infty} t^2\mathrm{e}^{-t}\,\mathrm{d}t$, which evaluates numerically to $6.283185\ldots = 2\pi$.

Common slips

- Forgetting the lower limit: $\ln 1 = 0$ kills most terms at $x = 1$, but not the constant $-2/x$, which supplies the whole limiting value 2π.
- Asserting $(\ln a)^2/a \to 0$ without any reason. A one-line comparison such as $\ln a < 4a^{1/4}$ is enough.

Beyond STEP

This is a cousin of "Gabriel's horn" (rotating $y = 1/x$), a solid with finite volume but infinite surface area: here even the plane region has infinite area. The substitution $x = \mathrm{e}^t$ turns the volume into $\pi\int_0^{\infty} t^2\mathrm{e}^{-t}\,\mathrm{d}t = \pi\,\Gamma(3) = 2!\,\pi$, a first glimpse of the gamma function, $\Gamma(n+1) = n!$. Making sense of improper integrals, and of comparisons such as "logs lose to powers", is a core topic of first-year analysis; Abbott's *Understanding Analysis* is a good place to read about it.

WORKED SOLUTION

STEP III 2002, Question 2

Pure • Inverse Tangent Series

STEP III 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q2

The key idea

The first result is the tangent addition formula in disguise. The only real work is checking that the two sides are angles in the same interval, where tan is one-to-one. Once we have that formula, the induction is one application of it plus some factorising. For the last sum, the trick is to notice that $r^2 - r + 1$ is just $r^2 + r + 1$ with r shifted by one. So the new series is the old one with a single extra term at the front.

Solution

The addition formula

Let $A = \arctan a$ and $B = \arctan b$. Then $\tan A = a$ and $\tan B = b$, and

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{a + b}{1 - ab},$$

which makes sense because $ab < 1$. This shows that $A + B$ and $\arctan\left(\frac{a+b}{1-ab}\right)$ have the same tangent. That alone doesn't make them equal, since tan repeats every π. So we need both angles in one interval on which tan is one-to-one.

? **Ask yourself** Why isn't $\tan(A + B) = \frac{a+b}{1-ab}$ already the end of the proof?

It isn't the end because arctan only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$. We must check that $A + B$ lies there too.

- Since $0 < a < 1$ and arctan is increasing, $0 < A < \frac{\pi}{4}$. Similarly $0 < B < \frac{\pi}{4}$, so $0 < A + B < \frac{\pi}{2}$.
- Since $a + b > 0$ and $1 - ab > 0$, the fraction $\frac{a+b}{1-ab}$ is positive. So its arctan also lies in $(0, \frac{\pi}{2})$.

The function tan is strictly increasing, and hence one-to-one, on $(0, \frac{\pi}{2})$. Two angles

there with the same tangent are therefore equal, which gives

$$\arctan a + \arctan b = \arctan\left(\frac{a+b}{1-ab}\right). \qquad \checkmark$$

The induction

Write $S_n = \sum_{r=1}^{n} \arctan\left(\frac{1}{r^2+r+1}\right)$. The claim is $S_n = \arctan\left(\frac{n}{n+2}\right)$.

Base case. When $n = 1$, both sides are $\arctan\frac{1}{3}$.

Inductive step. Suppose $S_n = \arctan\left(\frac{n}{n+2}\right)$ for some $n \geq 1$. The next term has

$$(n+1)^2 + (n+1) + 1 = n^2 + 3n + 3,$$

so

$$S_{n+1} = \arctan\left(\frac{n}{n+2}\right) + \arctan\left(\frac{1}{n^2+3n+3}\right).$$

Both arguments lie strictly between 0 and 1, so the addition formula applies with $a = \frac{n}{n+2}$ and $b = \frac{1}{n^2+3n+3}$. Multiplying the top and bottom of $\frac{a+b}{1-ab}$ by $(n+2)(n^2+3n+3)$ gives

$$\begin{aligned}\frac{a+b}{1-ab} &= \frac{n(n^2+3n+3)+(n+2)}{(n+2)(n^2+3n+3)-n} \\ &= \frac{n^3+3n^2+4n+2}{n^3+5n^2+8n+6}.\end{aligned}$$

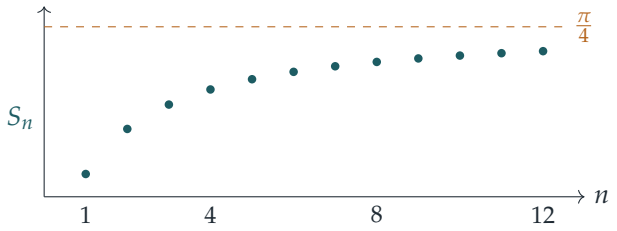
Both cubics vanish at a negative integer: the top at $n = -1$ and the bottom at $n = -3$. Factorising,

$$\frac{a+b}{1-ab} = \frac{(n+1)(n^2+2n+2)}{(n+3)(n^2+2n+2)} = \frac{n+1}{n+3},$$

where we can cancel because $n^2+2n+2 = (n+1)^2+1 > 0$. Hence $S_{n+1} = \arctan\left(\frac{n+1}{(n+1)+2}\right)$, which is the statement for $n+1$. By induction, the result holds for all $n \geq 1$. ✓

The infinite sum. As $n \to \infty$ we have $\frac{n}{n+2} \to 1$. Since arctan is continuous,

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2+r+1}\right) = \lim_{n\to\infty} \arctan\left(\frac{n}{n+2}\right) = \arctan 1 = \frac{\pi}{4}.$$



The final sum

Put $s = r - 1$. Then

$$r^2 - r + 1 = (s+1)^2 - (s+1) + 1 = s^2 + s + 1.$$

As r runs over $1, 2, 3, \ldots$, s runs over $0, 1, 2, \ldots$. The $s = 0$ term is $\arctan 1 = \frac{\pi}{4}$, and the remaining terms ($s \geq 1$) make up exactly the series we have just summed. Therefore

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2 - r + 1}\right) = \arctan 1 + \sum_{s=1}^{\infty} \arctan\left(\frac{1}{s^2 + s + 1}\right) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}. \qquad \checkmark$$

The second series converges because the first one does, so splitting off the first term is legitimate.

Checks

- When $n = 2$: $\arctan\frac{1}{3} + \arctan\frac{1}{7}$ has tangent $\frac{10/21}{20/21} = \frac{1}{2} = \frac{2}{4}$, as the formula predicts.
- Numerically, $S_{2000} \approx 0.784898$, while $\arctan\frac{2000}{2002}$ agrees to 10^{-12} and $\frac{\pi}{4} \approx 0.785398$.
- Summing the second series to $r = 10^6$ gives $1.5707953\ldots$, which is within 10^{-6} of $\frac{\pi}{2}$. That matches the gap after n terms, $\frac{\pi}{4} - \arctan\frac{n-1}{n+1} = \arctan\frac{1}{n} \approx \frac{1}{n}$.

Common slips

- Stopping at "tan of both sides is equal". Without the range argument, the proof is incomplete. (For example, with $a = b = 2$ the formula fails: the left side is positive but $\arctan(-\frac{4}{3})$ is negative.)
- Forgetting to check that both arguments in the inductive step lie in $(0, 1)$ before using the formula.
- Writing the final sum as $\frac{\pi}{4}$ again. The shift $r \mapsto r - 1$ adds the extra term $\arctan 1$ at the front.

Beyond STEP

There is a slicker route to the induction. Since $\frac{1}{r^2+r+1} = \frac{(r+1)-r}{1+r(r+1)}$, each term is $\arctan(r+1) - \arctan r$. The sum therefore *telescopes* to $\arctan(n+1) - \frac{\pi}{4}$, which is $\arctan\frac{n}{n+2}$ by the subtraction formula. Spotting a telescoping structure and justifying limits of partial sums by continuity are everyday tools in a first analysis course. Abbott's *Understanding Analysis* treats series and continuity carefully, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice of this kind.

WORKED SOLUTION

STEP III 2002, Question 3

Pure • Curve Sketching

STEP III 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q3

The key idea

Everything follows from the shape of the graph: it starts at $x = b$ with a vertical tangent going downwards, falls to a single minimum, and then rises without bound. Once we know the height of the starting point and the height of the minimum, counting solutions of $\mathrm{f}(x) = c$ is just a matter of drawing a horizontal line at height c and seeing how many times it meets the curve. The two numerical equations are then special cases whose values of c^2 land on the boundaries and in the interior of these ranges.

Solution

The graph

At the left-hand end, $\mathrm{f}(b) = a\sqrt{b}$. Differentiating, for $x > b$,

$$\mathrm{f}'(x) = \frac{a}{2\sqrt{x}} - \frac{1}{2\sqrt{x-b}}.$$

As $x \to b^+$ the second term tends to $-\infty$, so the curve leaves the point $(b, a\sqrt{b})$ heading straight down (a vertical tangent).

Since both square roots are positive,

$$\mathrm{f}'(x) < 0 \iff a\sqrt{x-b} < \sqrt{x} \iff a^2(x-b) < x \iff x < \frac{a^2 b}{a^2-1},$$

where the last step uses $a^2 - 1 > 0$. So f is strictly decreasing on $[b, x_0]$ and strictly increasing on $[x_0, \infty)$, where

$$x_0 = \frac{a^2 b}{a^2 - 1}.$$

At the minimum, $x_0 - b = \dfrac{b}{a^2-1}$, so

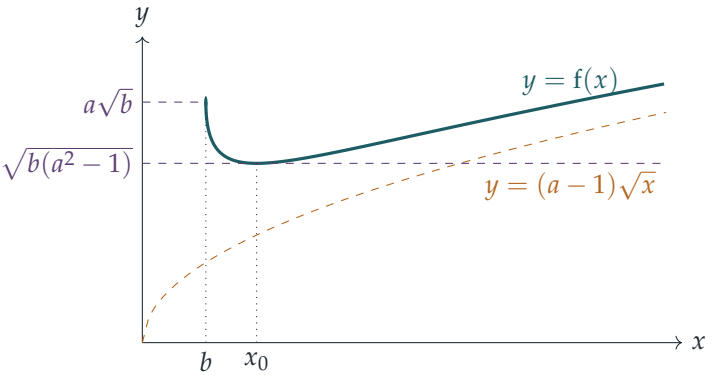
$$\mathrm{f}(x_0) = a \cdot \frac{a\sqrt{b}}{\sqrt{a^2-1}} - \frac{\sqrt{b}}{\sqrt{a^2-1}} = \frac{(a^2-1)\sqrt{b}}{\sqrt{a^2-1}} = \sqrt{b(a^2-1)}.$$

For large x, rationalising gives

$$\mathrm{f}(x) = (a-1)\sqrt{x} + \left(\sqrt{x} - \sqrt{x-b}\right) = (a-1)\sqrt{x} + \frac{b}{\sqrt{x}+\sqrt{x-b}},$$

so, because $a > 1$, $\mathrm{f}(x) \to \infty$, and the curve lies just above $y = (a-1)\sqrt{x}$, approaching it as $x \to \infty$.

Finally, the starting height is above the minimum: $a^2 b > (a^2-1)b$, so $a\sqrt{b} > \sqrt{b(a^2-1)}$.



Counting solutions

Write $m = \sqrt{b(a^2-1)}$ for the minimum value and $M = a\sqrt{b}$ for the starting value, so $0 < m < M$. On each of the two monotonic pieces, f is continuous and strictly monotonic, so it takes each value in its range exactly once:

- on $[b, x_0]$, f takes each value in $[m, M]$ exactly once;
- on $[x_0, \infty)$, f takes each value in $[m, \infty)$ exactly once.

The two pieces share only the point x_0, where the value is m.

Since $\mathrm{f}(x) \geq m$ for every $x \geq b$, the equation $\mathrm{f}(x) = c$ has no solution when $c < m$. For $c > 0$ this is the same as $c^2 < m^2$, that is,

$$c^2 < b(a^2-1) \quad \Longrightarrow \quad \text{no solution},$$

which is what we had to show. For the other cases:

Ask yourself Why does $c^2 = a^2 b$ give two solutions rather than one?

At $c = M$ the left piece contributes the endpoint $x = b$ itself, which is a perfectly good solution because the domain includes $x = b$; the right piece contributes another one further out. Only when $c > M$ does the left piece stop contributing. So, for $c > 0$:

- **exactly one solution** when $c^2 = b(a^2-1)$ (just $x = x_0$, where the line touches the minimum) or when $c^2 > a^2 b$ (only the rising branch reaches that high);

- **exactly two solutions** when $b(a^2-1) < c^2 \le a^2 b$ (one on each branch).

Part (i)

Here $a = 3$, $b = 2$, $c = 4$, so $b(a^2-1) = 16 = c^2$: the line touches the minimum, and there is exactly one solution,

$$x = x_0 = \frac{9 \cdot 2}{8} = \frac{9}{4}.$$

Indeed $3 \cdot \frac{3}{2} - \sqrt{\frac{1}{4}} = \frac{9}{2} - \frac{1}{2} = 4$. Directly: rearranging to $3\sqrt{x} - 4 = \sqrt{x-2}$ and squaring gives $8x - 24\sqrt{x} + 18 = 0$, that is, $(2\sqrt{x} - 3)^2 = 0$. The repeated root is the algebraic sign of tangency.

Part (ii)

Here $a = 3$, $b = 3$, $c = 5$, so $b(a^2-1) = 24 < 25 = c^2 < 27 = a^2 b$, and there are exactly two solutions. To find them, write the equation as $3\sqrt{x} - 5 = \sqrt{x-3}$ and square:

$$9x - 30\sqrt{x} + 25 = x - 3 \quad \Longrightarrow \quad 4x - 15\sqrt{x} + 14 = 0.$$

As a quadratic in $\sqrt{x}$, $(\sqrt{x} - 2)(4\sqrt{x} - 7) = 0$, so $\sqrt{x} = 2$ or $\sqrt{x} = \frac{7}{4}$, giving

$$x = 4 \qquad \text{or} \qquad x = \frac{49}{16}.$$

Squaring can introduce spurious roots, so we check both in the original equation. Both satisfy $x \ge 3$ (as $\frac{49}{16} = 3.0625$), and

$$3 \cdot 2 - \sqrt{1} = 5, \qquad 3 \cdot \tfrac{7}{4} - \sqrt{\tfrac{1}{16}} = \tfrac{21}{4} - \tfrac{1}{4} = 5.$$

So both are genuine, matching the count of two. As expected, one lies on each side of $x_0 = \frac{27}{8} = 3.375$.

Checks

- With $a = \frac{3}{2}$, $b = 2$ (the sketch), $x_0 = \frac{18}{5}$ and $f(\frac{18}{5}) = \frac{3}{2}\sqrt{3.6} - \sqrt{1.6} = \sqrt{2.5} \approx 1.581$, below $f(2) = \frac{3}{2}\sqrt{2} \approx 2.121$.
- A numerical scan of f over a fine grid for 200 random pairs (a, b) confirmed the minimum value $\sqrt{b(a^2-1)}$ at x_0, and counting crossings of $y = c$ gave 0, 2 and 1 solutions in the three ranges of c^2.
- In (i) the double root is consistent with c^2 sitting exactly at $b(a^2-1)$; in (ii) $c^2 = 25$ is strictly between 24 and 27, and we did find two roots.

Common slips

- Putting $c^2 = a^2 b$ in the "one solution" case: $x = b$ is in the domain, so it counts.

- Squaring without checking afterwards. Here both roots survive, but you must verify that $3\sqrt{x} - 5 \geq 0$ and $x \geq 3$ for each.
- Forgetting that $a > 1$ is what makes $f \to \infty$; if $a \leq 1$ there would be no minimum and the picture changes completely.

Beyond STEP

The counting argument rests on two facts from analysis: a continuous function on an interval takes every value between its values at the ends (the intermediate value theorem), and a strictly monotonic function takes each value at most once. Together they turn a sketch into a proof. Abbott's *Understanding Analysis* develops both carefully and shows why continuity is exactly the hypothesis needed.

WORKED SOLUTION

STEP III 2002, Question 4

Pure • Diophantine Inequalities

STEP III 2002, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q4

The key idea

The equation $x^3 + x^2 = y^3 - y^2$ pits a quantity that grows fast against one that is held back by the $-y^2$, so y has to "overtake" x, but only by less than 1. Those two inequalities squeeze y strictly between x and $x+1$, and no integer fits in that gap. For the last part, the extra solutions can only come from zero or negative values, and a neat symmetry, swapping (m, n) for $(-n, -m)$, cuts the casework in half.

Solution

The first result

Suppose $x, y > 0$ and $x^3 + x^2 = y^3 - y^2$, and suppose for contradiction that $x \geq y$. Since cubing and squaring are increasing on the positive reals,

$$x^3 + x^2 \geq y^3 + y^2 > y^3 - y^2,$$

where the strict step uses $y^2 > 0$. This contradicts the equation, so $x < y$.

The second result

Now let $0 < x \leq y - 1$, so $y \geq x + 1 > 1$. Write $f(t) = t^3 - t^2$. Then $f'(t) = 3t^2 - 2t = t(3t-2) > 0$ for $t > \frac{2}{3}$, so f is increasing on $[1, \infty)$. Since $y \geq x+1$ and both lie in that interval,

$$\begin{aligned} y^3 - y^2 = f(y) &\geq f(x+1) = (x+1)^2\big((x+1) - 1\big) \\ &= x(x+1)^2 = x^3 + 2x^2 + x \\ &= (x^3 + x^2) + (x^2 + x) \\ &> x^3 + x^2, \end{aligned}$$

because $x^2 + x > 0$ for $x > 0$. So $x^3 + x^2 < y^3 - y^2$, as required.

Together, the two results say that for positive x, any positive y with $x^3 + x^2 = y^3 - y^2$ satisfies

$$x < y < x + 1.$$

No positive integer pair

Suppose positive integers a and b have $a^3 - b^3 = a^2 + b^2$. The right-hand side is positive, so $a^3 > b^3$ and $a > b$. Rearranging,

$$b^3 + b^2 = a^3 - a^2,$$

which is the equation above with $x = b$, $y = a$, both positive. So $b < a < b + 1$.

Ask yourself Where does integrality actually get used?

Only here: there is no integer strictly between b and $b + 1$. Put another way, $a > b$ with a, b integers forces $a \geq b + 1$, that is $0 < b \leq a - 1$, and then the second result gives $b^3 + b^2 < a^3 - a^2$, contradicting the equation. So no such pair of positive integers exists. (For real numbers there are plenty of solutions, one $y \in (x, x + 1)$ for each $x > 0$; the integers simply miss them.)

All integer pairs

We now want all integers m, n with

$$m^3 - n^3 = m^2 + n^2. \qquad (*)$$

Two observations save work.

- If (m, n) is a solution, so is $(-n, -m)$: indeed $(-n)^3 - (-m)^3 = m^3 - n^3$, and the sum of squares is unchanged.
- The right-hand side of $(*)$ is non-negative, so $m^3 \geq n^3$, that is $m \geq n$, with equality only if $m^2 + n^2 = 0$, i.e. $m = n = 0$.

One of them is zero. If $n = 0$, then $m^3 = m^2$, so $m = 0$ or $m = 1$. If $m = 0$, then $-n^3 = n^2$, so $n = 0$ or $n = -1$. This gives $(0, 0)$, $(1, 0)$ and $(0, -1)$ (the last is the image of $(1, 0)$ under the symmetry).

Both positive. Impossible, by the previous part.

Both negative. Then $-n$ and $-m$ are both positive, and $(-n, -m)$ would be a solution with both entries positive, which is impossible.

Opposite signs. Since $m > n$, we have $m > 0 > n$. Write $n = -p$ with $p \geq 1$. Then $(*)$ becomes

$$m^3 + p^3 = m^2 + p^2, \qquad \text{i.e.} \qquad m^2(m - 1) + p^2(p - 1) = 0.$$

For positive integers both terms are ≥ 0, so both are zero, forcing $m = p = 1$. This gives $(1, -1)$, and indeed $1 - (-1) = 2 = 1 + 1$.

So the integer solutions (m, n) of $m^3 - n^3 = m^2 + n^2$ are exactly

$$(m, n) = (0, 0), \quad (1, 0), \quad (0, -1), \quad (1, -1).$$

If "pair" is meant to exclude the trivial pair, drop $(0, 0)$.

Checks

Each pair can be checked by hand: for $(1, 0)$, $1 - 0 = 1 = 1 + 0$; for $(0, -1)$, $0 - (-1) = 1 = 0 + 1$; for $(1, -1)$, $2 = 2$. A computer search over all $|m|, |n| \leq 300$ finds exactly these four solutions. As a check on the first two results, numerically solving $y^3 - y^2 = x^3 + x^2$ for random $x > 0$ always gave $x < y < x + 1$; for example $x = 1$ gives $y \approx 1.696$.

Common slips

- In the final part, forgetting zero and negative integers, which is where all the solutions live.
- Assuming which of the two integers is "first". The difference of cubes is positive here, so the larger number must come first; say so.
- Proving the second result by just "expanding" without noting that $t^3 - t^2$ is increasing for $t \geq 1$; the step $f(y) \geq f(x + 1)$ needs it.

Beyond STEP

The argument "the real solution is trapped strictly between two consecutive integers, so there is no integer solution" is a standard tool in the study of Diophantine equations, and often the first thing to try on a polynomial equation in integers. Bounding a solution in this way is also how one proves many such equations have only finitely many solutions, which can then be found by a finite search. Davenport's *The Higher Arithmetic* is a lovely introduction to this kind of number theory.

WORKED SOLUTION

STEP III 2002, Question 5

Pure • Factorising Quartic Equations

STEP III 2002, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q5

The key idea

A quadratic is a constant multiple of a perfect square exactly when its discriminant vanishes. That is the whole of the first part, and it is also the engine of the rest: if we can choose t so that a square of a quadratic minus the quartic is a multiple of a perfect square, then the quartic is a *difference of two squares*, and it factorises at once. This is Ferrari's classical method for the quartic, and it trades the quartic for a cubic in t.

Solution

The quadratic condition

Suppose $p \neq 0$. Completing the square,

$$px^2 + qx + r = p\left(x + \frac{q}{2p}\right)^2 + r - \frac{q^2}{4p}.$$

This has the form $p(x+h)^2$ for some h exactly when the constant $r - \frac{q^2}{4p}$ is zero, that is, when

$$q^2 = 4pr,$$

and then $h = \frac{q}{2p}$. (Conversely, expanding $p(x+h)^2 = px^2 + 2phx + ph^2$ gives $q^2 = 4p^2h^2 = 4pr$, so the condition is necessary as well as sufficient.) If $p = 0$, the only possibility is $q = r = 0$; the condition $q^2 = 4pr$ still forces $q = 0$, but we also need $r = 0$.

The equation for t

Expanding,

$$\left(x^2 + \tfrac{1}{2}bx + t\right)^2 = x^4 + bx^3 + \left(\tfrac{1}{4}b^2 + 2t\right)x^2 + btx + t^2.$$

The choice of $\frac{1}{2}b$ is exactly what makes the x^3 terms cancel (and the x^4 terms cancel too) when we subtract the quartic:

$$\left(x^2 + \tfrac{1}{2}bx + t\right)^2 - \left(x^4 + bx^3 + cx^2 + dx + e\right) = \left(\tfrac{1}{4}b^2 + 2t - c\right)x^2 + (bt - d)x + \left(t^2 - e\right).$$

This is a quadratic with $p = \frac{1}{4}b^2 + 2t - c$, $q = bt - d$ and $r = t^2 - e$, and we want it to equal $k(x+h)^2$, a multiple (namely $k = p$) of a perfect square. By the first part, t must satisfy

$$(bt-d)^2 = 4\left(\tfrac{1}{4}b^2 + 2t - c\right)\left(t^2 - e\right).$$

This is a cubic equation in t (the t^3 term on the right has coefficient 8, while the left is only quadratic in t), so it always has at least one real root.

Factorising the quartic

Here $b = 6$, $c = 9$, $d = -2$, $e = -7$, so

$$\tfrac{1}{4}b^2 + 2t - c = 2t, \qquad bt - d = 6t + 2, \qquad t^2 - e = t^2 + 7.$$

The equation for t becomes

$$\begin{aligned}(6t+2)^2 &= 8t\left(t^2+7\right)\\ 36t^2 + 24t + 4 &= 8t^3 + 56t\\ 0 &= 2t^3 - 9t^2 + 8t - 1.\end{aligned}$$

Ask yourself How do we find a root of this cubic quickly?

Try small values first: the coefficients add up to $2 - 9 + 8 - 1 = 0$, so $t = 1$ is a root. Any root will do, because each one gives a valid factorisation; the simplest root gives the simplest one.

With $t = 1$ the difference is

$$2x^2 + 8x + 8 = 2(x+2)^2,$$

so $k = 2$ and $h = 2$, and indeed $8^2 = 4 \cdot 2 \cdot 8$. Hence

$$\left(x^2 + 3x + 1\right)^2 - \left(x^4 + 6x^3 + 9x^2 - 2x - 7\right) = 2(x+2)^2,$$

that is,

$$x^4 + 6x^3 + 9x^2 - 2x - 7 = \left(x^2 + 3x + 1\right)^2 - \left(\sqrt{2}\,(x+2)\right)^2.$$

This is a difference of two squares, so

$$\begin{aligned}x^4 + 6x^3 + 9x^2 - 2x - 7 &= \left(x^2 + 3x + 1 - \sqrt{2}(x+2)\right)\left(x^2 + 3x + 1 + \sqrt{2}(x+2)\right)\\ &= \left(x^2 + (3-\sqrt{2})x + 1 - 2\sqrt{2}\right)\left(x^2 + (3+\sqrt{2})x + 1 + 2\sqrt{2}\right).\end{aligned}$$

Because $k = 2 > 0$, its square root is real, so these are two quadratic factors with real coefficients, as required.

The other two roots of the cubic, $t = \frac{1}{4}\left(7 \pm \sqrt{41}\right)$ from $2t^2 - 7t + 1 = 0$, also have $k = 2t > 0$, and they give the other two ways of pairing up the four (real) roots of the quartic into real quadratic factors. Their coefficients are much messier, which is why $t = 1$ is the one to use.

Checks

- Multiply out: the product of the two factors has x^3 coefficient $(3-\sqrt{2})+(3+\sqrt{2})=6$ and constant term $(1-2\sqrt{2})(1+2\sqrt{2})=1-8=-7$, both correct.
- At $x=1$ the quartic is $1+6+9-2-7=7$, and the factorisation gives $(5-\sqrt{2}-2\sqrt{2})(5+\sqrt{2}+2\sqrt{2})=25-18=7$.
- The discriminants of the factors are $7+2\sqrt{2}\approx 9.83$ and $7-2\sqrt{2}\approx 4.17$, both positive, so the quartic has four real roots, consistent with all three roots of the cubic giving real factorisations. We also checked the factorisation numerically at 200 random points.

Common slips

- Getting the x^2 coefficient of $(x^2+\frac{1}{2}bx+t)^2$ wrong: it is $\frac{1}{4}b^2+2t$, since the cross term $2\cdot x^2\cdot t$ contributes as well as $(\frac{1}{2}bx)^2$.
- Writing the discriminant condition as $q^2=pr$ or forgetting that k is the leading coefficient $\frac{1}{4}b^2+2t-c$, which must be positive for a real factorisation.

Beyond STEP

The cubic in t is called the *resolvent cubic*, and this is Ferrari's sixteenth-century method: every quartic reduces to a cubic, which Cardano's formula then solves. The deeper reason it works, and the reason there is no such formula for general quintics, belongs to Galois theory, which studies how the roots of a polynomial can be permuted. For a first look at the algebra of fields and polynomials, try Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP III 2002, Question 6

Pure • Nonlinear Differential Equations

STEP III 2002, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q6

The key idea

The equation looks fearsome because of the fourth powers, but $y\frac{\mathrm{d}y}{\mathrm{d}x}$ is exactly half the derivative of y^2. So substitute $u = y^2 - 1$: the equation collapses to one about u and u' alone, which separates once we take a square root. The only care needed is with signs: the sign of u splits the plane into the strip $|y| < 1$ and the regions $|y| > 1$, and they give different families of curves.

Solution

Put $u = y^2 - 1$, so that $u' = 2y\frac{\mathrm{d}y}{\mathrm{d}x}$ and $y^4\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^4 = \left(\frac{1}{2}u'\right)^4$. The equation becomes

$$\tfrac{1}{16}(u')^4 = u^2 \quad \Longleftrightarrow \quad (u')^2 = 4|u|,$$

taking the non-negative square root of both sides (both $(u')^2$ and $4|u|$ are non-negative, so nothing is lost).

Note first that a solution curve can never meet $y = 0$: there the left-hand side of the original equation is 0 but the right-hand side is 1. So on any solution through either given point, $y > 0$ throughout.

Ask yourself How do we solve $(u')^2 = 4u$ without worrying about which sign of u' to take?

Where $u > 0$ we may write $u = v^2$ with $v = \sqrt{u} > 0$. Then $u' = 2vv'$, so $(u')^2 = 4u$ becomes $4v^2(v')^2 = 4v^2$, i.e. $(v')^2 = 1$. Since v' is continuous and never zero it has a constant sign, so $v' = \pm 1$ and $v = \pm(x - a)$ for some constant a. Either way,

$$u = v^2 = (x - a)^2.$$

The sign ambiguity simply disappears on squaring. The same argument with $u = -v^2$ handles $u < 0$.

Part (i): through $(0, \frac{1}{2}\sqrt{3})$

Here $y^2 = \frac{3}{4} < 1$, so near this point $u < 0$, i.e. $|y| < 1$. Write $1 - y^2 = -u = v^2$ with $v > 0$; then $(u')^2 = -4u$ gives $(v')^2 = 1$ exactly as above, so $1 - y^2 = (x - a)^2$, that is

$$(x - a)^2 + y^2 = 1 :$$

circles of radius 1 centred on the x-axis. At $x = 0$ we need $a^2 = 1 - \frac{3}{4} = \frac{1}{4}$, so $a = \pm\frac{1}{2}$. Since $y > 0$, the solution curves through $(0, \frac{1}{2}\sqrt{3})$ are the upper semicircles

$$y = \sqrt{1 - \left(x - \tfrac{1}{2}\right)^2} \qquad \text{and} \qquad y = \sqrt{1 - \left(x + \tfrac{1}{2}\right)^2}.$$

Checking directly: on $(x - a)^2 + y^2 = 1$, implicit differentiation gives $y\frac{\mathrm{d}y}{\mathrm{d}x} = -(x - a)$, so $y^4\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^4 = (x - a)^4 = (1 - y^2)^2 = (y^2 - 1)^2$. ✓

Part (ii): through $(0, \frac{1}{2}\sqrt{5})$

Now $y^2 = \frac{5}{4} > 1$, so $u > 0$ and the argument above gives $y^2 - 1 = (x - a)^2$, that is

$$y^2 - (x - a)^2 = 1 :$$

rectangular hyperbolas with centre $(a, 0)$. At $x = 0$, $a^2 = \frac{5}{4} - 1 = \frac{1}{4}$, so again $a = \pm\frac{1}{2}$, and with $y > 0$ the solution curves are the upper branches

$$y = \sqrt{1 + \left(x - \tfrac{1}{2}\right)^2} \qquad \text{and} \qquad y = \sqrt{1 + \left(x + \tfrac{1}{2}\right)^2},$$

defined for all x. Directly: $y\frac{\mathrm{d}y}{\mathrm{d}x} = x - a$, so the left-hand side is $(x - a)^4 = (y^2 - 1)^2$. ✓

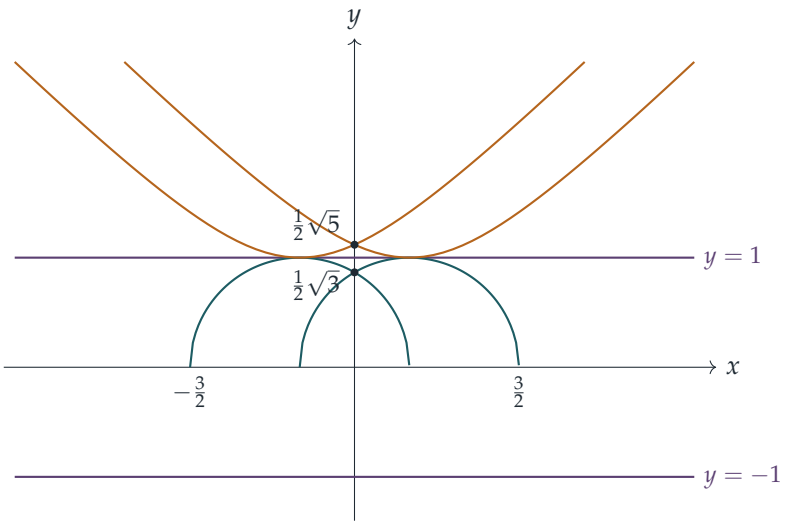
The lines $y = \pm 1$

If $y = 1$ or $y = -1$ for all x, then $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the left-hand side is 0; and $y^2 - 1 = 0$, so the right-hand side is 0 too. Both lines are solutions.

? **Ask yourself** How do these lines relate to the circles and hyperbolas?

Each circle $(x - a)^2 + y^2 = 1$ and each hyperbola $y^2 - (x - a)^2 = 1$ touches $y = 1$ at $(a, 1)$ with zero gradient (and likewise $y = -1$ at $(a, -1)$). So $y = \pm 1$ are the *envelopes* of both families, and they are not members of either family: they come from $u = 0$, which our division by v excluded. A consequence is that solutions are not unique through points on $y = 1$: for instance, following the circle with $a = \frac{1}{2}$ from $(0, \frac{1}{2}\sqrt{3})$ up to its top $(\frac{1}{2}, 1)$, then along $y = 1$, then up a hyperbola $y = \sqrt{1 + (x - b)^2}$ for $x \geq b$ (any $b \geq \frac{1}{2}$), gives a curve that is smooth enough (continuous gradient) and satisfies the equation everywhere. The four curves above are the "basic" solutions through the given points; every other solution through them is built from pieces of these, of $y = 1$, and of further circles and hyperbolas touching $y = 1$.

Sketch



The semicircles (teal) have centres $(\pm\frac{1}{2}, 0)$ and radius 1, meeting the x-axis vertically at $x = \pm\frac{1}{2} \pm 1$ and touching $y = 1$ at $x = \pm\frac{1}{2}$. The hyperbola branches (amber) touch $y = 1$ at the same two points and have asymptotes $y = \pm(x \mp \frac{1}{2})$, i.e. they grow like $|x|$. Each pair is symmetric about the y-axis, and the two curves of each pair cross on it at the given point.

Checks

- At $(0, \frac{1}{2}\sqrt{3})$ on the circle with $a = \frac{1}{2}$: $y' = -(x - a)/y = 1/\sqrt{3}$, so the left-hand side is $\frac{9}{16} \cdot \frac{1}{9} = \frac{1}{16}$ and the right-hand side is $(\frac{3}{4} - 1)^2 = \frac{1}{16}$.
- Numerically integrating $y' = \pm\sqrt{|y^2 - 1|}/y$ from $(0, \frac{1}{2}\sqrt{5})$ with the plus sign gives $y(1.5) = 2.2360680$, matching $\sqrt{1 + 2^2} = \sqrt{5}$ on the hyperbola with $a = -\frac{1}{2}$.

Common slips

- Taking the square root as $(u')^2 = 4u$ in both regions; for $|y| < 1$ this has no real solutions, and the circles are lost.
- Keeping only one sign of the square root at each stage, and so finding only one curve through each point instead of two.
- Forgetting that $y = \pm 1$ are not in either family, even though they touch every member.

Beyond STEP

The lines $y = \pm 1$ are *singular solutions*: envelopes of the general solution that no choice of the constant a produces. They appear exactly where the uniqueness theorem for first-order equations (Picard–Lindelöf) fails, because $\sqrt{|y^2 - 1|}$ is not Lipschitz at $y = \pm 1$, and that is why solutions can branch there. Envelopes of this kind are typical of first-order equations that are not linear in y' (Clairaut's equation is the classic example); Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness and when it breaks down.

WORKED SOLUTION

STEP III 2002, Question 7

Pure • Coordinate Geometry Identities

STEP III 2002, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q7

The key idea

The expression $s^2 + b^2 - a^2$ should look familiar: it is what the cosine rule produces in triangle OAB, where $OA = s$ and $OB = b$. So the first identity is really the opening result about $\cos^2\alpha + \cos^2\beta$, applied to the two angles that OB makes with the axes. After that, the second identity is pure algebra built on the first, and the inequality falls out because a sum of two squares is at least each square.

Solution

The angle result

Suppose first that $\alpha + \beta = \frac{1}{2}\pi$. Then $\cos\beta = \cos(\frac{1}{2}\pi - \alpha) = \sin\alpha$, so

$$\cos^2\alpha + \cos^2\beta = \cos^2\alpha + \sin^2\alpha = 1.$$

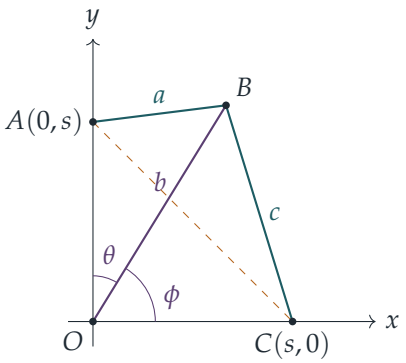
Conversely, suppose $\cos^2\alpha + \cos^2\beta = 1$. Then $\cos^2\beta = 1 - \cos^2\alpha = \sin^2\alpha$. Both α and β are acute, so $\cos\beta > 0$ and $\sin\alpha > 0$, and taking positive square roots gives

$$\cos\beta = \sin\alpha = \cos(\tfrac{1}{2}\pi - \alpha).$$

Now β and $\frac{1}{2}\pi - \alpha$ both lie in $(0, \frac{1}{2}\pi)$, where cosine is strictly decreasing and so takes each value only once. Hence $\beta = \frac{1}{2}\pi - \alpha$, that is, $\alpha + \beta = \frac{1}{2}\pi$. This proves the "if and only if".

The first identity

Let $\theta = \angle AOB$ and $\phi = \angle BOC$.



The cosine rule in triangle OAB (sides $OA = s$, $OB = b$, $AB = a$) and in triangle OCB (sides $OC = s$, $OB = b$, $CB = c$) gives

$$\cos\theta = \frac{s^2 + b^2 - a^2}{2sb}, \qquad \cos\phi = \frac{s^2 + b^2 - c^2}{2sb}.$$

Because B lies in the open first quadrant, the ray OB lies strictly between the positive y-axis (along OA) and the positive x-axis (along OC). So θ and ϕ are both acute and $\theta + \phi = \frac{1}{2}\pi$. By the "only if" direction above, $\cos^2\theta + \cos^2\phi = 1$, that is,

$$\frac{(s^2 + b^2 - a^2)^2}{4s^2b^2} + \frac{(s^2 + b^2 - c^2)^2}{4s^2b^2} = 1.$$

Multiplying by $4s^2b^2$ gives the first identity:

$$(s^2 + b^2 - a^2)^2 + (s^2 + b^2 - c^2)^2 = 4s^2b^2.$$

(For a direct check, write $B = (x, y)$. Then $a^2 = x^2 + (y - s)^2$, $b^2 = x^2 + y^2$ and $c^2 = (x - s)^2 + y^2$, so $s^2 + b^2 - a^2 = 2sy$ and $s^2 + b^2 - c^2 = 2sx$, and the left-hand side is $4s^2(x^2 + y^2) = 4s^2b^2$.)

The second identity

To keep the algebra tidy, write

$$P = s^2 + b^2 - a^2, \qquad Q = s^2 + b^2 - c^2, \qquad S = s^2 + b^2, \qquad D = s^2 - b^2,$$

so that $a^2 = S - P$ and $c^2 = S - Q$. The first identity says $P^2 + Q^2 = 4s^2b^2$. Since $4s^2b^2 = (s^2 + b^2)^2 - (s^2 - b^2)^2 = S^2 - D^2$, it can be rewritten as

$$D^2 = S^2 - P^2 - Q^2. \tag{†}$$

Now $a^2 + c^2 = 2S - P - Q$ and $2s^2 = S + D$, $2b^2 = S - D$, so

$$2s^2 - a^2 - c^2 = (P + Q - S) + D, \qquad 2b^2 - a^2 - c^2 = (P + Q - S) - D.$$

Ask yourself Why is this split into $(P+Q-S)\pm D$ worth making?

Because $(u+D)^2+(u-D)^2=2u^2+2D^2$: the cross terms cancel, and D then appears only as D^2, which (†) lets us replace by something involving P, Q and S alone. With $u=P+Q-S$,

$$\begin{aligned}&(2s^2-a^2-c^2)^2+(2b^2-a^2-c^2)^2\\&\quad=2(P+Q-S)^2+2D^2\\&\quad=2(P^2+Q^2+S^2+2PQ-2PS-2QS)+2(S^2-P^2-Q^2)\\&\quad=4S^2-4PS-4QS+4PQ\\&\quad=4(S-P)(S-Q)\\&\quad=4a^2c^2,\end{aligned}$$

using (†) in the second line. This is the second identity.

The inequality

Both squares on the left of the second identity are non-negative, so

$$(2b^2-a^2-c^2)^2\le 4a^2c^2,\qquad\text{so}\qquad |2b^2-a^2-c^2|\le 2ac,$$

taking non-negative square roots (note $a,c>0$). This says

$$-2ac\le 2b^2-a^2-c^2\le 2ac,$$

that is,

$$a^2-2ac+c^2\le 2b^2\le a^2+2ac+c^2,$$

which is $(a-c)^2\le 2b^2\le(a+c)^2$, as required.

Equality holds exactly when $2s^2=a^2+c^2$. Since $AC^2=2s^2$, by the converse of Pythagoras that means $\angle ABC$ is a right angle: B lies on the circle with diameter AC.

Checks

- Take $s=2$ and $B=(1,3)$. Then $a^2=2$, $b^2=c^2=10$. The first identity reads $12^2+4^2=160=4\cdot4\cdot10$; the second reads $(-4)^2+8^2=80=4\cdot2\cdot10$; and the inequality is $3.06\le 20\le 20.94$.
- The point $B=(s,s)$ is the fourth corner of the square, so $\angle ABC$ is right and $a=c=s$, $b=s\sqrt{2}$: then $2b^2=4s^2=(a+c)^2$, the equality case.
- A random test of all three results at 2×10^5 points B found no failures.

Common slips

- Leaving out the acute condition in the converse: without it, $\cos^2\beta=\sin^2\alpha$ does not force $\beta=\frac{1}{2}\pi-\alpha$ (for example, $\beta=\frac{1}{2}\pi+\alpha$ also works).

- Expanding the second identity in a, b, c, s by brute force without using the first. It can be done, but it is easy to drop a term; the substitution S, D keeps it under control.
- Taking square roots of $(2b^2 - a^2 - c^2)^2 \leq 4a^2c^2$ without the modulus, and so getting only one half of the inequality.

Beyond STEP

The quantity $2b^2 - a^2 - c^2$ is $2s(x + y - s)$, which is ± 4 times the area of triangle ABC, so the second identity is the cosine rule and the area formula $\frac{1}{2}ac \sin B$ combined into $\cos^2 B + \sin^2 B = 1$. The upper bound is also a case of Ptolemy's inequality: for any four points, $OB \cdot AC \leq OA \cdot BC + OC \cdot AB$, with equality for a cyclic quadrilateral, and here this reads $b\,s\sqrt{2} \leq s(a + c)$. Relations between distances like this lead on to the study of metric spaces and of distance geometry; Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP III 2002, Question 8

Pure • Ptolemy's Theorem

STEP III 2002, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q8

The key idea

The difference of two points on the unit circle factorises beautifully if we pull out the "average" angle: $e^{i\alpha} - e^{i\beta} = e^{i(\alpha+\beta)/2}\left(e^{i(\alpha-\beta)/2} - e^{-i(\alpha-\beta)/2}\right)$, and the bracket is 2i times a sine. That gives both the modulus and the argument of every chord at once. For the last part, two complex numbers with the same argument point in the same direction, so their moduli simply add; all we then need is an algebraic identity linking the three products.

Solution

The argument of $u_1 - u_2$

Write $u_k = e^{i\theta_k}$. Taking out a factor $e^{i(\theta_1+\theta_2)/2}$,

$$\begin{aligned} u_1 - u_2 &= e^{i(\theta_1+\theta_2)/2}\left(e^{i(\theta_1-\theta_2)/2} - e^{-i(\theta_1-\theta_2)/2}\right) \\ &= e^{i(\theta_1+\theta_2)/2} \cdot 2i \sin \tfrac{1}{2}(\theta_1 - \theta_2) \\ &= 2\sin \tfrac{1}{2}(\theta_2 - \theta_1) \cdot (-i)\, e^{i(\theta_1+\theta_2)/2} \\ &= 2\sin \tfrac{1}{2}(\theta_2 - \theta_1)\, e^{i(\theta_1+\theta_2-\pi)/2}, \end{aligned}$$

using $-i = e^{-i\pi/2}$ in the last line.

Ask yourself Why can we read the argument straight off this, and why do we need the sine in the form $\sin \frac{1}{2}(\theta_2 - \theta_1)$ rather than $\sin \frac{1}{2}(\theta_1 - \theta_2)$?

A number of the form $re^{i\phi}$ has argument ϕ (up to multiples of 2π) only when $r > 0$. Here $0 < \theta_2 - \theta_1 < 2\pi$, so $0 < \frac{1}{2}(\theta_2 - \theta_1) < \pi$ and the sine is strictly positive. Written the other way round the sine would be negative, and the argument would be out by π.

So $|u_1 - u_2| = 2\sin \frac{1}{2}(\theta_2 - \theta_1)$ and $\frac{1}{2}(\theta_1 + \theta_2 - \pi)$ is an argument of $u_1 - u_2$. Taking

arg to be the principal value in $(-\pi, \pi]$: since $-\pi < \theta_1 < \theta_2 < \pi$,

$$-\tfrac{3\pi}{2} < \tfrac{1}{2}(\theta_1 + \theta_2 - \pi) < \tfrac{\pi}{2}.$$

If this value is greater than $-\pi$ it is already the principal argument ($n = 0$); otherwise it lies in $(-\frac{3\pi}{2}, -\pi]$ and adding 2π brings it into range ($n = 1$). Hence

$$\arg(u_1 - u_2) = \tfrac{1}{2}(\theta_1 + \theta_2 - \pi) + 2n\pi, \qquad n = 0 \text{ or } 1,$$

as required.

The two products have the same argument

Nothing above used anything about u_1, u_2 except that $\theta_1 < \theta_2$ and both lie in $(-\pi, \pi)$. So for any $j < k$,

$$\begin{aligned} u_j - u_k &= 2\sin\tfrac{1}{2}(\theta_k - \theta_j)\, \mathrm{e}^{\mathrm{i}(\theta_j + \theta_k - \pi)/2}, \\ u_k - u_j = -(u_j - u_k) &= 2\sin\tfrac{1}{2}(\theta_k - \theta_j)\, \mathrm{e}^{\mathrm{i}(\theta_j + \theta_k + \pi)/2}, \end{aligned}$$

using $-1 = \mathrm{e}^{\mathrm{i}\pi}$, and in each case the real factor in front is positive. Write $s_{jk} = \sin\frac{1}{2}(\theta_k - \theta_j) > 0$ for $j < k$, and $\Sigma = \theta_1 + \theta_2 + \theta_3 + \theta_4$. Multiplying (arguments add),

$$\begin{aligned} (u_1 - u_2)(u_4 - u_3) &= 4s_{12}s_{34}\, \mathrm{e}^{\mathrm{i}[(\theta_1 + \theta_2 - \pi) + (\theta_3 + \theta_4 + \pi)]/2} = 4s_{12}s_{34}\, \mathrm{e}^{\mathrm{i}\Sigma/2}, \\ (u_1 - u_4)(u_3 - u_2) &= 4s_{14}s_{23}\, \mathrm{e}^{\mathrm{i}[(\theta_1 + \theta_4 - \pi) + (\theta_2 + \theta_3 + \pi)]/2} = 4s_{14}s_{23}\, \mathrm{e}^{\mathrm{i}\Sigma/2}. \end{aligned}$$

Both are positive multiples of $\mathrm{e}^{\mathrm{i}\Sigma/2}$, so each has argument $\frac{1}{2}\Sigma$ up to a multiple of 2π. Their principal arguments can therefore differ only by a multiple of 2π:

$$\arg\big((u_1 - u_2)(u_4 - u_3)\big) = \arg\big((u_1 - u_4)(u_3 - u_2)\big) + 2n\pi$$

for some integer n. (In fact $n = 0$, since two principal arguments that differ by a multiple of 2π must be equal, but the weaker statement is all that was asked.)

The final part

Call the three products

$$A = (u_1 - u_2)(u_4 - u_3), \qquad B = (u_1 - u_4)(u_3 - u_2), \qquad C = (u_1 - u_3)(u_4 - u_2).$$

Expanding,

$$\begin{aligned} A + B &= (u_1u_4 - u_1u_3 - u_2u_4 + u_2u_3) + (u_1u_3 - u_1u_2 - u_3u_4 + u_2u_4) \\ &= u_1u_4 + u_2u_3 - u_1u_2 - u_3u_4, \\ C &= u_1u_4 - u_1u_2 - u_3u_4 + u_2u_3, \end{aligned}$$

so $A + B = C$. This is pure algebra and holds for any four complex numbers.

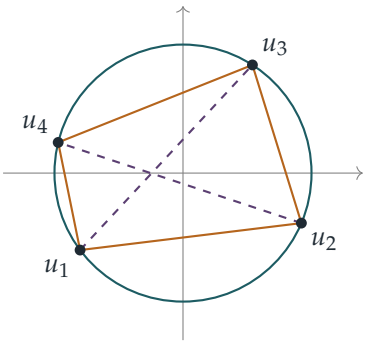
Now use the previous part: $A = a\,\mathrm{e}^{\mathrm{i}\Sigma/2}$ and $B = b\,\mathrm{e}^{\mathrm{i}\Sigma/2}$ with $a = 4s_{12}s_{34} > 0$ and $b = 4s_{14}s_{23} > 0$. Then $|A| = a$, $|B| = b$, and

$$|C| = |A + B| = \left|(a + b)\,\mathrm{e}^{\mathrm{i}\Sigma/2}\right| = a + b = |A| + |B|,$$

because $a+b>0$ and $|\mathrm{e}^{\mathrm{i}\Sigma/2}|=1$. That is,

$$|(u_1-u_2)(u_4-u_3)|+|(u_1-u_4)(u_3-u_2)|=|(u_1-u_3)(u_4-u_2)|.$$

The point is that the triangle inequality $|A+B|\le|A|+|B|$ becomes an equality exactly when A and B point the same way, and that is what the middle part established.



Geometrically, $|u_j-u_k|$ is the length of a chord, and because the arguments increase, $u_1u_2u_3u_4$ is a quadrilateral inscribed in the circle with its vertices in order. The identity says that the products of opposite sides add up to the product of the diagonals (dashed): this is Ptolemy's theorem.

Checks

- A square: $\theta_k=-\frac{3\pi}{4},-\frac{\pi}{4},\frac{\pi}{4},\frac{3\pi}{4}$. Each side is $\sqrt{2}$ and each diagonal is 2, and indeed $2+2=4$.
- With $\theta=(-2.5,-0.4,1.0,2.9)$ we get $|A|\approx 2.8223$, $|B|\approx 1.1013$ and $|C|\approx 3.9236=|A|+|B|$.
- Over 2×10^5 random ordered choices of the θ_k, the formula for $\arg(u_1-u_2)$ held with $n=0$ or 1 every time (both values occur), and the identity held to within 2×10^{-15}.

Common slips

- Writing $u_1-u_2=2\mathrm{i}\sin\frac{1}{2}(\theta_1-\theta_2)\,\mathrm{e}^{\mathrm{i}(\theta_1+\theta_2)/2}$ and reading off the argument as $\frac{1}{2}(\theta_1+\theta_2+\pi)$, forgetting that the sine is negative.
- Treating u_4-u_3 as if it had the same argument as u_3-u_4: reversing a difference adds π to the argument.
- Asserting $|A|+|B|=|A+B|$ without saying why; it needs A and B to have the same argument.

Beyond STEP

Ptolemy's theorem has a converse, and a more general form: for any four points in the plane, $|A| + |B| \geq |C|$ (just the triangle inequality applied to $A + B = C$), with equality exactly when the points lie in order on a circle or a line. Behind this lies the *cross-ratio* $\frac{(u_1-u_2)(u_3-u_4)}{(u_1-u_4)(u_3-u_2)}$, which is real precisely when four points are concyclic and is preserved by Möbius transformations, a central object in first-year complex analysis and geometry. Needham's *Visual Complex Analysis* develops these ideas with wonderful pictures.

WORKED SOLUTION

STEP III 2002, Question 9

Mechanics • Centre of Mass and Toppling

STEP III 2002, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q9

The key idea

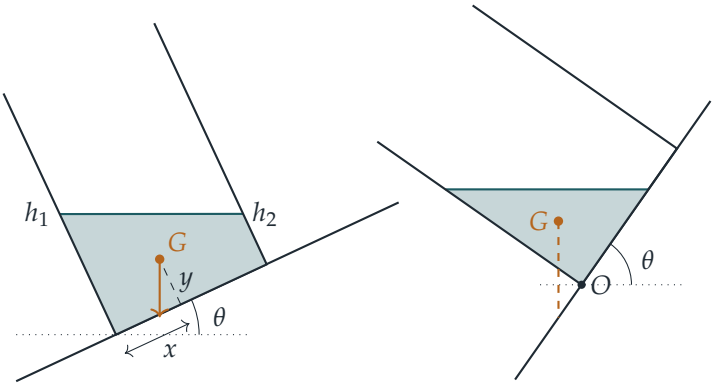
The free surface of the fluid is horizontal, so in the frame of the container (which is tilted at θ) it is a straight line sloping at angle θ to the base. A cross-section through the fluid is therefore a trapezium if the surface meets both side walls, and a right-angled triangle if it reaches the base before the upper wall. Finding the centre of mass is then a two-dimensional centroid calculation. For toppling, the container is light, so everything depends on where the vertical through the fluid's centre of mass meets the plane.

Solution

Setting up

We model the container as a light prism with walls of negligible thickness, the fluid as a uniform continuum at rest relative to the container, and the plane as rough enough to prevent slipping. Each vertical cross-section perpendicular to the horizontal base edges is the same, and the fluid is uniform, so the centre of mass of the fluid is the centroid of one cross-section. That cross-section has width a and area $ka^3/a = ka^2$.

Measure X along the base from the lower side wall (up the slope) and Y perpendicular to the base. The base rises at angle θ to the horizontal as X increases, so the horizontal surface falls at angle θ relative to the base: its height above the base drops by $\tan\theta$ for every unit of X. Write $d = a\tan\theta$ for the total drop across the width.



The left picture shows the case $k > \frac{1}{2}\tan\theta$ (the fluid touches both walls); the right shows $k < \frac{1}{2}\tan\theta$ with $\theta > 45°$, where the vertical through G misses the base.

The case $k > \frac{1}{2}\tan\theta$

Let the depths at the lower and upper walls be h_1 and h_2. Then $h_1 - h_2 = d$, and the trapezium has area $\frac{1}{2}a(h_1 + h_2) = ka^2$, so $h_1 + h_2 = 2ka$. Hence

$$h_2 = ka - \tfrac{1}{2}d = a\big(k - \tfrac{1}{2}\tan\theta\big),$$

which is positive exactly when $k > \frac{1}{2}\tan\theta$. So this is the case where the surface meets both walls.

Split the trapezium into a rectangle $a \times h_2$ and, on top of it, a right-angled triangle with legs a (along $Y = h_2$) and d (up the lower wall).

- Rectangle: area ah_2, centroid $\big(\frac{1}{2}a,\ \frac{1}{2}h_2\big)$.
- Triangle: vertices $(0, h_2)$, (a, h_2), $(0, h_2 + d)$, so area $\frac{1}{2}ad$ and centroid (the average of the vertices) $\big(\frac{1}{3}a,\ h_2 + \frac{1}{3}d\big)$.

Taking moments about the lower wall, with total area ka^2,

$$ka^2\,x = ah_2 \cdot \tfrac{1}{2}a + \tfrac{1}{2}ad \cdot \tfrac{1}{3}a \quad\Longrightarrow\quad kx = \tfrac{1}{2}h_2 + \tfrac{1}{6}d = \tfrac{1}{2}ka - \tfrac{1}{4}d + \tfrac{1}{6}d = \tfrac{1}{2}ka - \tfrac{1}{12}d.$$

Dividing by ka and using $d = a\tan\theta$,

$$\frac{x}{a} = \frac{1}{2} - \frac{\tan\theta}{12k}.$$

Taking moments about the base,

$$ka^2\,y = ah_2 \cdot \tfrac{1}{2}h_2 + \tfrac{1}{2}ad\big(h_2 + \tfrac{1}{3}d\big),$$
$$ka\,y = \tfrac{1}{2}h_2^2 + \tfrac{1}{2}dh_2 + \tfrac{1}{6}d^2 = \tfrac{1}{2}h_2(h_2 + d) + \tfrac{1}{6}d^2.$$

Now $h_2 = ka - \frac{1}{2}d$ and $h_2 + d = ka + \frac{1}{2}d$, so $h_2(h_2 + d) = k^2a^2 - \frac{1}{4}d^2$ and

$$ka\,y = \tfrac{1}{2}k^2a^2 - \tfrac{1}{8}d^2 + \tfrac{1}{6}d^2 = \tfrac{1}{2}k^2a^2 + \tfrac{1}{24}d^2.$$

Dividing by ka^2,

$$\frac{y}{a} = \frac{k}{2} + \frac{\tan^2\theta}{24k},$$

as required.

The case $k < \frac{1}{2}\tan\theta$

Now h_2 would be negative, so the surface meets the base at some distance $b < a$ from the lower wall, and the fluid's cross-section is a right-angled triangle with legs b along the base and h up the lower wall. The surface slope gives $h = b\tan\theta$, and the area gives $\frac{1}{2}bh = ka^2$. So $b^2\tan\theta = 2ka^2$, and

$$b = a\sqrt{\frac{2k}{\tan\theta}}, \qquad h = a\sqrt{2k\tan\theta}.$$

(Indeed $b < a$ precisely because $2k < \tan\theta$; the container is tall, so h causes no trouble.) The centroid of the triangle is a third of the way along each leg, so

$$\frac{x}{a} = \frac{1}{3}\sqrt{\frac{2k}{\tan\theta}} = \frac{1}{3}\sqrt{2k\cot\theta}, \qquad \frac{y}{a} = \frac{1}{3}\sqrt{2k\tan\theta}.$$

At $k = \frac{1}{2}\tan\theta$ both pairs of formulae give $x/a = \frac{1}{3}$ and $y/a = \frac{1}{3}\tan\theta$, so the two cases join up.

Toppling

> **Ask yourself** What decides whether the container tips over the moment it is let go?

The container is light, so the only weight is that of the fluid, acting vertically through its centre of mass G. The plane can push only on the base, so the normal reaction acts somewhere on the base, between the lower edge O and the upper edge. If the vertical through G meets the plane downhill of O, the weight has a moment about O that no such reaction can balance, and the container rotates about O: it topples.

Take O as origin, with x up the slope and y perpendicular to it. The vertical direction downwards has components $(-\sin\theta, -\cos\theta)$ in these axes, so going down the vertical from $G = (x, y)$ through a distance $y\sec\theta$ reaches the plane at

$$x - y\tan\theta.$$

So the container topples if $x < y\tan\theta$.

Suppose $k < \frac{1}{2}$ and $\theta > 45°$. Then $\tan\theta > 1 > 2k$, so $k < \frac{1}{2}\tan\theta$ and we are in the triangular case. There

$$y\tan\theta - x = \frac{a}{3}\left(\sqrt{2k\tan\theta}\,\tan\theta - \sqrt{\frac{2k}{\tan\theta}}\right) = \frac{a}{3}\sqrt{\frac{2k}{\tan\theta}}\left(\tan^2\theta - 1\right) > 0,$$

because $\tan\theta > 1$. Hence $x < y\tan\theta$: the vertical through G passes below the lower edge, and the container topples, whatever the value of $k < \frac{1}{2}$. (In the triangular case the same calculation shows the container is stable if $\theta < 45°$, so $45°$ is exactly the threshold there.) This is the result. The condition $k < \frac{1}{2}$ is what guarantees the triangular case when $\theta > 45°$.

Checks

- With $\theta = 0$ the first case gives $x = \frac{1}{2}a$ and $y = \frac{1}{2}ka$, the centre of a block of fluid of depth ka.
- The two cases agree at $k = \frac{1}{2}\tan\theta$, where the trapezium degenerates into the triangle.
- A numerical check: integrating the fluid region directly for 300 random pairs (k, θ) matched both pairs of formulae to within 1.4×10^{-6} (with $a = 1$). For example, $\theta = 55°$, $k = 0.3$ gives $x \approx 0.216a$, $y \approx 0.309a$, and $y\tan\theta \approx 0.441a > x$, so it topples.

Common slips

- Drawing the surface parallel to the base, or sloping the wrong way: the fluid piles up against the *lower* wall.
- Forgetting that the container is light, and adding a weight for the container: its centre of mass is that of the fluid.
- Testing the vertical through G against the centre of the base rather than against the lower edge O.

Beyond STEP

The test "does the vertical through the centre of mass meet the base?" is the simplest case of a stability analysis. Floating ships are treated the same way, but there the buoyancy acts through the centre of mass of the displaced water, which moves as the ship heels; this leads to the idea of the metacentre. Moving fluid (sloshing after release) is a much harder problem in fluid dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body equilibrium and rotation about an axis carefully.

WORKED SOLUTION

STEP III 2002, Question 10

Mechanics • Rotation with Friction

STEP III 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q10

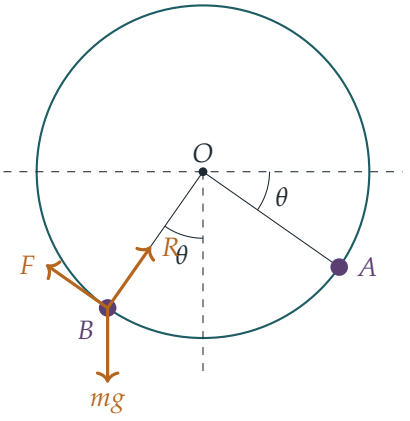
The key idea

As long as the second particle does not slip, it moves with the cylinder, so the whole system is a rigid body turning about a fixed axis: two equal masses at radius a, a quarter-turn apart. Energy (or moments about the axis) then gives the motion in one line. To find when slipping starts, we look at the loose particle on its own and use Newton's second law along and across the surface to find the friction F and the normal reaction R it needs. It slips when F first reaches μR.

Solution

Setting up

Model both particles as point masses on a light cylinder, with a smooth axle. Let O be the axis, A the fixed particle and B the loose one. Released from rest, A falls, so the cylinder turns (clockwise in the figure) and carries B up the side away from A. After a turn through θ, OA is at angle θ below the horizontal and OB is at angle θ to the downward vertical, on the other side.



The equation of motion

While B does not slip, both particles move on circles of radius a with angular speed $\dot\theta$, and the cylinder itself has no mass. Taking the level of O as zero potential energy, A is at height $-a\sin\theta$ and B at height $-a\cos\theta$. The system starts from rest at $\theta = 0$, and no work is done by the smooth axle, while friction between B and the cylinder does no net work because there is no relative motion. So energy is conserved:

$$2\cdot\tfrac{1}{2}ma^2\dot\theta^2 - mga\sin\theta - mga\cos\theta = -mga,$$

that is,

$$\dot\theta^2 = \frac{g}{a}(\sin\theta + \cos\theta - 1). \qquad (*)$$

Differentiating with respect to time and dividing by $2\dot\theta$ (which is non-zero once motion has started) gives

$$\ddot\theta = \frac{g}{2a}(\cos\theta - \sin\theta),$$

as required. Equivalently, taking moments about O: the weight of A has moment $mga\cos\theta$ in the direction of rotation, that of B has moment $mga\sin\theta$ against it, and the moment of inertia is $2ma^2$.

When does B slip?

Now look at B alone. The forces on it are its weight mg, the normal reaction R towards O, and friction F along the surface, which we take as positive in B's direction of motion (up the side). B moves on a circle of radius a, so it has acceleration $a\dot\theta^2$ towards O and $a\ddot\theta$ along its direction of motion. The weight has component $mg\cos\theta$ directly away from O and $mg\sin\theta$ against the motion. Hence

$$\begin{aligned}\text{towards } O:&\quad R - mg\cos\theta = ma\dot\theta^2,\\ \text{along the motion:}&\quad F - mg\sin\theta = ma\ddot\theta.\end{aligned}$$

Using $(*)$ and the equation of motion,

$$\begin{aligned}R &= mg\cos\theta + mg(\sin\theta + \cos\theta - 1) = mg\,(2\cos\theta + \sin\theta - 1),\\ F &= mg\sin\theta + \tfrac{1}{2}mg(\cos\theta - \sin\theta) = \tfrac{1}{2}mg\,(\sin\theta + \cos\theta).\end{aligned}$$

For $0 \le \theta \le \frac{\pi}{3}$ both are positive. ($R > 0$ because $2\cos\theta + \sin\theta - 1 \ge \sin\theta > 0$ for $0 < \theta \le \frac{\pi}{3}$, using $2\cos\theta \ge 1$, and $R = mg$ at $\theta = 0$.) So B stays in contact, and friction pushes it up the side. The particle does not slip as long as $F \le \mu R$, that is, as long as

$$\rho(\theta) = \frac{F}{R} = \frac{\sin\theta + \cos\theta}{2(2\cos\theta + \sin\theta - 1)} \le \mu.$$

At $\theta = \frac{\pi}{3}$, where $\sin\theta = \frac{\sqrt3}{2}$ and $\cos\theta = \frac{1}{2}$,

$$\rho\left(\tfrac{\pi}{3}\right) = \frac{\frac{1}{2}(\sqrt3+1)}{2\cdot\frac{\sqrt3}{2}} = \frac{\sqrt3+1}{2\sqrt3} = \frac{3+\sqrt3}{6} = \mu.$$

Ask yourself Why can't the particle have started to slip earlier than this?

We need $\rho(\theta) < \mu$ for all $0 \le \theta < \frac{\pi}{3}$. Write $N = \sin\theta + \cos\theta$ and $D = 2\cos\theta + \sin\theta - 1$. Then

$$\begin{aligned} N'D - ND' &= (\cos\theta - \sin\theta)(2\cos\theta + \sin\theta - 1) - (\sin\theta + \cos\theta)(\cos\theta - 2\sin\theta) \\ &= \cos^2\theta + \sin^2\theta + \sin\theta - \cos\theta = 1 + \sin\theta - \cos\theta, \end{aligned}$$

which is positive for $0 < \theta < \pi$. Since $D > 0$ on $[0, \frac{\pi}{3}]$, $\rho' = (N'D - ND')/(2D^2) > 0$ there, so ρ is strictly increasing from $\rho(0) = \frac{1}{2}$. It therefore stays below μ until it reaches μ exactly at $\theta = \frac{\pi}{3}$. Also, by $(*)$, $\dot\theta^2 > 0$ for $0 < \theta < \frac{\pi}{2}$, so the cylinder really does keep turning and reaches this angle.

So the second particle starts to slip when the cylinder has turned through 60°, as required.

Checks

- At $\theta = 0$: B is at rest at the bottom, yet $F = \frac{1}{2}mg \neq 0$. That is right: A's weight is trying to turn the cylinder, and friction must accelerate B along with it. Indeed $F = ma\ddot\theta = \frac{1}{2}mg$ there, and $R = mg$ because B has no speed yet.
- At $\theta = \frac{\pi}{4}$, $\ddot\theta = 0$ and the angular speed is a maximum, as expected when the two weights balance about O.
- Numerically, integrating the equation of motion and computing the contact force on B directly from its acceleration reproduces the formulas for R and F, and F/R first reaches $(3+\sqrt{3})/6 \approx 0.7887$ at 60.00°.

Common slips

- Including a moment of inertia for the cylinder: it is light, so only the two particles contribute $2ma^2$.
- Forgetting the tangential acceleration $a\ddot\theta$ of B, which would give $F = mg\sin\theta$ and the wrong angle.
- Checking only that $F = \mu R$ at 60° without showing that F/R was smaller before then.

Beyond STEP

This is a one-degree-of-freedom system: once the constraint (no slipping) is imposed, a single coordinate θ describes everything, and energy gives the motion directly. The constraint force is found afterwards, and the constraint fails when the force it needs exceeds what the surface can supply. This is the same logic as a particle leaving a

sphere, and in Lagrangian mechanics it is made systematic with Lagrange multipliers. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and constraint forces in this spirit.

WORKED SOLUTION

STEP III 2002, Question 11

Mechanics • Oblique Impact with Walls

STEP III 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q11

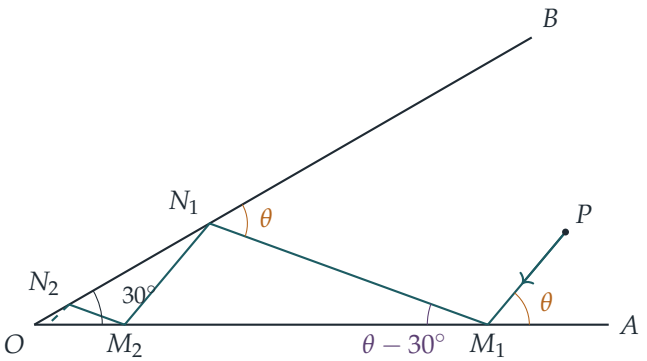
This question has a full guided treatment in *Beyond STEP*, Section 10.5, *Infinitely Many Bounces in Finite Time*: https://beyondstep.keithmcnulty.org/ch10/zeno-wedge.html

The key idea

At a smooth wall the impulse is perpendicular to the wall. So the velocity component along the wall survives unchanged and only the perpendicular component is multiplied by e. Combine this with the angles of the triangle OM_1N_1 and the first part reduces to one trigonometric equation. For the second part, the motion repeats itself on a smaller scale after each bounce. The legs get shorter, but the particle also slows down, so we have to add up the *times*, not the distances. That sum is a geometric series.

Solution

We model the particle as a point moving on a smooth horizontal surface, so between impacts it moves in a straight line at constant speed. The walls are smooth, and each impact obeys Newton's experimental law with coefficient e.



Finding e and v_1/u_1

Since angle $PM_1A = \theta$, the particle reaches M_1 travelling towards O, at angle θ to the wall. Just before impact, its velocity components are

$u_1\cos\theta$ along the wall (towards O), $\qquad$ $u_1\sin\theta$ perpendicular to the wall (into it).

The wall is smooth, so the impulse on the particle is perpendicular to OA. Hence:

along the wall: $u_1\cos\theta$ (unchanged),
perpendicular: $e\,u_1\sin\theta$ (reversed and multiplied by e).

So the particle leaves M_1 still moving towards O, at an angle φ to the wall, where

$$\tan\varphi = \frac{e\,u_1\sin\theta}{u_1\cos\theta} = e\tan\theta, \qquad v_1\cos\varphi = u_1\cos\theta. \tag{$*$}$$

Next we find φ from the geometry of triangle OM_1N_1. The points O, N_1, B lie on one line, and angle $M_1N_1B = \theta$, so angle $ON_1M_1 = 180^\circ - \theta$. The angle at O is 30°, so

$$\varphi = \angle OM_1N_1 = 180^\circ - 30^\circ - (180^\circ - \theta) = \theta - 30^\circ.$$

For φ to be positive we need $\theta > 30^\circ$, which this configuration implicitly assumes. Then $(*)$ gives

$$\boxed{e = \frac{\tan(\theta - 30^\circ)}{\tan\theta}}$$

and

$$\boxed{\frac{v_1}{u_1} = \frac{\cos\theta}{\cos(\theta - 30^\circ)} = \frac{2}{\sqrt{3} + \tan\theta}},$$

where the last form comes from expanding $\cos(\theta - 30^\circ) = \frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta$. If you prefer, write $e = \dfrac{\sqrt{3}\tan\theta - 1}{\tan\theta\,(\sqrt{3} + \tan\theta)}$. Since $0 < \theta - 30^\circ < \theta < 90^\circ$, we get $0 < e < 1$, as the question requires.

Reaching O in a finite time

At N_1 the particle arrives travelling towards O, at angle θ to the wall OB (because angle $M_1N_1B = \theta$). This is exactly the situation at M_1, with the roles of the walls swapped. So the same calculation applies: it rebounds at angle $\theta - 30^\circ$ to N_1O, and its speed is multiplied by

$$r = \frac{\cos\theta}{\cos(\theta - 30^\circ)}.$$

Ask yourself Does the particle arrive at M_2 at angle θ again, so that the pattern really repeats?

Yes. In triangle ON_1M_2 the angle at O is 30° and the angle at N_1 is $\theta - 30^\circ$. So angle $OM_2N_1 = 180^\circ - \theta$, which means angle $N_1M_2A = \theta$. By induction, every impact happens at angle θ to the wall. Every leg leaves its wall at angle $\theta - 30^\circ$. And every triangle OM_nN_n, ON_nM_{n+1} has angles 30°, $\theta - 30^\circ$, $180^\circ - \theta$, so they are all similar.

Lengths. By the sine rule in triangle OM_1N_1,

$$\frac{ON_1}{\sin(\theta - 30^\circ)} = \frac{OM_1}{\sin(180^\circ - \theta)} = \frac{M_1N_1}{\sin 30^\circ}.$$

So each new triangle is the previous one scaled by

$$k = \frac{ON_1}{OM_1} = \frac{\sin(\theta - 30^\circ)}{\sin\theta}.$$

Call the legs M_1N_1, N_1M_2, $M_2N_2, \ldots$ legs $1, 2, 3, \ldots$. Leg n has length $\ell_1 k^{n-1}$, where $\ell_1 = M_1N_1 = \dfrac{OM_1}{2\sin\theta}$.

Speeds. Leg n is travelled at constant speed $v_1 r^{n-1}$.

Times. Leg n therefore takes time

$$\tau_n = \frac{\ell_1 k^{n-1}}{v_1 r^{n-1}} = \frac{\ell_1}{v_1} q^{n-1}, \qquad q = \frac{k}{r} = \frac{\sin(\theta - 30^\circ)\cos(\theta - 30^\circ)}{\sin\theta\cos\theta} = \frac{\sin(2\theta - 60^\circ)}{\sin 2\theta}.$$

Ask yourself The legs shrink geometrically, but the particle is slowing down too. Why does that not settle the matter?

The times are what we add up, and the time ratio q compares the shrinking of the legs (k) with the slowing of the particle (r). We need $q < 1$. Now suppose $30^\circ < \theta < 60^\circ$. Then

$$0^\circ < 2\theta - 60^\circ < 60^\circ \;\Rightarrow\; \sin(2\theta - 60^\circ) < \tfrac{\sqrt{3}}{2}, \qquad 60^\circ < 2\theta < 120^\circ \;\Rightarrow\; \sin 2\theta > \tfrac{\sqrt{3}}{2},$$

so $0 < q < 1$. The total time is then a convergent geometric series:

$$T = \frac{PM_1}{u_1} + \sum_{n=1}^{\infty} \frac{\ell_1}{v_1} q^{n-1} = \frac{PM_1}{u_1} + \frac{\ell_1}{v_1(1-q)}.$$

This is finite. To simplify, use $\sin 2\theta - \sin(2\theta - 60^\circ) = 2\cos(2\theta - 30^\circ)\sin 30^\circ$, which gives $1 - q = \dfrac{\cos(2\theta - 30^\circ)}{\sin 2\theta}$. Then the time from M_1 onwards is

$$\frac{\ell_1}{v_1(1-q)} = \frac{OM_1\cos(\theta - 30^\circ)}{u_1\cos(2\theta - 30^\circ)},$$

which is positive and finite for $30^\circ < \theta < 60^\circ$.

Finally, the particle really does arrive at O. After the nth impact its distance from O is $OM_1k^n \to 0$, because $0 < k < 1$. During a leg, its distance from O is at most the larger of the distances of the leg's two ends. So as $t \to T$ the particle's distance from O tends to 0: it reaches O at the finite time T, after infinitely many impacts. ■

Checks

- At $\theta = 45°$: $e = \tan 15° = 2 - \sqrt{3} \approx 0.268$, $v_1/u_1 = \sqrt{3} - 1 \approx 0.732$, and $q = \sin 30° / \sin 90° = \frac{1}{2}$, so each leg takes half as long as the one before. With $OM_1 = u_1 = 1$, the time from M_1 to O is $\cos 15° / \cos 60° = 2\cos 15° \approx 1.932$. A step-by-step simulation of the impacts (sixty bounces) gives 1.931852, in agreement.
- At $\theta = 60°$: $q = 1$, so every leg takes the same time and the total is infinite. For $\theta > 60°$, $q > 1$. So $60°$ is exactly the borderline, and the formula for T blows up there through $\cos(2\theta - 30°) = \cos 90° = 0$.
- As $\theta \to 30°^+$, $e \to 0$: the particle barely bounces and slides almost straight to O, as you would expect.

Common slips

- Getting the triangle wrong: angle M_1N_1B is the *exterior* angle at N_1, so the interior angle is $180° - \theta$ and the rebound angle at M_1 is $\theta - 30°$, not $\theta + 30°$.
- Arguing that the particle reaches O just because the distances $OM_n \to 0$. Infinitely many legs need a convergent series of *times*, and the particle slows down at each impact.
- Reversing the tangential component at impact. On a smooth wall only the normal component changes.

Beyond STEP

Infinitely many impacts in a finite time is called *Zeno behaviour*. The familiar example is a ball bouncing on the floor with $e < 1$, and here the wedge plays the role of the floor. Mathematicians studying 'hybrid' systems, which mix smooth motion with sudden jumps, must decide what happens after the Zeno time. In granular materials the same effect, called *inelastic collapse*, causes real trouble in simulations. For collisions and impulse treated carefully at first-year level, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2002, Question 12

Probability and Statistics • Expectation by Conditioning

STEP III 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q12

The key idea

Condition on how the game *starts*. Every game begins with H, with TH, or with TT. In the first two cases nothing useful has been achieved (the run of tails has been broken), so the game simply starts afresh, and its expected remaining length is E again. This "renewal" gives a linear equation for E that we can solve. The same idea, splitting by when the first head appears, handles r tails.

Solution

Throughout, p is the probability of a head, $q = 1 - p$ is the probability of a tail, and tosses are independent. We assume $0 < p < 1$, so that the game ends with probability 1 and every expectation below is finite.

Explaining the equation for E

Look at the first one or two tosses. Exactly one of these happens:

- **H first** (probability p). One toss has been used and there is no run of tails in progress. Because the tosses are independent, the rest of the game is a fresh copy of the whole game, with expected length E. Expected total: $1 + E$.
- **T then H** (probability qp). Two tosses used, and the head has again wiped out the run of tails, so the game restarts. Expected total: $2 + E$.
- **T then T** (probability q^2). The game ends after exactly 2 tosses.

These three cases are exclusive and exhaustive (their probabilities add to $p + qp + q^2 = p + q = 1$). By the law of total expectation,

$$E = p(1 + E) + (1 - p)p(2 + E) + 2(1 - p)^2,$$

which is the given equation.

Finding E

Collect the E terms:

$$E\,(1-p-qp) = p + 2qp + 2q^2.$$

On the left, $1-p-qp = q-qp = q(1-p) = q^2$. On the right, $p+2q(p+q) = p+2q = 1+q$. Since $q \neq 0$,

$$E = \frac{1+q}{q^2} = \frac{2-p}{(1-p)^2}.$$

Expected winnings

Let W be the expected number of heads, which is the expected winnings in pounds. Use the same three cases. After H we have won £1 and the game restarts; after TH we have also won £1 and the game restarts; after TT we have won nothing and the game is over. So

$$W = p(1+W) + qp(1+W) + q^2 \cdot 0.$$

> **Ask yourself** Why is it $1+W$, not W, in the TH case?

The second toss in that case *is* a head, so it earns £1 before the game restarts. Forgetting it is the easiest way to go wrong here.

Collecting terms, $W(1-p-qp) = p(1+q)$, so $Wq^2 = p(1+q)$ and

$$W = \frac{p(1+q)}{q^2} = \frac{p(2-p)}{(1-p)^2} \text{ pounds.}$$

Notice that $W = pE$: on average a proportion p of the tosses are heads, even though the game stops at a moment that depends on the tosses.

The game with r successive tails

Now let E be the expected number of tosses when the game stops at r successive tails. Condition on the first head. Either the first r tosses are all tails, or the first head appears at toss $j+1$ for some j with $0 \leq j \leq r-1$:

- j **tails then a head** (probability $q^j p$). That uses $j+1$ tosses, the run of tails has been broken, and the game restarts. Expected total: $j+1+E$.
- r **tails** (probability q^r). The game ends after exactly r tosses.

So

$$E = \sum_{j=0}^{r-1} q^j p\,(j+1+E) + rq^r.$$

The coefficient of E on the right is $p\sum_{j=0}^{r-1} q^j = p \cdot \dfrac{1-q^r}{1-q} = 1-q^r$, since $1-q = p$. Moving it to the left,

$$q^r E = p\sum_{j=0}^{r-1}(j+1)q^j + rq^r. \qquad (*)$$

? Ask yourself How do we sum $\sum(j+1)q^j$ without a formula sheet?

Write $S = \sum_{j=0}^{r-1}(j+1)q^j = 1 + 2q + 3q^2 + \cdots + rq^{r-1}$ and subtract $qS = q + 2q^2 + \cdots + (r-1)q^{r-1} + rq^r$:

$$(1-q)S = 1 + q + q^2 + \cdots + q^{r-1} - rq^r = \frac{1-q^r}{p} - rq^r.$$

Hence $pS = \dfrac{1-q^r}{p} - rq^r$, and substituting into $(*)$, the rq^r terms cancel:

$$q^r E = \frac{1-q^r}{p} - rq^r + rq^r = \frac{1-q^r}{p}.$$

Therefore

$$E = \frac{1-q^r}{pq^r},$$

as required. For $r = 2$ this is $\dfrac{(1-q)(1+q)}{pq^2} = \dfrac{1+q}{q^2}$, agreeing with the first game.

Checks

- With a fair coin, $p = q = \frac{1}{2}$: the first game lasts $E = \frac{3/2}{1/4} = 6$ tosses on average and wins $W = £3$, half of E as it should be.
- $r = 1$: the game stops at the first tail, so the length is geometric with mean $1/q$; the formula gives $\dfrac{1-q}{pq} = \dfrac{1}{q}$.
- As $q \to 0$ (almost every toss a head) both E and W blow up, as they should.
- We confirmed these formulae exactly with fractions by solving the chain of states "current run of k tails" for $r = 1, \ldots, 7$ and several values of p. A simulation of 2×10^5 games with $p = 0.4$ gave a mean of 4.438 tosses (theory 4.444) and winnings £1.772 (theory £1.778) for $r = 2$, and 9.076 tosses (theory 9.074) for $r = 3$.

Common slips

- In the winnings equation, writing $qp\,W$ instead of $qp(1+W)$, which gives the wrong answer p/q^2.
- Conditioning on only the first toss: after a single T the game does *not* restart, because one tail of the run has already been banked. That is why the given equation looks two tosses ahead.

Beyond STEP

The argument "once a head appears, the future looks exactly like the start" is the *renewal* or *first-step* method for Markov chains, and it is how expected hitting times are computed in general. The neat fact $W = pE$ is a case of Wald's identity: the expected sum of a random number N of independent, identically distributed terms is $\mathbb{E}[N]$ times the mean of one term, provided N is a stopping time. You can read about both in Grimmett and Welsh, *Probability: An Introduction*, or in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 2002, Question 13

Probability and Statistics • Minimum of Exponentials

STEP III 2002, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q13

The key idea

"At least one of them is less than x" is the complement of "both are at least x", and the probability of the second event factorises by independence. Exponential tails are so simple, $\mathrm{P}(X_i \geq x) = \mathrm{e}^{-\lambda_i x}$, that the product is again an exponential tail, and that is the whole of the proof about the minimum. The buses are then a direct application: the wait for the next bus is the smaller of two exponential gaps.

Solution

The probability that one or both is less than x

First the tail of a single exponential variable. For $x \geq 0$,

$$\mathrm{P}(X_i \geq x) = \int_x^\infty \lambda_i \mathrm{e}^{-\lambda_i t}\,\mathrm{d}t = \left[-\mathrm{e}^{-\lambda_i t}\right]_x^\infty = \mathrm{e}^{-\lambda_i x}.$$

Let E be the event that $X_1 < x$ or $X_2 < x$ (or both). Its complement is the event that *both* $X_1 \geq x$ and $X_2 \geq x$. Since X_1 and X_2 are independent,

$$\mathrm{P}(X_1 \geq x \text{ and } X_2 \geq x) = \mathrm{P}(X_1 \geq x)\,\mathrm{P}(X_2 \geq x) = \mathrm{e}^{-\lambda_1 x}\mathrm{e}^{-\lambda_2 x},$$

so, for $x \geq 0$,

$$\boxed{\mathrm{P}(E) = 1 - \mathrm{e}^{-(\lambda_1+\lambda_2)x}.}$$

(For $x < 0$ the probability is 0, since neither variable can be negative.)

Ask yourself Why not just add $\mathrm{P}(X_1 < x)$ and $\mathrm{P}(X_2 < x)$?

Because the events overlap: both can be less than x at once, and adding would count that case twice. You can do it by inclusion–exclusion, $(1 - \mathrm{e}^{-\lambda_1 x}) + (1 - \mathrm{e}^{-\lambda_2 x}) - (1 - \mathrm{e}^{-\lambda_1 x})(1 - \mathrm{e}^{-\lambda_2 x})$, which simplifies to the same answer, but the complement is quicker.

The minimum is exponential

Let $X = \min(X_1, X_2)$. The smaller of two numbers is less than x exactly when at least one of them is less than x, so $\{X < x\}$ is the event E above. Hence the distribution function of X is

$$F(x) = \mathrm{P}(X \le x) = 1 - \mathrm{e}^{-(\lambda_1+\lambda_2)x} \qquad (x \ge 0),$$

with $F(x) = 0$ for $x < 0$. (For a continuous variable $\mathrm{P}(X < x) = \mathrm{P}(X \le x)$.) Differentiating, the density of X is

$$f(x) = F'(x) = (\lambda_1 + \lambda_2)\,\mathrm{e}^{-(\lambda_1+\lambda_2)x} \qquad (0 \le x < \infty),$$

which is the exponential density with parameter $\lambda_1 + \lambda_2$. So X has an exponential distribution, as required.

The buses

An exponential distribution with parameter λ has mean $\int_0^\infty t\lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1/\lambda$ (integrating by parts). Measuring time in minutes, the mean gaps of 15 and 30 minutes give parameters

$$\lambda_A = \tfrac{1}{15}, \qquad \lambda_B = \tfrac{1}{30}.$$

A bus on each route has just left, so the time until the next Route A bus is a full gap T_A between Route A buses, and likewise T_B for Route B. These are independent, exponential with parameters λ_A and λ_B. Either route takes me to college, so my wait is $W = \min(T_A, T_B)$, and by the result just proved W is exponential with parameter

$$\lambda_A + \lambda_B = \tfrac{1}{15} + \tfrac{1}{30} = \tfrac{1}{10}.$$

The median m satisfies $\mathrm{P}(W \le m) = \frac{1}{2}$:

$$1 - \mathrm{e}^{-m/10} = \tfrac{1}{2} \iff \mathrm{e}^{-m/10} = \tfrac{1}{2} \iff m = 10\ln 2.$$

Since $\ln 2 \approx 0.693$, the median wait is $m \approx 6.93$ minutes, which is approximately 7 minutes, as required.

Checks

- If $\lambda_2 \to 0$ (Route B buses almost never come), the answer becomes $1 - \mathrm{e}^{-\lambda_1 x}$, just the distribution of X_1, as it should.
- The mean wait is 10 minutes, shorter than either route's mean of 15 or 30, and the median $10\ln 2$ is below the mean, as usual for a right-skewed distribution.
- A simulation of 2×10^5 pairs of independent exponential gaps with means 15 and 30 gave a median minimum of about 6.91 minutes, against $10\ln 2 = 6.931\ldots$.

Common slips

- Using the means as the parameters: an exponential with mean 15 has $\lambda = \frac{1}{15}$, not 15.
- Adding the two probabilities $\mathrm{P}(X_1 < x) + \mathrm{P}(X_2 < x)$, which double-counts and can exceed 1.
- Averaging the routes' medians, or taking the median of route A alone ($15\ln 2 \approx 10.4$): having two routes shortens the wait.

Beyond STEP

Exponential gaps between buses mean the buses arrive as a *Poisson process*, and the result here says that merging two independent Poisson processes gives another one whose rate is the sum of the rates. Together with the memoryless property, $\mathrm{P}(X > s + t \mid X > s) = \mathrm{P}(X > t)$, this is the foundation of queueing theory and continuous-time Markov chains, met in first-year probability courses. Grimmett and Welsh, *Probability: An Introduction*, covers exponential variables and the Poisson process.

WORKED SOLUTION

STEP III 2002, Question 14

Probability and Statistics • Variance and Covariance

STEP III 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q14

The key idea

The variance identity is just the definition of variance, expanded using linearity of expectation. The chess part is cleverer than it looks: we never compute $\mathrm{E}(WL)$ directly. Instead, we know the variance of $W + L$ from a binomial, we know $\mathrm{Var}(W)$ and $\mathrm{Var}(L)$ from binomials too, and the identity we have just proved hands us the covariance for free.

Solution

The variance identity

Write $\mu_X = \mathrm{E}(X)$ and $\mu_Y = \mathrm{E}(Y)$. We use the definitions

$$\mathrm{Var}(X) = \mathrm{E}\big[(X-\mu_X)^2\big], \qquad \mathrm{Cov}(X,Y) = \mathrm{E}\big[(X-\mu_X)(Y-\mu_Y)\big],$$

and linearity of expectation, $\mathrm{E}(aU + bV) = a\,\mathrm{E}(U) + b\,\mathrm{E}(V)$, which for discrete random variables follows by splitting the sum $\sum(au+bv)\,\mathrm{P}(U=u, V=v)$ into two sums and summing out the other variable in each.

By linearity, $\mathrm{E}(X+Y) = \mu_X + \mu_Y$, so

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathrm{E}\big[\big((X-\mu_X)+(Y-\mu_Y)\big)^2\big] \\
&= \mathrm{E}\big[(X-\mu_X)^2 + (Y-\mu_Y)^2 + 2(X-\mu_X)(Y-\mu_Y)\big] \\
&= \mathrm{E}\big[(X-\mu_X)^2\big] + \mathrm{E}\big[(Y-\mu_Y)^2\big] + 2\,\mathrm{E}\big[(X-\mu_X)(Y-\mu_Y)\big] \\
&= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y),
\end{aligned}$$

using linearity again in the third line. This is the required result. Note that no independence was needed: the identity holds for any X and Y.

Part (i)

Every game is a win, a loss or a draw, so $W + L + D = m$ whatever happens. A random variable that always takes the same value equals its mean, so its variance is $\mathrm{E}\big[(m-m)^2\big] = 0$.

Part (ii)

Call a game *decisive* if it is won or lost. Each game is decisive with probability $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$, the games are independent, and there are a fixed number m of them. So $W + L$, the number of decisive games, counts successes in m independent trials with the same success probability:

$$W + L \sim \mathrm{B}\big(m, \tfrac{2}{3}\big).$$

The correlation

In exactly the same way, W counts wins in m independent trials, each a win with probability $\frac{1}{3}$, so $W \sim \mathrm{B}(m, \frac{1}{3})$, and likewise $L \sim \mathrm{B}(m, \frac{1}{3})$. Using $\mathrm{Var} = np(1-p)$ for a binomial,

$$\mathrm{Var}(W) = \mathrm{Var}(L) = m \cdot \tfrac{1}{3} \cdot \tfrac{2}{3} = \frac{2m}{9}, \qquad \mathrm{Var}(W+L) = m \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} = \frac{2m}{9}.$$

Ask yourself How do we find $\mathrm{Cov}(W, L)$ without working out $\mathrm{E}(WL)$?

Run the identity backwards. With $X = W$ and $Y = L$,

$$\frac{2m}{9} = \frac{2m}{9} + \frac{2m}{9} + 2\,\mathrm{Cov}(W, L), \qquad \text{so} \qquad \mathrm{Cov}(W, L) = -\frac{m}{9}.$$

Therefore, for $m \geq 1$ (so that the variances are non-zero),

$$\frac{\mathrm{Cov}(W, L)}{\sqrt{\mathrm{Var}(W)\,\mathrm{Var}(L)}} = \frac{-m/9}{2m/9} = -\frac{1}{2}.$$

The answer does not depend on m. It is negative, as it should be: with the number of games fixed, more wins leave less room for losses.

Part (i) gives a second route. Applying the identity twice, $\mathrm{Var}(W + L + D)$ is the sum of the three variances plus twice the sum of the three pairwise covariances. By symmetry all three covariances are equal to some c, so $0 = 3 \cdot \frac{2m}{9} + 6c$, giving $c = -\frac{m}{9}$ again.

Checks

- Take $m = 1$. Then W and L are indicators with $\mathrm{E}(W) = \mathrm{E}(L) = \frac{1}{3}$ and $WL = 0$ always, so $\mathrm{Cov}(W, L) = 0 - \frac{1}{9} = -\frac{1}{9}$, matching $-m/9$.
- The two routes (via part (ii) and via part (i)) agree.
- An exact enumeration of all 3^m outcomes for $m = 1, \ldots, 7$ gives $\mathrm{Var}(W) = 2m/9$, $\mathrm{Cov}(W, L) = -m/9$ and a correlation of exactly $-\frac{1}{2}$ each time.

Common slips

- Assuming W and L are independent because the *games* are. They are not: they share a fixed total with D.
- Taking $\mathrm{Var}(W+L) = \mathrm{Var}(W) + \mathrm{Var}(L) = 4m/9$, which forgets the covariance term and contradicts the binomial answer $2m/9$.

Beyond STEP

(W, L, D) has a *multinomial* distribution, the natural generalisation of the binomial to more than two outcomes per trial. In general, if the outcome probabilities are $p_1, \ldots, p_k$, the counts N_i and N_j have covariance $-mp_ip_j$ and correlation $-\sqrt{p_ip_j/\big((1-p_i)(1-p_j)\big)}$, which gives $-\frac{1}{2}$ here. Covariance matrices and the bilinearity of covariance are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

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